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Fourier Analysis: An Introduction

Chapter 1. The Genesis of Fourier Analysis

1. Consider the Dirichlet problem illustrated in Figure 11.

More precisely, we look for a solution of the steady-state heat equation $\triangle u = 0$ in the rectangle $R = \{(x, y) : \ 0 \leq x \leq \pi, \ \ 0 \leq y \leq 1\}$ that vanishes on the vertical sides of R, and so that

$$u(x, 0) = f_0(x) \qquad \text{and} \qquad u(x, 1) = f_1(x)\,,$$

where f_0 and f_1 are initial data which fix the temperature distribution on the horizontal sides of the rectangle.

Use separation of variables to show that if f_0 and f_1 have Fourier expansions

$$f_0(x) = \sum_{k=1}^{\infty} A_k \sin kx \qquad \text{and} \qquad f_1(x) = \sum_{k=1}^{\infty} B_k \sin kx,$$

then

$$u(x, y) = \sum_{k=1}^{\infty} \left(\frac{\sinh k(1-y)}{\sinh k} A_k + \frac{\sinh ky}{\sinh k} B_k \right) \sin kx.$$

We recall the definitions of the hyperbolic sine and cosine functions:

$$\sinh x = \frac{e^x - e^{-x}}{2} \qquad \text{and} \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Compare this result with the solution of the Dirichlet problem in the strip obtained in Problem 3, Chapter 5.

Chapter 2. Basic Properties of Fourier Series

1. One can construct Riemann integrable functions on $[0,1]$ that have a dense set of discontinuities as follows.

(a) Let $f(x) = 0$ when $x < 0$, and $f(x) = 1$ if $x \geq 0$. Choose a countable dense sequence $\{r_n\}$ in $[0,1]$. Then, show that the function

$$F(x) = \sum_{n=1}^{\infty} \frac{1}{n^2} f(x - r_n)$$

is integrable and has discontinuities at all points of the sequence $\{r_n\}$. [Hint: F is monotonic and bounded.]

(b) Consider next

$$F(x) = \sum_{n=1}^{\infty} 3^{-n} g(x - r_n),$$

where $g(x) = \sin 1/x$ when $x \neq 0$, and $g(0) = 0$. Then F is integrable, discontinuous at each $x = r_n$, and fails to be monotonic in any subinterval of $[0,1]$. [Hint: Use the fact that $3^{-k} > \sum_{n>k} 3^{-n}$.]

(c) The original example of Riemann is the function

$$F(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2},$$

where $(x) = x$ for $x \in (-1/2, 1/2]$ and (x) is continued to $\mathbb{R}$ by periodicity, that is, $(x+1) = (x)$. It can be shown that F is discontinuous whenever $x = m/2n$, where $m, n \in \mathbb{Z}$ with m odd and $n \neq 0$.

2. Let D_N denote the Dirichlet kernel

$$D_N(\theta) = \sum_{k=-N}^{N} e^{ik\theta} = \frac{\sin((N+1/2)\theta)}{\sin(\theta/2)},$$

and define

$$L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)|\, d\theta.$$

(a) Prove that

$$L_N \geq c \log N$$

for some constant $c > 0$. [Hint: Show that $|D_N(\theta)| \geq c\frac{\sin((N+1/2)\theta)}{|\theta|}$, change variables, and prove that

$$L_N \geq c \int_{\pi}^{N\pi} \frac{|\sin\theta|}{|\theta|}\, d\theta + O(1).$$

Write the integral as a sum $\sum_{k=1}^{N-1} \int_{k\pi}^{(k+1)\pi}$. To conclude, use the fact that $\sum_{k=1}^{n} 1/k \geq c \log n$.] A more careful estimate gives

$$L_N = \frac{4}{\pi^2} \log N + O(1).$$

(b) Prove the following as a consequence: for each $n \geq 1$, there exists a continuous function f_n such that $|f_n| \leq 1$ and $|S_n(f_n)(0)| \geq c' \log n$. [Hint: The function g_n which is equal to 1 when D_n is positive and -1 when D_n is negative has the desired property but is not continuous. Approximate g_n in the integral norm (in the sense of Lemma 3.2) by continuous functions h_k satisfying $|h_k| \leq 1$.]

3.* Littlewood provided a refinement of Tauber's theorem:

(a) If $\sum c_n$ is Abel summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s.

(b) As a consequence, if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s.

These results may be applied to Fourier series. By Exercise 17, they imply that if f is an integrable function that satisfies $\hat{f}(\nu) = O(1/|\nu|)$, then:

(i) If f is continuous at θ, then

$$S_N(f)(\theta) \to f(\theta) \qquad \text{as } N \to \infty.$$

(ii) If f has a jump discontinuity at θ, then

$$S_N(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } N \to \infty.$$

(iii) If f is continuous on $[-\pi, \pi]$, then $S_N(f) \to f$ uniformly.

For the simpler assertion (b), hence a proof of (i), (ii), and (iii), see Problem 5 in Chapter 4.

Chapter 3. Convergence of Fourier Series

1. For each $0 < \alpha < 1$ the series

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n^{\alpha}}$$

converges for every x but is not the Fourier series of a Riemann integrable function.

(a) If the **conjugate Dirichlet kernel** is defined by

$$\tilde{D}_N(x) = \sum_{|n|\leq N} \operatorname{sign}(x)\, e^{inx} \qquad \text{where } \operatorname{sign}(x) = \begin{cases} 1 & \text{if } n > 0 \\ 0 & \text{if } n = 0 \\ -1 & \text{if } n < 0, \end{cases}$$

then show that

$$\tilde{D}_N(x) = \frac{\cos(x/2) - \cos((N+1/2)x)}{\sin(x/2)},$$

and

$$\int_{-\pi}^{\pi} |\tilde{D}_N(x)|\, dx \leq c \log N.$$

(b) As a result, if f is Riemann integrable, then

$$(f * \tilde{D}_N)(0) = O(\log N).$$

(c) In the present case, this leads to

$$\sum_{n=1}^{N} \frac{1}{n^{\alpha}} = O(\log N),$$

which is a contradiction.

2. An important fact we have proved is that the family $\{e^{inx}\}_{n\in\mathbb{Z}}$ is orthonormal in $\mathcal{R}$ and it is also complete, in the sense that the Fourier series of f converges to f in the norm. In this exercise, we consider another family possessing these same properties.

On $[-1,1]$ define

$$L_n(x) = \frac{d^n}{dx^n}(x^2-1)^n, \qquad n = 0, 1, 2, \ldots.$$

Then L_n is a polynomial of degree n which is called the nth **Legendre polynomial**.

(a) Show that if f is indefinitely differentiable on $[-1,1]$, then

$$\int_{-1}^{1} L_n(x) f(x)\, dx = (-1)^n \int_{-1}^{1} (x^2-1)^n f^{(n)}(x)\, dx.$$

In particular, show that L_n is orthogonal to x^m whenever $m < n$. Hence $\{L_n\}_{n=0}^{\infty}$ is an orthogonal family.

(b) Show that

$$\|L_n\|^2 = \int_{-1}^{1} |L_n(x)|^2\, dx = \frac{(n!)^2 2^{2n+1}}{2n+1}.$$

[Hint: First, note that $\|L_n\|^2 = (-1)^n(2n)!\int_{-1}^{1}(x^2-1)^n\, dx$. Write $(x^2-1)^n = (x-1)^n(x+1)^n$ and integrate by parts n times to calculate this last integral.]

(c) Prove that any polynomial of degree n that is orthogonal to $1, x, x^2, \ldots, x^{n-1}$ is a constant multiple of L_n.

(d) Let $\mathcal{L}_n = L_n/\|L_n\|$, which are the normalized Legendre polynomials. Prove that $\{\mathcal{L}_n\}$ is the family obtained by applying the "Gram-Schmidt process" to $\{1, x, \ldots, x^n, \ldots\}$, and conclude that every Riemann integrable function f on $[-1,1]$ has a **Legendre expansion**

$$\sum_{n=0}^{\infty} \langle f, \mathcal{L}_n\rangle \mathcal{L}_n$$

which converges to f in the mean-square sense.

3. Let α be a complex number not equal to an integer.

(a) Calculate the Fourier series of the 2π-periodic function defined on $[-\pi, \pi]$ by $f(x) = \cos(\alpha x)$.

(b) Prove the following formulas due to Euler:

$$\sum_{n=1}^{\infty} \frac{1}{n^2 - \alpha^2} = \frac{1}{2\alpha^2} - \frac{\pi}{2\alpha \tan(\alpha\pi)}.$$

For all $u \in \mathbb{C} - \pi\mathbb{Z}$,

$$\cot u = \frac{1}{u} + 2\sum_{n=1}^{\infty} \frac{u}{u^2 - n^2\pi^2}.$$

(c) Show that for all $\alpha \in \mathbb{C} - \mathbb{Z}$ we have

$$\frac{\alpha\pi}{\sin(\alpha\pi)} = 1 + 2\alpha^2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2 - \alpha^2}.$$

(d) For all $0 < \alpha < 1$, show that

$$\int_0^{\infty} \frac{t^{\alpha-1}}{t+1}\, dt = \frac{\pi}{\sin(\alpha\pi)}.$$

[Hint: Split the integral as $\int_0^1 + \int_1^{\infty}$ and change variables $t = 1/u$ in the second integral. Now both integrals are of the form

$$\int_0^1 \frac{t^{\gamma-1}}{1+t}\, dt, \qquad 0 < \gamma < 1,$$

which one can show is equal to $\sum_{k=0}^{\infty} \frac{(-1)^k}{k+\gamma}$. Use part (c) to conclude the proof.]

4. In this problem, we find the formula for the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^k}$$

where k is any even integer. These sums are expressed in terms of the Bernoulli numbers; the related Bernoulli polynomials are discussed in the next problem.

Define the **Bernoulli numbers** B_n by the formula

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n.$$

(a) Show that $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_3 = 0$, $B_4 = -1/30$, and $B_5 = 0$.

(b) Show that for $n \geq 1$ we have

$$B_n = -\frac{1}{n+1} \sum_{k=0}^{n-1} \binom{n+1}{k} B_k.$$

(c) By writing

$$\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{n=2}^{\infty} \frac{B_n}{n!} z^n,$$

show that $B_n = 0$ if n is odd and > 1. Also prove that

$$z \cot z = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n}.$$

(d) The **zeta function** is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \text{for all } s > 1.$$

Deduce from the result in (c), and the expression for the cotangent function obtained in the previous problem, that

$$x \cot x = 1 - 2 \sum_{m=1}^{\infty} \frac{\zeta(2m)}{\pi^{2m}} x^{2m}.$$

(e) Conclude that

$$2\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{(2m)!} B_{2m}.$$

5. Define the **Bernoulli polynomials** $B_n(x)$ by the formula

$$\frac{ze^{xz}}{e^z-1} = \sum_{n=0}^{\infty} \frac{B_n(x)}{n!} z^n.$$

(a) The functions $B_n(x)$ are polynomials in x and

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}.$$

Show that $B_0(x) = 1$, $B_1(x) = x - 1/2$, $B_2(x) = x^2 - x + 1/6$, and $B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$.

(b) If $n \geq 1$, then

$$B_n(x+1) - B_n(x) = nx^{n-1},$$

and if $n \geq 2$, then

$$B_n(0) = B_n(1) = B_n.$$

(c) Define $S_m(n) = 1^m + 2^m + \cdots + (n-1)^m$. Show that

$$(m+1)S_m(n) = B_{m+1}(n) - B_{m+1}.$$

(d) Prove that the Bernoulli polynomials are the only polynomials that satisfy

(i) $B_0(x) = 1$,

(ii) $B_n'(x) = nB_{n-1}(x)$ for $n \geq 1$,

(iii) $\int_0^1 B_n(x)\,dx = 0$ for $n \geq 1$, and show that from (b) one obtains

$$\int_x^{x+1} B_n(t)\,dt = x^n.$$

(e) Calculate the Fourier series of $B_1(x)$ to conclude that for $0 < x < 1$ we have

$$B_1(x) = x - 1/2 = \frac{-1}{\pi} \sum_{k=1}^{\infty} \frac{\sin(2\pi kx)}{k}.$$

Integrate and conclude that

$$B_{2n}(x) = (-1)^{n+1} \frac{2(2n)!}{(2\pi)^{2n}} \sum_{k=1}^{\infty} \frac{\cos(2\pi kx)}{k^{2n}},$$

$$B_{2n+1}(x) = (-1)^{n+1} \frac{2(2n+1)!}{(2\pi)^{2n+1}} \sum_{k=1}^{\infty} \frac{\sin(2\pi kx)}{k^{2n+1}}.$$

Finally, show that for $0 < x < 1$,

$$B_n(x) = -\frac{n!}{(2\pi i)^n} \sum_{k \neq 0} \frac{e^{2\pi ikx}}{k^n}.$$

We observe that the Bernoulli polynomials are, up to normalization, successive integrals of the sawtooth function.

Chapter 4. Some Applications of Fourier Series

1.* This problem explores another relationship between the geometry of a curve and Fourier series. The diameter of a closed curve Γ parametrized by $\gamma(t) = (x(t), y(t))$ on $[-\pi, \pi]$ is defined by

$$d = \sup_{P,\, Q \in \Gamma} |P - Q| = \sup_{t_1,\, t_2 \in [-\pi,\pi]} |\gamma(t_1) - \gamma(t_2)|.$$

If a_n is the nth Fourier coefficient of $\gamma(t) = x(t) + iy(t)$ and ℓ denotes the length of Γ, then

(a) $2|a_n| \leq d$ for all $n \neq 0$.

(b) $\ell \leq \pi d$, whenever Γ is convex.

Property (a) follows from the fact that $2a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\gamma(t) - \gamma(t + \pi/n)] e^{-int}\, dt$.

The equality $\ell = \pi d$ is satisfied when Γ is a circle, but surprisingly, this is not the only case. In fact, one finds that $\ell = \pi d$ is equivalent to $2|a_1| = d$. We re-parametrize γ so that for each t in $[-\pi, \pi]$ the tangent to the curve makes an angle t with the y-axis. Then, if $a_1 = 1$ we have

$$\gamma'(t) = ie^{it}(1 + r(t)),$$

where r is a real-valued function which satisfies $r(t) + r(t + \pi) = 0$, and $|r(t)| \leq 1$. Figure 7 (a) shows the curve obtained by setting $r(t) = \cos 5t$. Also, Figure 7 (b) consists of the curve where $r(t) = h(3t)$, with $h(s) = -1$ if $-\pi \leq s \leq 0$ and $h(s) = 1$ if $0 < s < \pi$. This curve (which is only piecewise of class C^1) is known as the Reuleaux triangle and is the classical example of a convex curve of constant width which is not a circle.

2.* Here we present an estimate of Weyl which leads to some interesting results.

(a) Let $S_N = \sum_{n=1}^{N} e^{2\pi i f(n)}$. Show that for $H \leq N$, one has

$$|S_N|^2 \leq c\,\frac{N}{H}\,\sum_{h=0}^{H}\left|\sum_{n=1}^{N-h} e^{2\pi i(f(n+h)-f(n))}\right|,$$

for some constant $c > 0$ independent of N, H, and f.

(b) Use this estimate to show that the sequence $\langle n^2\gamma\rangle$ is equidistributed in $[0,1)$ whenever γ is irrational.

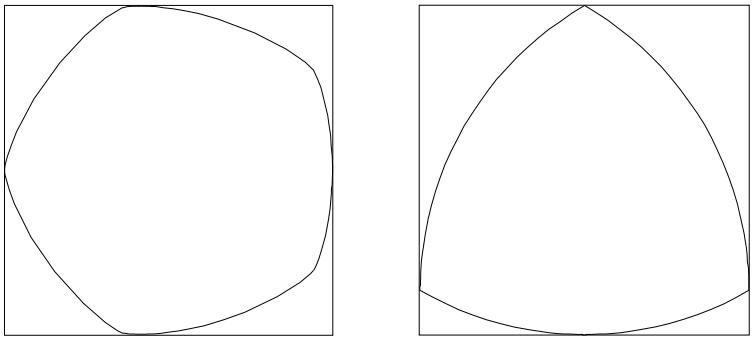


Figure 7. Some curves with maximal length for a given diameter

(c) More generally, show that if $\{\xi_n\}$ is a sequence of real numbers so that for all positive integers h the difference $\langle \xi_{n+h} - \xi_n\rangle$ is equidistributed in $[0,1)$, then $\langle \xi_n\rangle$ is also equidistributed in $[0,1)$.

(d) Suppose that $P(x) = c_n x^n + \cdots + c_0$ is a polynomial with real coefficients, where at least one of $c_1, \ldots, c_n$ is irrational. Then the sequence $\langle P(n)\rangle$ is equidistributed in $[0,1)$.

[Hint: For (a), let $a_n = e^{2\pi i f(n)}$ when $1 \leq n \leq N$ and 0 otherwise. Then write $H\sum_n a_n = \sum_{k=1}^{H}\sum_n a_{n+k}$ and apply the Cauchy-Schwarz inequality. For (b), note that $(n+h)^2\gamma - n^2\gamma = 2nh\gamma + h^2\gamma$, and use the fact that for each integer h, the sequence $\langle 2nh\gamma\rangle$ is equidistributed. Finally, to prove (d), assume first that $P(x) = Q(x) + c_1 x + c_0$ where c_1 is irrational, and estimate the exponential sum $\sum_{n=1}^{N} e^{2\pi i k P(n)}$. Then, argue by induction on the highest degree term which has an irrational coefficient, and use part (c).]

3.* If $\sigma > 0$ is not an integer and $a \neq 0$, then $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$. See also Exercise 8.

4. An elementary construction of a continuous but nowhere differentiable function is obtained by "piling up singularities," as follows.

On $[-1, 1]$ consider the function

$$\varphi(x) = |x|$$

and extend φ to $\mathbb{R}$ by requiring it to be periodic of period 2. Clearly, φ is continuous on $\mathbb{R}$ and $|\varphi(x)| \leq 1$ for all x so the function f defined by

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$$

is continuous on $\mathbb{R}$.

(a) Fix $x_0 \in \mathbb{R}$. For every positive integer m, let $\delta_m = \pm\frac{1}{2}4^{-m}$ where the sign is chosen so that no integer lies in between $4^m x_0$ and $4^m(x_0 + \delta_m)$. Consider the quotient

$$\gamma_n = \frac{\varphi(4^n(x_0 + \delta_m)) - \varphi(4^n x_0)}{\delta_m}.$$

Prove that if $n > m$, then $\gamma_n = 0$, and for $0 \leq n \leq m$ one has $|\gamma_n| \leq 4^n$ with $|\gamma_m| = 4^m$.

(b) From the above observations prove the estimate

$$\left|\frac{f(x_0 + \delta_m) - f(x_0)}{\delta_m}\right| \geq \frac{1}{2}(3^m + 1),$$

and conclude that f is not differentiable at x_0.

5. Let f be a Riemann integrable function on the interval $[-\pi, \pi]$. We define the generalized delayed means of the Fourier series of f by

$$\sigma_{N,K} = \frac{S_N + \cdots + S_{N+K-1}}{K}.$$

Note that in particular

$$\sigma_{0,N} = \sigma_N, \qquad \sigma_{N,1} = S_N \qquad \text{and} \qquad \sigma_{N,N} = \Delta_N,$$

where Δ_N are the specific delayed means used in Section 3.

(a) Show that

$$\sigma_{N,K} = \frac{1}{K}\left((N+K)\sigma_{N+K} - N\sigma_N\right),$$

and

$$\sigma_{N,K} = S_N + \sum_{N+1 \le |\nu| \le N+K-1} \left(1 - \frac{|\nu| - N}{K}\right) \hat{f}(\nu) e^{i\nu\theta}.$$

From this last expression for $\sigma_{N,K}$ conclude that

$$|\sigma_{N,K} - S_M| \le \sum_{N+1 \le |\nu| \le N+K-1} |\hat{f}(\nu)|$$

for all $N \le M < N + K$.

(b) Use one of the above formulas and Fejér's theorem to show that with $N = kn$ and $K = n$, then

$$\sigma_{kn,n}(f)(\theta) \to f(\theta) \qquad \text{as } n \to \infty$$

whenever f is continuous at θ, and also

$$\sigma_{kn,n}(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } n \to \infty$$

at a jump discontinuity (refer to the preceding chapters and their exercises for the appropriate definitions and results). In the case when f is continuous on $[-\pi, \pi]$, show that $\sigma_{kn,n}(f) \to f$ uniformly as $n \to \infty$.

(c) Using part (a), show that if $\hat{f}(\nu) = O(1/|\nu|)$ and $kn \le m < (k+1)n$, we get

$$|\sigma_{kn,n} - S_m| \le \frac{C}{k} \qquad \text{for some constant } C > 0.$$

(d) Suppose that $\hat{f}(\nu) = O(1/|\nu|)$. Prove that if f is continuous at θ then

$$S_N(f)(\theta) \to f(\theta) \qquad \text{as } N \to \infty,$$

and if f has a jump discontinuity at θ then

$$S_N(f)(\theta) \to \frac{f(\theta^+) + f(\theta^-)}{2} \qquad \text{as } N \to \infty.$$

Also, show that if f is continuous on $[-\pi, \pi]$, then $S_N(f) \to f$ uniformly.

(e) The above arguments show that if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s. This is a weak version of Littlewood's theorem (Problem 3, Chapter 2).

6. Dirichlet's theorem states that the Fourier series of a real continuous periodic function f which has only a finite number of relative maxima and minima converges everywhere to f (and uniformly).

Prove this theorem by showing that such a function satisfies $\hat{f}(n) = O(1/|n|)$. [Hint: Argue as in Exercise 17, Chapter 3; then use conclusion (d) in Problem 5 above.]

Chapter 5. The Fourier Transform on R

1. The equation

$$x^2 \frac{\partial^2 u}{\partial x^2} + ax \frac{\partial u}{\partial x} = \frac{\partial u}{\partial t} \tag{17}$$

with $u(x,0) = f(x)$ for $0 < x < \infty$ and $t > 0$ is a variant of the heat equation which occurs in a number of applications. To solve (17), make the change of variables $x = e^{-y}$ so that $-\infty < y < \infty$. Set $U(y,t) = u(e^{-y},t)$ and $F(y) = f(e^{-y})$. Then the problem reduces to the equation

$$\frac{\partial^2 U}{\partial y^2} + (1-a)\frac{\partial U}{\partial y} = \frac{\partial U}{\partial t},$$

with $U(y,0) = F(y)$. This can be solved like the usual heat equation (the case $a = 1$) by taking the Fourier transform in the y variable. One must then compute the integral $\int_{-\infty}^{\infty} e^{(-4\pi^2\xi^2 + (1-a)2\pi i\xi)t} e^{2\pi i \xi v}\, d\xi$. Show that the solution of the original problem is then given by

$$u(x,t) = \frac{1}{(4\pi t)^{1/2}} \int_0^\infty e^{-(\log(v/x)+(1-a)t)^2/(4t)} f(v)\, \frac{dv}{v}.$$

2. The **Black-Scholes** equation from finance theory is

$$\frac{\partial V}{\partial t} + rs\frac{\partial V}{\partial s} + \frac{\sigma^2 s^2}{2}\frac{\partial^2 V}{\partial s^2} - rV = 0, \qquad 0 < t < T, \tag{18}$$

subject to the "final" boundary condition $V(s,T) = F(s)$. An appropriate change of variables reduces this to the equation in Problem 1. Alternatively, the substitution $V(s,t) = e^{ax+b\tau}U(x,\tau)$ where $x = \log s$, $\tau = \frac{\sigma^2}{2}(T-t)$, $a = \frac{1}{2} - \frac{r}{\sigma^2}$, and $b = -\left(\frac{1}{2} + \frac{r}{\sigma^2}\right)^2$ reduces (18) to the one-dimensional heat equation with the initial condition $U(x,0) = e^{-ax}F(e^x)$. Thus a solution to the Black-Scholes equation is

$$V(s,t) = \frac{e^{-r(T-t)}}{\sqrt{2\pi\sigma^2(T-t)}} \int_0^\infty e^{-\frac{(\log(s/s^*)+(r-\sigma^2/2)(T-t))^2}{2\sigma^2(T-t)}} F(s^*)\, ds^*.$$

3. * **The Dirichlet problem in a strip**. Consider the equation $\triangle u = 0$ in the horizontal strip

$$\{(x, y) : 0 < y < 1,\ -\infty < x < \infty\}$$

with boundary conditions $u(x, 0) = f_0(x)$ and $u(x, 1) = f_1(x)$, where f_0 and f_1 are both in the Schwartz space.

(a) Show (formally) that if u is a solution to this problem, then

$$\hat{u}(\xi, y) = A(\xi)e^{2\pi\xi y} + B(\xi)e^{-2\pi\xi y}.$$

Express A and B in terms of $\widehat{f_0}$ and $\widehat{f_1}$, and show that

$$\hat{u}(\xi, y) = \frac{\sinh(2\pi(1-y)\xi)}{\sinh(2\pi\xi)}\widehat{f_0}(\xi) + \frac{\sinh(2\pi y\xi)}{\sinh(2\pi\xi)}\widehat{f_0}(\xi).$$

(b) Prove as a result that

$$\int_{-\infty}^{\infty} |u(x, y) - f_0(x)|^2\, dx \to 0 \qquad \text{as } y \to 0$$

and

$$\int_{-\infty}^{\infty} |u(x, y) - f_1(x)|^2\, dx \to 0 \qquad \text{as } y \to 1.$$

(c) If $\Phi(\xi) = (\sinh 2\pi a\xi)/(\sinh 2\pi\xi)$, with $0 \le a < 1$, then Φ is the Fourier transform of φ where

$$\varphi(x) = \frac{\sin \pi a}{2} \cdot \frac{1}{\cosh \pi x + \cos \pi a}.$$

This can be shown, for instance, by using contour integration and the residue formula from complex analysis (see Book II, Chapter 3).

(d) Use this result to express u in terms of Poisson-like integrals involving f_0 and f_1 as follows:

$$u(x, y) = \frac{\sin \pi y}{2}\left(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y}\, dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y}\, dt\right).$$

(e) Finally, one can check that the function $u(x, y)$ defined by the above expression is harmonic in the strip, and converges uniformly to $f_0(x)$ as $y \to 0$, and to $f_1(x)$ as $y \to 1$. Moreover, one sees that $u(x, y)$ vanishes at infinity, that is, $\lim_{|x|\to\infty} u(x, y) = 0$, uniformly in y.

In Exercise 12, we gave an example of a function that satisfies the heat equation in the upper half-plane, with boundary value 0, but which was not identically 0. We observed in this case that u was in fact not continuous up to the boundary.

In Problem 4 we exhibit examples illustrating non-uniqueness, but this time with continuity up to the boundary $t = 0$. These examples satisfy a growth condition at infinity, namely $|u(x, t)| \le Ce^{cx^{2+\epsilon}}$, for any $\epsilon > 0$. Problems 5 and 6 show that under the more restrictive growth condition $|u(x, t)| \le Ce^{cx^2}$, uniqueness does hold.

4.* If g is a smooth function on $\mathbb{R}$, define the formal power series

$$u(x,t) = \sum_{n=0}^{\infty} g^{(n)}(t) \frac{x^{2n}}{(2n!)}. \tag{19}$$

(a) Check formally that u solves the heat equation.

(b) For $a > 0$, consider the function defined by

$$g(t) = \begin{cases} e^{-t^{-a}} & \text{if } t > 0 \\ 0 & \text{if } t \leq 0. \end{cases}$$

One can show that there exists $0 < \theta < 1$ depending on a so that

$$|g^{(k)}(t)| \leq \frac{k!}{(\theta t)^k} e^{-\frac{1}{2}t^{-a}} \qquad \text{for } t > 0.$$

(c) As a result, for each x and t the series (19) converges; u solves the heat equation; u vanishes for $t = 0$; and u satisfies the estimate $|u(x,t)| \leq Ce^{c|x|^{2a/(a-1)}}$ for some constants $C, c > 0$.

(d) Conclude that for every $\epsilon > 0$ there exists a non-zero solution to the heat equation which is continuous for $x \in \mathbb{R}$ and $t \geq 0$, which satisfies $u(x,0) = 0$ and $|u(x,t)| \leq Ce^{c|x|^{2+\epsilon}}$.

5.* The following "maximum principle" for solutions of the heat equation will be used in the next problem.

Theorem. *Suppose that $u(x,t)$ is a real-valued solution of the heat equation in the upper half-plane, which is continuous on its closure. Let R denote the rectangle*

$$R = \{(x,y) \in \mathbb{R}^2 : \ a \le x \le b, \ 0 \le t \le c\}$$

and $\partial' R$ be the part of the boundary of R which consists of the two vertical sides and its base on the line $t = 0$ (see Figure 3*). Then*

$$\min_{(x,t)\in\partial' R} u(x,t) = \min_{(x,t)\in R} u(x,t) \qquad \textit{and} \qquad \max_{(x,t)\in\partial' R} u(x,t) = \max_{(x,t)\in R} u(x,t).$$

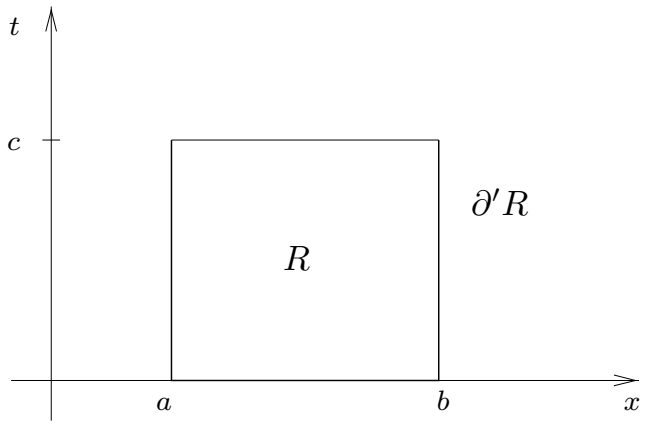


Figure 3. The rectangle R and part of its boundary $\partial' R$

The steps leading to a proof of this result are outlined below.

(a) Show that it suffices to prove that if $u \ge 0$ on $\partial' R$, then $u \ge 0$ in R.

(b) For $\epsilon > 0$, let

$$v(x,t) = u(x,t) + \epsilon t.$$

Then, v has a minimum on R, say at (x_1, t_1). Show that $x_1 = a$ or b, or else $t_1 = 0$. To do so, suppose on the contrary that $a < x_1 < b$ and $0 < t_1 \le c$, and prove that $v_{xx}(x_1,t_1) - v_t(x_1,t_1) \le -\epsilon$. However, show also that the left-hand side must be non-negative.

(c) Deduce from (b) that $u(x,t) \ge \epsilon(t_1 - t)$ for any $(x,t) \in R$ and let $\epsilon \to 0$.

6.* The examples in Problem 4 are optimal in the sense of the following uniqueness theorem due to Tychonoff.

Theorem. *Suppose $u(x,t)$ satisfies the following conditions:*

(i) *$u(x,t)$ solves the heat equation for all $x \in \mathbb{R}$ and and all $t > 0$.*

(ii) *$u(x,t)$ is continuous for all $x \in \mathbb{R}$ and $0 \le t \le c$.*

(iii) *$u(x,0) = 0$.*

(iv) *$|u(x,t)| \le Me^{ax^2}$ for some M, a, and all $x \in \mathbb{R}$, $0 \le t < c$.*

Then u is identically equal to 0.

7.* The **Hermite functions** $h_k(x)$ are defined by the generating identity

$$\sum_{k=0}^{\infty} h_k(x)\frac{t^k}{k!} = e^{-(x^2/2-2tx+t^2)}.$$

(a) Show that an alternate definition of the Hermite functions is given by the formula

$$h_k(x) = (-1)^k e^{x^2/2}\left(\frac{d}{dx}\right)^k e^{-x^2}.$$

[Hint: Write $e^{-(x^2/2-2tx+t^2)} = e^{x^2/2}e^{-(x-t)^2}$ and use Taylor's formula.] Conclude from the above expression that each $h_k(x)$ is of the form $P_k(x)e^{-x^2/2}$, where P_k is a polynomial of degree k. In particular, the Hermite functions belong to the Schwartz space and $h_0(x) = e^{-x^2/2}$, $h_1(x) = 2xe^{-x^2/2}$.

(b) Prove that the family $\{h_k\}_{k=0}^{\infty}$ is complete in the sense that if f is a Schwartz function, and

$$(f, h_k) = \int_{-\infty}^{\infty} f(x)h_k(x)\,dx = 0 \qquad \text{for all } k \geq 0,$$

then $f = 0$. [Hint: Use Exercise 8.]

(c) Define $h_k^*(x) = h_k((2\pi)^{1/2}x)$. Then

$$\widehat{h_k^*}(\xi) = (-i)^k h_k^*(\xi).$$

Therefore, each h_k^* is an eigenfunction for the Fourier transform.

(d) Show that h_k is an eigenfunction for the operator defined in Exercise 23, and in fact, prove that

$$Lh_k = (2k+1)h_k.$$

In particular, we conclude that the functions h_k are mutually orthogonal for the L^2 inner product on the Schwartz space.

(e) Finally, show that $\int_{-\infty}^{\infty}[h_k(x)]^2\,dx = \pi^{1/2}2^k k!$. [Hint: Square the generating relation.]

8.* To refine the results in Chapter 4, and to prove that

$$f_\alpha(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{2\pi i 2^n x}$$

is nowhere differentiable even in the case $\alpha = 1$, we need to consider a variant of the delayed means $\triangle_N$, which in turn will be analyzed by the Poisson summation formula.

(a) Fix an indefinitely differentiable function Φ satisfying

$$\Phi(\xi) = \begin{cases} 1 & \text{when } |\xi| \le 1, \\ 0 & \text{when } |\xi| \ge 2. \end{cases}$$

By the Fourier inversion formula, there exists $\varphi \in \mathcal{S}$ so that $\hat{\varphi}(\xi) = \Phi(\xi)$. Let $\varphi_N(x) = N\varphi(Nx)$ so that $\widehat{\varphi_N}(\xi) = \Phi(\xi/N)$. Finally, set

$$\tilde{\triangle}_N(x) = \sum_{n=-\infty}^{\infty} \varphi_N(x+n).$$

Observe by the Poisson summation formula that $\tilde{\triangle}_N(x) = \sum_{n=-\infty}^{\infty} \Phi(n/N) e^{2\pi i n x}$, thus $\tilde{\triangle}_N$ is a trigonometric polynomial of degree $\le 2N$, with terms whose coefficients are 1 when $|n| \le N$. Let

$$\tilde{\triangle}_N(f) = f * \tilde{\triangle}_N.$$

Note that

$$S_N(f_\alpha) = \tilde{\triangle}_{N'}(f_\alpha)$$

where N' is the largest integer of the form 2^k with $N' \le N$.

(b) If we set $\tilde{\triangle}_N(x) = \varphi_N(x) + E_N(x)$ where

$$E_N(x) = \sum_{|n|\ge 1} \varphi_N(x+n),$$

then one sees that:

(i) $\sup_{|x|\le 1/2} |E_N'(x)| \to 0$ as $N \to \infty$.

(ii) $|\tilde{\triangle}_N'(x)| \le cN^2$.

(iii) $|\tilde{\triangle}_N'(x)| \le c/(N|x|^3)$, for $|x| \le 1/2$.

Moreover, $\int_{|x|\le 1/2} \tilde{\triangle}_N'(x)\,dx = 0$, and $-\int_{|x|\le 1/2} x\tilde{\triangle}_N'(x)\,dx \to 1$ as $N \to \infty$.

(c) The above estimates imply that if $f'(x_0)$ exists, then

$$(f * \tilde{\triangle}_N')(x_0 + h_N) \to f'(x_0) \qquad \text{as } N \to \infty,$$

whenever $|h_N| \le C/N$. Then, conclude that both the real and imaginary parts of f_1 are nowhere differentiable, as in the proof given in Section 3, Chapter 4.

Chapter 6. The Fourier Transform on $\mathbf{R}^d$

1. Let J_n denote the nth order Bessel function, for $n \in \mathbb{Z}$. Prove that

(a) $J_n(\rho)$ is real for all real ρ.

(b) $J_{-n}(\rho) = (-1)^n J_n(\rho)$.

(c) $2J_n'(\rho) = J_{n-1}(\rho) - J_{n+1}(\rho)$.

(d) $\left(\frac{2n}{\rho}\right) J_n(\rho) = J_{n-1}(\rho) + J_{n+1}(\rho)$.

(e) $(\rho^{-n} J_n(\rho))' = -\rho^{-n} J_{n+1}(\rho)$.

(f) $(\rho^n J_n(\rho))' = \rho^n J_{n-1}(\rho)$.

(g) $J_n(\rho)$ satisfies the second order differential equation

$$J_n''(\rho) + \rho^{-1} J_n'(\rho) + (1 - n^2/\rho^2) J_n(\rho) = 0.$$

(h) Show that

$$J_n(\rho) = \left(\frac{\rho}{2}\right)^n \sum_{m=0}^{\infty} (-1)^m \frac{\rho^{2m}}{2^{2m} m!(n+m)!}.$$

(i) Show that for all integers n and all real numbers a and b we have

$$J_n(a+b) = \sum_{\ell \in \mathbb{Z}} J_\ell(a) J_{n-\ell}(b).$$

2. Another formula for $J_n(\rho)$ that allows one to define Bessel functions for non-integral values of n, $(n > -1/2)$ is

$$J_n(\rho) = \frac{(\rho/2)^n}{\Gamma(n+1/2)\sqrt{\pi}} \int_{-1}^{1} e^{i\rho t}(1-t^2)^{n-(1/2)}\, dt.$$

(a) Check that the above formula agrees with the definition of $J_n(\rho)$ for integral $n \geq 0$. [Hint: Verify it for $n = 0$ and then check that both sides satisfy the recursion formula (e) in Problem 1.]

(b) Note that $J_{1/2}(\rho) = \sqrt{\frac{2}{\pi}}\rho^{-1/2} \sin \rho$.

(c) Prove that

$$\lim_{n \to -1/2} J_n(\rho) = \sqrt{\frac{2}{\pi}}\rho^{-1/2} \cos \rho.$$

(d) Observe that the formulas we have proved in the text giving F_0 in terms of f_0 (when describing the Fourier transform of a radial function) take the form

$$F_0(\rho) = 2\pi\rho^{-(d/2)+1} \int_0^\infty J_{(d/2)-1}(2\pi\rho r) f_0(r) r^{d/2}\, dr, \tag{17}$$

for $d = 1, 2$, and 3, if one uses the formulas above with the understanding that $J_{-1/2}(\rho) = \lim_{n \to -1/2} J_n(\rho)$. It turns out that the relation between F_0 and f_0 given by (17) is valid in all dimensions d.

3. We observed that the solution $u(x,t)$ of the Cauchy problem for the wave equation given by formula (3) depends only on the initial data on the base on the backward light cone. It is natural to ask if this property is shared by *any* solution of the wave equation. An affirmative answer would imply uniqueness of the solution.

Let $B(x_0, r_0)$ denote the closed ball in the hyperplane $t = 0$ centered at x_0 and of radius r_0. The **backward light cone** with base $B(x_0, r_0)$ is defined by

$$\mathcal{L}_{B(x_0,r_0)} = \{(x,t) \in \mathbb{R}^d \times \mathbb{R} : |x - x_0| \leq r_0 - t, \quad 0 \leq t \leq r_0\}.$$

Theorem *Suppose that $u(x,t)$ is a C^2 function on the closed upper half-plane $\{(x,t) : \ x \in \mathbb{R}^d, t \geq 0\}$ that solves the wave equation*

$$\frac{\partial^2 u}{\partial t^2} = \triangle u.$$

If $u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0$ for all $x \in B(x_0, r_0)$, then $u(x,t) = 0$ for all $(x,t) \in \mathcal{L}_{B(x_0,r_0)}$.

In words, if the initial data of the Cauchy problem for the wave equation vanishes on a ball B, then *any* solution u of the problem vanishes in the backward light cone with base B. The following steps outline a proof of the theorem.

(a) Assume that u is real. For $0 \leq t \leq r_0$ let $B_t(x_0, r_0) = \{x : |x - x_0| \leq r_0 - t\}$, and also define

$$\nabla u(x,t) = \left(\frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_d}, \frac{\partial u}{\partial t} \right).$$

Now consider the energy integral

$$\begin{aligned} E(t) &= \frac{1}{2} \int_{B_t(x_0,r_0)} |\nabla u|^2 \, dx \\ &= \frac{1}{2} \int_{B_t(x_0,r_0)} \left(\frac{\partial u}{\partial t} \right)^2 + \sum_{j=1}^{d} \left(\frac{\partial u}{\partial x_j} \right)^2 \, dx. \end{aligned}$$

Observe that $E(t) \geq 0$ and $E(0) = 0$. Prove that

$$E'(t) = \int_{B_t(x_0,r_0)} \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial t^2} + \sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_j \partial t} \, dx - \frac{1}{2} \int_{\partial B_t(x_0,r_0)} |\nabla u|^2 \, d\sigma(\gamma).$$

(b) Show that

$$\frac{\partial}{\partial x_j} \left[\frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \right] = \frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_j \partial t} + \frac{\partial^2 u}{\partial x_j^2} \frac{\partial u}{\partial t}.$$

(c) Use the last identity, the divergence theorem, and the fact that u solves the wave equation to prove that

$$E'(t) = \int_{\partial B_t(x_0,r_0)} \sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \nu_j \, d\sigma(\gamma) - \frac{1}{2} \int_{\partial B_t(x_0,r_0)} |\nabla u|^2 \, d\sigma(\gamma),$$

where ν_j denotes the jth coordinate of the outward normal to $B_t(x_0, r_0)$.

(d) Use the Cauchy-Schwarz inequality to conclude that

$$\sum_{j=1}^{d} \frac{\partial u}{\partial x_j} \frac{\partial u}{\partial t} \nu_j \leq \frac{1}{2} |\nabla u|^2,$$

and as a result, $E'(t) \leq 0$. Deduce from this that $E(t) = 0$ and $u = 0$.

4.* There exist formulas for the solution of the Cauchy problem for the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2} \qquad \text{with } u(x,0) = f(x) \text{ and } \frac{\partial u}{\partial t}(x,0) = g(x)$$

in $\mathbb{R}^d \times \mathbb{R}$ in terms of spherical means which generalize the formula given in the text for $d = 3$. In fact, the solution for even dimensions is deduced from that for odd dimensions, so we discuss this case first.

Suppose that $d > 1$ is odd and let $h \in \mathcal{S}(\mathbb{R}^d)$. The spherical mean of h on the ball centered at x of radius t is defined by

$$M_r h(x) = Mh(x,r) = \frac{1}{A_d} \int_{S^{d-1}} h(x - r\gamma)\, d\sigma(\gamma),$$

where A_d denotes the area of the unit sphere S^{d-1} in $\mathbb{R}^d$.

(a) Show that

$$\triangle_x Mh(x,r) = \left[\partial_r^2 + \frac{d-1}{r}\right] Mh(x,r),$$

where $\triangle_x$ denotes the Laplacian in the space variables x, and $\partial_r = \partial/\partial r$.

(b) Show that a twice differentiable function $u(x,t)$ satisfies the wave equation if and only if

$$\left[\partial_r^2 + \frac{d-1}{r}\right] Mu(x,r,t) = \partial_t^2 Mu(x,r,t),$$

where $Mu(x,r,t)$ denote the spherical means of the function $u(x,t)$.

(c) If $d = 2k+1$, define $T\varphi(r) = (r^{-1}\partial_r)^{k-1}[r^{2k-1}\varphi(r)]$, and let $\tilde{u} = TMu$. Then this function solves the one-dimensional wave equation for each fixed x:

$$\partial_t^2 \tilde{u}(x,r,t) = \partial_r^2 \tilde{u}(x,r,t).$$

One can then use d'Alembert's formula to find the solution $\tilde{u}(x,r,t)$ of this problem expressed in terms of the initial data.

(d) Now show that

$$u(x,t) = Mu(x,0,t) = \lim_{r \to 0} \frac{\tilde{u}(x,r,t)}{\alpha r}$$

where $\alpha = 1 \cdot 3 \cdots (d-2)$.

(e) Conclude that the solution of the Cauchy problem for the d-dimensional wave equation, when $d > 1$ is odd, is

$$u(x,t) = \frac{1}{1 \cdot 3 \cdots (d-2)} \left[\partial_t (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} M_t f(x)\right) + \right.$$

$$\left. (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} M_t g(x)\right)\right].$$

5.* The method of descent can be used to prove that the solution of the Cauchy problem for the wave equation in the case when d is even is given by the formula

$$u(x,t) = \frac{1}{1 \cdot 3 \cdots (d-2)} \left[\partial_t (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} \widetilde{M}_t f(x) \right) + \right.$$

$$\left. (t^{-1}\partial_t)^{(d-3)/2} \left(t^{d-2} \widetilde{M}_t g(x) \right) \right],$$

where $\widetilde{M}_t$ denotes the modified spherical means defined by

$$\widetilde{M}_t h(x) = \frac{2}{A_{d+1}} \int_{B^d} \frac{f(x+ty)}{\sqrt{1-|y|^2}}\, dy.$$

6.* Given initial data f and g of the form

$$f(x) = F(x \cdot \gamma) \qquad \text{and} \qquad g(x) = G(x \cdot \gamma),$$

check that the plane wave given by

$$u(x,t) = \frac{F(x \cdot \gamma + t) + F(x \cdot \gamma - t)}{2} + \frac{1}{2} \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} G(s)\, ds$$

is a solution of the Cauchy problem for the d-dimensional wave equation.

In general, the solution is given as a superposition of plane waves. For the case $d = 3$, this can be expressed in terms of the Radon transform as follows. Let

$$\tilde{\mathcal{R}}(f)(t,\gamma) = -\frac{1}{8\pi^2} \left(\frac{d}{dt}\right)^2 R(f)(t,\gamma).$$

Then $u(x,t) =$

$$\frac{1}{2} \int_{S^2} \left[\tilde{\mathcal{R}}(f)(x \cdot \gamma - t, \gamma) + \tilde{\mathcal{R}}(f)(x \cdot \gamma + t, \gamma) + \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} \tilde{\mathcal{R}}(g)(s,\gamma)\, ds \right] d\sigma(\gamma).$$

7. For every real number $a > 0$, define the operator $(-\triangle)^a$ by the formula

$$(-\triangle)^a f(x) = \int_{\mathbb{R}^d} (2\pi|\xi|)^{2a} \hat{f}(\xi) e^{2\pi i \xi \cdot x}\, d\xi$$

whenever $f \in \mathcal{S}(\mathbb{R}^d)$.

(a) Check that $(-\triangle)^a$ agrees with the usual definition of the ath power of $-\triangle$ (that is, a compositions of minus the Laplacian) when a is a positive integer.

(b) Verify that $(-\triangle)^a(f)$ is indefinitely differentiable.

(c) Prove that if a is not an integer, then in general $(-\triangle)^a(f)$ is not rapidly decreasing.

(d) Let $u(x, y)$ be the solution of the steady-state heat equation

$$\frac{\partial^2 u}{\partial y^2} + \sum_{j=1}^{d} \frac{\partial^2 u}{\partial x_j^2} = 0, \qquad \text{with } u(x, 0) = f(x),$$

given by convolving f with the Poisson kernel (see Exercise 8). Check that

$$(-\triangle)^{1/2} f(x) = -\lim_{y \to 0} \frac{\partial u}{\partial y}(x, y),$$

and more generally that

$$(-\triangle)^{k/2} f(x) = (-1)^k \lim_{y \to 0} \frac{\partial^k u}{\partial y^k}(x, y)$$

for any positive integer k.

8.* The reconstruction formula for the Radon transform in $\mathbb{R}^d$ is as follows:

(a) When $d = 2$,

$$\frac{(-\triangle)^{1/2}}{4\pi}\mathcal{R}^*(\mathcal{R}(f)) = f,$$

where $(-\triangle)^{1/2}$ is defined in Problem 7.

(b) If the Radon transform and its dual are defined by analogy to the cases $d = 2$ and $d = 3$, then for general d,

$$\frac{(2\pi)^{1-d}}{2}(-\triangle)^{(d-1)/2}\mathcal{R}^*(\mathcal{R}(f)) = f.$$

Chapter 7. Finite Fourier Analysis

1. Prove that if n and m are two positive integers that are relatively prime, then

$$\mathbb{Z}(nm) \approx \mathbb{Z}(n) \times \mathbb{Z}(m).$$

[Hint: Consider the map $\mathbb{Z}(nm) \to \mathbb{Z}(n) \times \mathbb{Z}(m)$ given by $k \mapsto (k \bmod n, k \bmod m)$, and use the fact that there exist integers x and y such that $xn + ym = 1$.]

2.* Every finite abelian group G is isomorphic to a direct product of cyclic groups. Here are two more precise formulations of this theorem.

- If $p_1, \ldots, p_s$ are the distinct primes appearing in the factorization of the order of G, then

$$G \approx G(p_1) \times \cdots \times G(p_s),$$

  where each $G(p)$ is of the form $G(p) = \mathbb{Z}(p^{r_1}) \times \cdots \times \mathbb{Z}(p^{r_\ell})$, with $0 \leq r_1 \leq \cdots \leq r_\ell$ (this sequence of integers depends on p of course). This decomposition is unique.

- There exist unique integers $d_1, \ldots, d_k$ such that

$$d_1 | d_2, \quad d_2 | d_3, \quad \cdots, \quad d_{k-1} | d_k$$

  and

$$G \approx \mathbb{Z}(d_1) \times \cdots \times \mathbb{Z}(d_k).$$

Deduce the second formulation from the first.

3. Let $\hat{G}$ denote the collection of distinct characters of the finite abelian group G.

(a) Note that if $G = \mathbb{Z}(N)$, then $\hat{G}$ is isomorphic to G.

(b) Prove that $\widehat{G_1 \times G_2} = \hat{G}_1 \times \hat{G}_2$.

(c) Prove using Problem 2 that if G is a finite abelian group, then $\hat{G}$ is isomorphic to G.

4.* When p is prime the group $\mathbb{Z}^*(p)$ is cyclic, and $\mathbb{Z}^*(p) \approx \mathbb{Z}(p-1)$.

Chapter 8. Dirichlet's Theorem

1.* Here are other series that can be summed by the methods in Exercise 15.

(a) For the non-trivial Dirichlet character modulo 6, $L(1,\chi)$ equals

$$\frac{\pi}{2\sqrt{3}} = 1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \frac{1}{13} + \cdots .$$

(b) If χ is the odd Dirichlet character modulo 8, then $L(1,\chi)$ equals

$$\frac{\pi}{2\sqrt{2}} = 1 + \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \frac{1}{11} \cdots .$$

(c) For an odd Dirichlet character modulo 7, $L(1,\chi)$ equals

$$\frac{\pi}{\sqrt{7}} = 1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} - \frac{1}{6} \cdots .$$

(d) For an even Dirichlet character modulo 8, $L(1,\chi)$ equals

$$\frac{\log(1+\sqrt{2})}{\sqrt{2}} = 1 - \frac{1}{3} - \frac{1}{5} + \frac{1}{7} + \frac{1}{9} - \frac{1}{11} \cdots .$$

(e) For an even Dirichlet character modulo 5, $L(1,\chi)$ equals

$$\frac{2}{\sqrt{5}} \log\left(\frac{1+\sqrt{5}}{2}\right) = 1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \frac{1}{9} + \frac{1}{11} \cdots .$$

2. Let $d(k)$ denote the number of positive divisors of k.

(a) Show that if $k = p_1^{a_1} \cdots p_n^{a_n}$ is the prime factorization of k, then

$$d(k) = (a_1 + 1) \cdots (a_n + 1).$$

Although Theorem 3.12 shows that on "average" $d(k)$ is of the order of $\log k$, prove the following on the basis of (a):

(b) $d(k) = 2$ for infinitely many k.

(c) For any positive integer N, there is a constant $c > 0$ so that $d(k) \geq c(\log k)^N$ for infinitely many k. [Hint: Let $p_1, \ldots, p_N$ be N distinct primes, and consider k of the form $(p_1 p_2 \cdots p_N)^m$ for $m = 1, 2, \ldots$.]

3. Show that if p is relatively prime to q, then

$$\prod_{\chi} \left(1 - \frac{\chi(p)}{p^s}\right) = \left(\frac{1}{1 - p^{fs}}\right)^g,$$

where $g = \varphi(q)/f$, and f is the order of p in $\mathbb{Z}^*(q)$ (that is, the smallest n for which $p^n \equiv 1 \mod q$). Here the product is taken over all Dirichlet characters modulo q.

4. Prove as a consequence of the previous problem that

$$\prod_{\chi} L(s,\chi) = \sum_{n\geq 1} \frac{a_n}{n^s},$$

where $a_n \geq 0$, and the product is over all Dirichlet characters modulo q.

Complex Analysis

Chapter 2. Cauchy's Theorem and Its Applications

1. Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let

f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C. A point w on C is said to be *regular* for f if there is an open neighborhood U of w and an analytic function g on U, so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f.

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \qquad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc. [Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \to \infty$ as $r \to 1$.]

(b) * Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \qquad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle. [Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

2.* Let

$$F(z) = \sum_{n=1}^{\infty} d(n) z^n \qquad \text{for } |z| < 1$$

where $d(n)$ denotes the number of divisors of n. Observe that the radius of convergence of this series is 1. Verify the identity

$$\sum_{n=1}^{\infty} d(n) z^n = \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n}.$$

Using this identity, show that if $z = r$ with $0 < r < 1$, then

$$|F(r)| \geq \ c \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Similarly, if $\theta = 2\pi p/q$ where p and q are positive integers and $z = re^{i\theta}$, then

$$|F(re^{i\theta})| \geq \ c_{p/q} \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Conclude that F cannot be continued analytically past the unit disc.

3. Morera's theorem states that if f is continuous in $\mathbb{C}$, and $\int_T f(z)\,dz = 0$ for all triangles T, then f is holomorphic in $\mathbb{C}$. Naturally, we may ask if the conclusion still holds if we replace triangles by other sets.

(a) Suppose that f is continuous on $\mathbb{C}$, and

$$\int_C f(z)\,dz = 0 \tag{16}$$

for every circle C. Prove that f is holomorphic.

(b) More generally, let Γ be any toy contour, and $\mathcal{F}$ the collection of all translates and dilates of Γ. Show that if f is continuous on $\mathbb{C}$, and

$$\int_\gamma f(z)\,dz = 0 \qquad \text{for all } \gamma \in \mathcal{F}$$

then f is holomorphic. In particular, Morera's theorem holds under the weaker assumption that $\int_T f(z)\,dz = 0$ for all equilateral triangles.

[Hint: As a first step, assume that f is twice real differentiable, and write $f(z) = f(z_0) + a(z - z_0) + b(\overline{z - z_0}) + O(|z - z_0|^2)$ for z near z_0. Integrating this expansion over small circles around z_0 yields $\partial f/\partial \overline{z} = b = 0$ at z_0. Alternatively, suppose only that f is differentiable and apply Green's theorem to conclude that the real and imaginary parts of f satisfy the Cauchy-Riemann equations.

In general, let $\varphi(w) = \varphi(x, y)$ (when $w = x + iy$) denote a smooth function with $0 \le \varphi(w) \le 1$, and $\int_{\mathbb{R}^2} \varphi(w)\,dV(w) = 1$, where $dV(w) = dxdy$, and $\int$ denotes the usual integral of a function of two variables in $\mathbb{R}^2$. For each $\epsilon > 0$, let $\varphi_\epsilon(z) = \epsilon^{-2}\varphi(\epsilon^{-1}z)$, as well as

$$f_\epsilon(z) = \int_{\mathbb{R}^2} f(z - w)\varphi_\epsilon(w)\,dV(w),$$

where the integral denotes the usual integral of functions of two variables, with $dV(w)$ the area element of $\mathbb{R}^2$. Then f_ϵ is smooth, satisfies condition (16), and $f_\epsilon \to f$ uniformly on any compact subset of $\mathbb{C}$.]

4. Prove the converse to Runge's theorem: if K is a compact set whose complement if not connected, then there exists a function f holomorphic in a neighborhood of K which cannot be approximated uniformly by polynomial on K.

[Hint: Pick a point z_0 in a bounded component of K^c, and let $f(z) = 1/(z - z_0)$. If f can be approximated uniformly by polynomials on K, show that there exists a polynomial p such that $|(z - z_0)p(z) - 1| < 1$. Use the maximum modulus principle (Chapter 3) to show that this inequality continues to hold for all z in the component of K^c that contains z_0.]

5.* There exists an entire function F with the following "universal" property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{n\to\infty} F(z+N_k) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$.

(a) Let $p_1, p_2, \ldots$ denote an enumeration of the collection of polynomials whose coefficients have rational real and imaginary parts. Show that it suffices to find an entire function F and an increasing sequence $\{M_n\}$ of positive integers, such that

$$|F(z) - p_n(z-M_n)| < \frac{1}{n} \qquad \text{whenever } z \in D_n, \tag{17}$$

where D_n denotes the disc centered at M_n and of radius n. [Hint: Given h entire, there exists a sequence $\{n_k\}$ such that $\lim_{k\to\infty} p_{n_k}(z) = h(z)$ uniformly on every compact subset of $\mathbb{C}$.]

(b) Construct F satisfying (17) as an infinite series

$$F(z) = \sum_{n=1}^{\infty} u_n(z)$$

where $u_n(z) = p_n(z-M_n)e^{-c_n(z-M_n)^2}$, and the quantities $c_n > 0$ and $M_n > 0$ are chosen appropriately with $c_n \to 0$ and $M_n \to \infty$. [Hint: The function e^{-z^2} vanishes rapidly as $|z|\to\infty$ in the sectors $\{|\arg z| < \pi/4 - \delta\}$ and $\{|\pi - \arg z| < \pi/4 - \delta\}$.]

In the same spirit, there exists an alternate "universal" entire function G with the following property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{k\to\infty} D^{N_k} G(z) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$. Here $D^j G$ denotes the jth (complex) derivative of G.

Chapter 3. Meromorphic Functions and the Logarithm

1.* Consider a holomorphic map on the unit disc $f : \mathbb{D} \to \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for *all* $f : \mathbb{D} \to \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

(a) Show that with no further restrictions on f, no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f_n'(0)$, and discuss.

(b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition.

[Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

(c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$.

[Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \to 1$.]

(d) If $f(z) = z + a_2 z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$.

[Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

(e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta} z)^2} \qquad \text{for some } \theta \in \mathbb{R}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

(f) If $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2, show that $|z_1 - z_2| \leq 4$.

[Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

(g) Complete the proof of the theorem. [Hint: If f avoids w, then $1/f$ avoids 0 and $1/w$.]

2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |\, z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta})\, d\theta \qquad \text{for } |z_0| < 1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1 - |\, z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r\cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1 - \overline{z_0} z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0, and make a change of variables in the integral.]

3. If $f(z)$ is holomorphic in the deleted neighborhood $\{0 < |z - z_0| < r\}$ and has a pole of order k at z_0, then we can write

$$f(z) = \frac{a_{-k}}{(z - z_0)^k} + \cdots + \frac{a_{-1}}{(z - z_0)} + g(z)$$

where g is holomorphic in the disc $\{|z - z_0| < r\}$. There is a generalization of this expansion that holds even if z_0 is an essential singularity. This is a special case of the **Laurent series expansion**, which is valid in an even more general setting.

Let f be holomorphic in a region containing the annulus $\{z : r_1 \leq | z - z_0| \leq r_2\}$ where $0 < r_1 < r_2$. Then,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where the series converges absolutely in the interior of the annulus. To prove this, it suffices to write

$$f(z) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

when $r_1 < |z - z_0| < r_2$, and argue as in the proof of Theorem 4.4, Chapter 2. Here C_{r_1} and C_{r_2} are the circles bounding the annulus.

4.* Suppose Ω is a bounded region. Let L be a (two-way infinite) line that intersects Ω. Assume that $\Omega \cap L$ is an interval I. Choosing an orientation for L, we can define Ω_l and Ω_r to be the subregions of Ω lying strictly to the left or right of L, with $\Omega = \Omega_l \cup I \cup \Omega_r$ a disjoint union. If Ω_l and Ω_r are simply connected, then Ω is simply connected.

5.* Let

$$g(z) = \frac{1}{2\pi i} \int_{-M}^{M} \frac{h(x)}{x - z}\, dx$$

where h is continuous and supported in $[-M, M]$.

(a) Prove that the function g is holomorphic in $\mathbb{C} - [-M, M]$, and vanishes at infinity, that is, $\lim_{|z|\to\infty} |g(z)| = 0$. Moreover, the "jump" of g across $[-M, M]$ is h, that is,

$$h(x) = \lim_{\epsilon \to 0, \epsilon > 0} g(x + i\epsilon) - g(x - i\epsilon).$$

[Hint: Express the difference $g(x + i\epsilon) - g(x - i\epsilon)$ in terms of a convolution of h with the Poisson kernel.]

(b) If h satisfies a mild smoothness condition, for instance a Hölder condition with exponent α, that is, $|h(x) - h(y)| \leq C|x - y|^{\alpha}$ for some $C > 0$ and all $x, y \in [-M, M]$, then $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly to functions $g_+(x)$ and $g_-(x)$ as $\epsilon \to 0$. Then, g can be characterized as the unique holomorphic function that satisfies:

(i) g is holomorphic outside $[-M, M]$,

(ii) g vanishes at infinity,

(iii) $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly as $\epsilon \to 0$ to functions $g_+(x)$ and $g_-(x)$ with

$$g_+(x) - g_-(x) = h(x).$$

[Hint: If G is another function satisfying these conditions, $g - G$ is entire.]

Chapter 4. The Fourier Transform

1. Suppose $\hat{f}(\xi) = O(e^{-a|\xi|^p})$ as $|\xi| \to \infty$, for some $p > 1$. Then f is holomorphic for all z and satisfies the growth condition

$$|f(z)| \leq \ Ae^{a|z|^q}$$

where $1/p + 1/q = 1$.

Note that on the one hand, when $p \to \infty$ then $q \to 1$, and this limiting case can be interpreted as part of Theorem 3.3. On the other hand, when $p \to 1$ then $q \to \infty$, and this limiting case in a sense brings us back to Theorem 2.1.

[Hint: To prove the result, use the inequality $-\xi^p + \xi u \leq u^q$, which is valid when ξ and u are non-negative. To establish this inequality, examine separately the cases $\xi^p \geq \xi u$ and $\xi^p < \xi u$; note also that the functions $\xi = u^{q-1}$ and $u = \xi^{p-1}$ are inverses of each other because $(p-1)(q-1) = 1$.]

2. The problem is to solve the differential equation

$$a_n \frac{d^n}{dt^n} u(t) + a_{n-1} \frac{d^{n-1}}{dt^{n-1}} u(t) + \cdots + a_0 u(t) = f(t)\,,$$

where $a_0, a_1, \ldots, a_n$ are complex constants, and f is a given function. Here we suppose that f has bounded support and is smooth (say of class C^2).

(a) Let

$$\hat{f}(z) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i z t}\, dt.$$

Observe that $\hat{f}$ is an entire function, and using integration by parts show that

$$|\hat{f}(x+iy)| \leq \frac{A}{1+x^2}$$

if $|y| \leq a$ for any fixed $a \geq 0$.

(b) Write

$$P(z) = a_n (2\pi i z)^n + a_{n-1} (2\pi i z)^{n-1} + \cdots + a_0.$$

Find a real number c so that $P(z)$ does not vanish on the line

$$L = \{z : z = x + ic, \quad x \in \mathbb{R}\}.$$

(c) Set

$$u(t) = \int_L \frac{e^{2\pi i z t}}{P(z)} \hat{f}(z)\, dz.$$

Check that

$$\sum_{j=0}^{n} a_j \left(\frac{d}{dt}\right)^j u(t) = \int_L e^{2\pi i z t} \hat{f}(z)\, dz$$

and

$$\int_L e^{2\pi i z t} \hat{f}(z)\, dz = \int_{-\infty}^{\infty} e^{2\pi i x t} \hat{f}(x)\, dx.$$

Conclude by the Fourier inversion theorem that

$$\sum_{j=0}^{n} a_j \left(\frac{d}{dt}\right)^j u(t) = f(t).$$

Note that the solution u depends on the choice c.

3.* In this problem, we investigate the behavior of certain bounded holomorphic functions in an infinite strip. The particular result described here is sometimes called the three-lines lemma.

(a) Suppose $F(z)$ is holomorphic and bounded in the strip $0 < \mathrm{Im}(z) < 1$ and continuous on its closure. If $|F(z)| \leq 1$ on the boundary lines, then $|F(z)| \leq 1$ throughout the strip.

(b) For the more general F, let $\sup_{x\in\mathbb{R}} |F(x)| = M_0$ and $\sup_{x\in\mathbb{R}} |F(x+i)| = M_1$. Then,

$$\sup_{x\in\mathbb{R}} |F(x+iy)| \leq M_0^{1-y} M_1^y, \qquad \text{if } 0 \leq y \leq 1.$$

(c) As a consequence, prove that $\log \sup_{x\in\mathbb{R}} |F(x+iy)|$ is a convex function of y when $0 \leq y \leq 1$.

[Hint: For part (a), apply the maximum modulus principle to $F_\epsilon(z) = F(z)e^{-\epsilon z^2}$. For part (b), consider $M_0^{z-1} M_1^{-z} F(z)$.]

4.* There is a relation between the Paley-Wiener theorem and an earlier representation due to E. Borel.

(a) A function $f(z)$, holomorphic for all z, satisfies $|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}$ for all ϵ if and only if it is representable in the form

$$f(z) = \int_C e^{2\pi i z w} g(w)\, dw$$

where g is holomorphic outside the circle of radius M centered at the origin, and g vanishes at infinity. Here C is any circle centered at the origin of radius larger than M. In fact, if $f(z) = \sum a_n z^n$, then $g(w) = \sum_{n=0}^{\infty} A_n w^{-n-1}$ with $a_n = A_n (2\pi i)^{n+1}/n!$.

(b) The connection with Theorem 3.3 is as follows. For these functions f (for which in addition f and $\hat{f}$ are of moderate decrease on the real axis), one can assert that the g above is holomorphic in the larger region, which consists of the slit plane $\mathbb{C} - [-M, M]$. Moreover, the relation between g and the Fourier transform $\hat{f}$ is

$$g(z) = \frac{1}{2\pi i} \int_{-M}^{M} \frac{\hat{f}(\xi)}{\xi - z}\, d\xi$$

so that $\hat{f}$ represents the jump of g across the segment $[-M, M]$; that is,

$$\hat{f}(x) = \lim_{\epsilon \to 0, \epsilon > 0} g(x + i\epsilon) - g(x - i\epsilon).$$

See Problem 5 in Chapter 3.

Chapter 5. Entire Functions

1. Prove that if f is holomorphic in the unit disc, bounded and not identically zero, and $z_1, z_2, \ldots, z_n, \ldots$ are its zeros ($|z_k| < 1$), then

$$\sum_n (1 - |\, z_n|) < \infty.$$

[Hint: Use Jensen's formula.]

2.* In this problem, we discuss Blaschke products, which are bounded analogues in the disc of the Weierstrass products for entire functions.

(a) Show that for $0 < |\alpha| < 1$ and $|z| \leq r < 1$ the inequality

$$\left| \frac{\alpha + |\alpha| z}{(1 - \overline{\alpha} z)\alpha} \right| \leq \frac{1 + r}{1 - r}$$

holds.

(b) Let $\{\alpha_n\}$ be a sequence in the unit disc such that $\alpha_n \neq 0$ for all n and

$$\sum_{n=1}^{\infty} (1 - |\alpha_n|) < \infty.$$

Note that this will be the case if $\{\alpha_n\}$ are the zeros of a bounded holomorphic function on the unit disc (see Problem 1). Show that the product

$$f(z) = \prod_{n=1}^{\infty} \frac{\alpha_n - z}{1 - \overline{\alpha}_n z} \frac{|\alpha_n|}{\alpha_n}$$

converges uniformly for $|z| \leq r < 1$, and defines a holomorphic function on the unit disc having precisely the zeros α_n and no other zeros. Show that $|f(z)| \leq 1$.

3.* Show that $\sum \frac{z^n}{(n!)^\alpha}$ is an entire function of order $1/\alpha$.

4.* Let $F(z) = \sum_{n=0}^{\infty} a_n z^n$ be an entire function of finite order. Then the growth order of F is intimately linked with the growth of the coefficients a_n as $n \to \infty$. In fact:

(a) Suppose $|F(z)| \leq Ae^{a|z|^\rho}$. Then

$$\limsup_{n\to\infty} |a_n|^{1/n} n^{1/\rho} < \infty. \tag{8}$$

(b) Conversely, if (8) holds, then $|F(z)| \leq A_\epsilon e^{a_\epsilon |z|^{\rho+\epsilon}}$, for every $\epsilon > 0$.

[Hint: To prove (a), use Cauchy's inequality

$$|a_n| \leq \frac{A}{r^n} e^{ar^\rho},$$

and the fact that the function $u^{-n}e^{u^\rho}$, $0 < u < \rho$, attains its minimum value $e^{n/\rho}(\rho/n)^{n/\rho}$ at $u = n^{1/\rho}/\rho^{1/\rho}$. Then, choose r in terms of n to achieve this minimum.

To establish (b), note that for $|z| = r$,

$$|F(z)| \leq \sum \frac{c^n r^n}{n^{n/\rho}} \leq \sum \frac{c^n r^n}{(n!)^{1/\rho}}$$

for some constant c, since $n^n \geq n!$. This yields a reduction to Problem 3.]

Chapter 6. The Gamma and Zeta Functions

1. This problem provides further estimates for ζ and ζ' near $\mathrm{Re}(s) = 1$.

(a) Use Proposition 2.5 and its corollary to prove

$$\zeta(s) = \sum_{1 \le n < N} n^{-s} - \frac{N^{s-1}}{s-1} + \sum_{n \ge N} \delta_n(s)$$

for every integer $N \ge 2$, whenever $\mathrm{Re}(s) > 0$.

(b) Show that $|\zeta(1+it)| = O(\log|t|)$, as $|t| \to \infty$ by using the previous result with $N =$greatest integer in $|t|$.

(c) The second conclusion of Proposition 2.7 can be similarly refined.

(d) Show that if $t \neq 0$ and t is fixed, then the partial sums of the series $\sum_{n=1}^{\infty} 1/n^{1+it}$ are bounded, but the series does not converge.

2.* Prove that for $\mathrm{Re}(s) > 0$

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{x\}}{x^{s+1}}\,dx$$

where $\{x\}$ is the fractional part of x.

3.* If $Q(x) = \{x\} - 1/2$, then we can write the expression in the previous problem as

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s\int_1^\infty \frac{Q(x)}{x^{s+1}}\,dx.$$

Let us construct $Q_k(x)$ recursively so that

$$\int_0^1 Q_k(x)\,dx = 0, \qquad \frac{dQ_{k+1}}{dx} = Q_k(x), \qquad Q_0(x) = Q(x) \qquad \text{and} \qquad Q_k(x+1) = Q_k(x).$$

Then we can write

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s\int_1^\infty \left(\frac{d^k}{dx^k}Q_k(x)\right) x^{-s-1}\,dx\,,$$

and a k-fold integration by parts gives the analytic continuation for $\zeta(s)$ when $\mathrm{Re}(s) > -k$.

4.* The functions Q_k in the previous problem are related to the Bernoulli polynomials $B_k(x)$ by the formula

$$Q_k(x) = \frac{B_{k+1}(x)}{(k+1)!} \qquad \text{for } 0 \le x \le 1.$$

Also, if k is a positive integer, then

$$2\zeta(2k) = (-1)^{k+1}\frac{(2\pi)^{2k}}{(2k)!}B_{2k},$$

where $B_k = B_k(0)$ are the Bernoulli numbers. For the definition of $B_k(x)$ and B_k see Chapter 3 in Book I.

Chapter 7. The Zeta Function and Prime Number Theorem

1. Let $F(s) = \sum_{n=1}^{\infty} a_n/n^s$, where $|a_n| \leq M$ for all n.

(a) Then

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |F(\sigma + it)|^2 \, dt = \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{2\sigma}} \qquad \text{if } \sigma > 1.$$

How is this reminiscent of the Parseval-Plancherel theorem? See e.g. Chapter 3 in Book I.

(b) Show as a consequence the uniqueness of Dirichlet series: If $F(s) = \sum_{n=1}^{\infty} a_n n^{-s}$, where the coefficients are assumed to satisfy $|a_n| \leq cn^k$ for some k, and $F(s) \equiv 0$, then $a_n = 0$ for all n.

Hint: For part (a) use the fact that

$$\frac{1}{2T} \int_{-T}^{T} (nm)^{-\sigma} n^{-it} m^{it} \, dt \to \begin{cases} n^{-2\sigma} & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$$

2.* One of the "explicit formulas" in the theory of primes is as follows: if ψ_1 is the integrated Tchebychev function considered in Section 2, then

$$\psi_1(x) = \frac{x^2}{2} - \sum_\rho \frac{x^\rho}{\rho(\rho+1)} - E(x)$$

where the sum is taken over all zeros ρ of the zeta function in the critical strip. The error term is given by $E(x) = c_1 x + c_0 + \sum_{k=1}^\infty x^{1-2k}/(2k(2k-1))$, where $c_1 = \zeta'(0)/\zeta(0)$ and $c_0 = \zeta'(-1)/\zeta(-1)$. Note that $\sum_\rho 1/|\rho|^{1+\epsilon} < \infty$ for every $\epsilon > 0$, because $(1-s)\zeta(s)$ has order of growth 1. (See Exercise 8.) Also, obviously $E(x) = O(x)$ as $x \to \infty$.

3.* Using the previous problem one can show that

$$\pi(x) - \mathrm{Li}(x) = O(x^{\alpha+\epsilon}) \qquad \text{as } x \to \infty$$

for every $\epsilon > 0$, where α is fixed and $1/2 \leq \alpha < 1$ if and only if $\zeta(s)$ has no zeros in the strip $\alpha < \mathrm{Re}(s) < 1$. The case $\alpha = 1/2$ corresponds to the Riemann hypothesis.

4.* One can combine ideas from the prime number theorem with the proof of Dirichlet's theorem about primes in arithmetic progression (given in Book I) to prove the following. Let q and ℓ be relatively prime integers. We consider the primes belonging to the arithmetic progression $\{qk + \ell\}_{k=1}^{\infty}$, and let $\pi_{q,\ell}(x)$ denote the number of such primes $\leq x$. Then one has

$$\pi_{q,\ell}(x) \sim \frac{x}{\varphi(q) \log x} \qquad \text{as } x \to \infty\,,$$

where $\varphi(q)$ denotes the number of positive integers less than q and relatively prime to q.

Chapter 8. Conformal Mappings

1. Let f be a complex-valued C^1 function defined in the neighborhood of a point z_0. There are several notions closely related to conformality at z_0. We say that f is **isogonal** at z_0 if whenever $\gamma(t)$ and $\eta(t)$ are two smooth curves with $\gamma(0) = \eta(0) = z_0$, that make an angle θ there ($|\theta| < \pi$), then $f(\gamma(t))$ and $f(\eta(t))$ make an angle of θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ. Also, f is said to be **isotropic** if it magnifies lengths by some factor for all directions emanating from z_0, that is, if the limit

$$\lim_{r \to 0} \frac{|f(z_0 + re^{i\theta}) - f(z_0)|}{r}$$

exists, is non-zero, and independent of θ.

Then f is isogonal at z_0 if and only if it is isotropic at z_0; moreover, f is isogonal at z_0 if and only if either $f'(z_0)$ exists and is non-zero, or the same holds for f replaced by $\overline{f}$.

2. The angle between two non-zero complex numbers z and w (taken in that order) is simply the oriented angle, in $(-\pi, \pi]$, that is formed between the two vectors in $\mathbb{R}^2$ corresponding to the points z and w. This oriented angle, say α, is uniquely determined by the two quantities

$$\frac{(z,w)}{|z|\,|w|} \quad \text{and} \quad \frac{(z,-iw)}{|z|\,|w|}$$

which are simply the cosine and sine of α, respectively. Here, the notation $(\cdot,\cdot)$ corresponds to the usual Euclidian inner product in $\mathbb{R}^2$, which in terms of complex numbers takes the form $(z,w) = \mathrm{Re}(z\overline{w})$.

In particular, we may now consider two smooth curves $\gamma : [a,b] \to \mathbb{C}$ and $\eta : [a,b] \to \mathbb{C}$, that intersect at z_0, say $\gamma(t_0) = \eta(t_0) = z_0$, for some $t_0 \in (a,b)$. If the quantities $\gamma'(t_0)$ and $\eta'(t_0)$ are non-zero, then they represent the tangents to the curves γ and η at the point z_0, and we say that the two curves intersect at z_0 at the angle formed by the two vectors $\gamma'(t_0)$ and $\eta'(t_0)$.

A holomorphic function f defined near z_0 is said to **preserve angles** at z_0 if for any two smooth curves γ and η intersecting at z_0, the angle formed between the curves γ and η at z_0 equals the angle formed between the curves $f \circ \gamma$ and $f \circ \eta$ at $f(z_0)$. (See Figure 12 for an illustration.) In particular, we assume that the tangents to the curves γ, η, $f \circ \gamma$, and $f \circ \eta$ at the point z_0 and $f(z_0)$ are all non-zero.



Figure 12. Preservation of angles at z_0

(a) Prove that if $f : \Omega \to \mathbb{C}$ is holomorphic, and $f'(z_0) \neq 0$, then f preserves angles at z_0. [Hint: Observe that

$$(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0),\eta'(t_0)).]$$

(b) Conversely, prove the following: suppose $f : \Omega \to \mathbb{C}$ is a complex-valued function, that is real-differentiable at $z_0 \in \Omega$, and $J_f(z_0) \neq 0$. If f preserves angles at z_0, then f is holomorphic at z_0 with $f'(z_0) \neq 0$.

3.* The Schwarz-Pick lemma (see Exercise 13) is the infinitesimal version of an important observation in complex analysis and geometry.

For complex numbers $w \in \mathbb{C}$ and $z \in \mathbb{D}$ we define the **hyperbolic length** of w at z by

$$\|w\|_z = \frac{|w|}{1 - |z|^2},$$

where $|w|$ and $|z|$ denote the usual absolute values. This length is sometimes referred to as the **Poincaré metric**, and as a Riemann metric it is written as

$$ds^2 = \frac{|dz|^2}{(1 - |z|^2)^2}.$$

The idea is to think of w as a vector lying in the tangent space at z. Observe that for a fixed w, its hyperbolic length grows to infinity as z approaches the boundary of the disc. We pass from the infinitesimal hyperbolic length of tangent vectors to the global hyperbolic distance between two points by integration.

(a) Given two complex numbers z_1 and z_2 in the disc, we define the **hyperbolic distance** between them by

$$d(z_1, z_2) = \inf_{\gamma} \int_0^1 \|\gamma'(t)\|_{\gamma(t)}\, dt,$$

where the infimum is taken over all smooth curves $\gamma : [0, 1] \to \mathbb{D}$ joining z_1 and z_2. Use the Schwarz-Pick lemma to prove that if $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then

$$d(f(z_1), f(z_2)) \leq d(z_1, z_2) \qquad \text{for any } z_1, z_2 \in \mathbb{D}.$$

In other words, holomorphic functions are distance-decreasing in the hyperbolic metric.

(b) Prove that automorphisms of the unit disc preserve the hyperbolic distance, namely

$$d(\varphi(z_1),\varphi(z_2)) = d(z_1, z_2), \qquad \text{for any } z_1, z_2 \in \mathbb{D}$$

and any automorphism φ. Conversely, if $\varphi : \mathbb{D} \to \mathbb{D}$ preserves the hyperbolic distance, then either φ or $\overline{\varphi}$ is an automorphism of $\mathbb{D}$.

(c) Given two points $z_1, z_2 \in \mathbb{D}$, show that there exists an automorphism φ such that $\varphi(z_1) = 0$ and $\varphi(z_2) = s$ for some s on the segment $[0, 1)$ on the real line.

(d) Prove that the hyperbolic distance between 0 and $s \in [0, 1)$ is

$$d(0, s) = \frac{1}{2} \log \frac{1 + s}{1 - s}.$$

(e) Find a formula for the hyperbolic distance between any two points in the unit disc.

4.* Consider the group of matrices of the form

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that satisfy the following conditions:

(i) a, b, c, and $d \in \mathbb{C}$,

(ii) the determinant of M is equal to 1,

(iii) the matrix M preserves the following hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$:

$$\langle Z, W \rangle = z_1 \overline{w}_1 - z_2 \overline{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. In other words, for all $Z, W \in \mathbb{C}^2$

$$\langle MZ, MW \rangle = \langle Z, W \rangle.$$

This group of matrices is denoted by $\mathrm{SU}(1,1)$.

(a) Prove that all matrices in $\mathrm{SU}(1,1)$ are of the form

$$\begin{pmatrix} a & b \\ \overline{b} & \overline{a} \end{pmatrix},$$

where $|a|^2 - |\, b|^2 = 1$. To do so, consider the matrix

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and observe that $\langle Z, W \rangle = {}^t\!W J Z$, where ${}^t\!W$ denotes the conjugate transpose of W.

(b) To every matrix in $\mathrm{SU}(1,1)$ we can associate a fractional linear transformation

$$\frac{az + b}{cz + d}.$$

Prove that the group $\mathrm{SU}(1,1)/\{\pm 1\}$ is isomorphic to the group of automorphisms of the disc. [Hint: Use the following association.]

$$e^{2i\theta}\, \frac{z - \alpha}{1 - \overline{\alpha} z} \longrightarrow \begin{pmatrix} \frac{e^{i\theta}}{\sqrt{1-|\alpha|^2}} & -\frac{\alpha e^{i\theta}}{\sqrt{1-|\alpha|^2}} \\ -\frac{\overline{\alpha} e^{-i\theta}}{\sqrt{1-|\alpha|^2}} & \frac{e^{-i\theta}}{\sqrt{1-|\alpha|^2}} \end{pmatrix}.$$

5. The following result is relevant to Problem 4 in Chapter 10 which treats modular functions.

(a) Suppose that $F : \mathbb{H} \to \mathbb{C}$ is holomorphic and bounded. Also, suppose that $F(z)$ vanishes when $z = ir_n$, $n = 1, 2, 3, \ldots$, where $\{r_n\}$ is a bounded sequence of positive numbers. Prove that if $\sum_{n=1}^{\infty} r_n = \infty$, then $F = 0$.

(b) If $\sum r_n < \infty$, it is possible to construct a bounded function on the upper half-plane with zeros precisely at the points ir_n.

For related results in the unit disc, see Problems 1 and 2 in Chapter 5.]

6.* The results of Exercise 18 extend to the case when γ is assumed merely to be closed, simple, and continuous. The proof, however, requires further ideas.

7.* Applying ideas of Carathéodory, Koebe gave a proof of the Riemann mapping theorem by constructing (more explicitly) a sequence of functions that converges to the desired conformal map.

Starting with a Koebe domain, that is, a simply connected domain $\mathcal{K}_0 \subset \mathbb{D}$ that is not all of $\mathbb{D}$, and which contains the origin, the strategy is to find an injective function f_0 such that $f_0(\mathcal{K}_0) = \mathcal{K}_1$ is a Koebe domain "larger" than $\mathcal{K}_0$. Then, one iterates this process, finally obtaining functions $F_n = f_n \circ \cdots \circ f_0 : \mathcal{K}_0 \to \mathbb{D}$ such that $F_n(\mathcal{K}_0) = \mathcal{K}_{n+1}$ and $\lim F_n = F$ is a conformal map from $\mathcal{K}_0$ to $\mathbb{D}$.

The **inner radius** of a region $\mathcal{K} \subset \mathbb{D}$ that contains the origin is defined by $r_{\mathcal{K}} = \sup\{\rho \geq 0 : D(0,\rho) \subset \mathcal{K}\}$. Also, a holomorphic injection $f : \mathcal{K} \to \mathbb{D}$ is said to be an **expansion** if $f(0) = 0$ and $|f(z)| > |z|$ for all $z \in \mathcal{K} - \{0\}$.

(a) Prove that if f is an expansion, then $r_{f(\mathcal{K})} \geq r_{\mathcal{K}}$ and $|f'(0)| > 1$. [Hint: Write $f(z) = zg(z)$ and use the maximum principle to prove that $|f'(0)| = |g(0)| > 1$.]

Suppose we begin with a Koebe domain $\mathcal{K}_0$ and a sequence of expansions $\{f_0, f_1, \ldots, f_n, \ldots\}$, so that $\mathcal{K}_{n+1} = f_n(\mathcal{K}_n)$ are also Koebe domains. We then define holomorphic maps $F_n : \mathcal{K}_0 \to \mathbb{D}$ by $F_n = f_n \circ \cdots \circ f_0$.

(b) Prove that for each n, the function F_n is an expansion. Moreover, $F_n'(0) = \prod_{k=0}^{n} f_k'(0)$, and conclude that $\lim_{n\to\infty} |f_n'(0)| = 1$. [Hint: Prove that the sequence $\{|F_n'(0)|\}$ has a limit by showing that it is bounded above and monotone increasing. Use the Schwarz lemma.]

(c) Show that if the sequence is osculating, that is, $r_{\mathcal{K}_n} \to 1$ as $n \to \infty$, then $\{F_n\}$ converges uniformly on compact subsets of $\mathcal{K}_0$ to a conformal map $F : \mathcal{K}_0 \to \mathbb{D}$. [Hint: If $r_{F(\mathcal{K}_0)} \geq 1$ then F is surjective.]

To construct the desired osculating sequence we shall use the automorphisms $\psi_\alpha = (\alpha - z)/(1 - \overline{\alpha}z)$.

(d) Given a Koebe domain $\mathcal{K}$, choose a point $\alpha \in \mathbb{D}$ on the boundary of $\mathcal{K}$ such that $|\alpha| = r_{\mathcal{K}}$, and also choose $\beta \in \mathbb{D}$ such that $\beta^2 = \alpha$. Let S denote the square root of ψ_α on $\mathcal{K}$ such that $S(0) = 0$. Why is such a function well defined? Prove that the function $f : \mathcal{K} \to \mathbb{D}$ defined by $f(z) = \psi_\beta \circ S \circ \psi_\alpha$

is an expansion. Moreover, show that $|f'(0)| = (1 + r_{\mathcal{K}})/2\sqrt{r_{\mathcal{K}}}$. [Hint: To prove that $|f(z)| > |z|$ on $\mathcal{K} - \{0\}$ apply the Schwarz lemma to the inverse function, namely $\psi_\alpha \circ g \circ \psi_\beta$ where $g(z) = z^2$.]

(e) Use part (d) to construct the desired sequence.

8.* Let f be an injective holomorphic function in the unit disc, with $f(0) = 0$ and $f'(0) = 1$. If we write $f(z) = z + a_2 z^2 + a_3 z^3 \cdots$, then Problem 1 in Chapter 3 shows that $|a_2| \leq 2$. Bieberbach conjectured that in fact $|a_n| \leq n$ for all $n \geq 2$; this was proved by deBranges. This problem outlines an argument to prove the conjecture under the additional assumption that the coefficients a_n are real.

(a) Let $z = re^{i\theta}$ with $0 < r < 1$, and show that if $v(r, \theta)$ denotes the imaginary part of $f(re^{i\theta})$, then

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin n\theta \, d\theta.$$

(b) Show that for $0 \leq \theta \leq \pi$ and $n = 1, 2, \ldots$ we have $|\sin n\theta| \leq n \sin \theta$.

(c) Use the fact that $a_n \in \mathbb{R}$ to show that $f(\mathbb{D})$ is symmetric with respect to the real axis, and use this fact to show that f maps the upper half-disc into either the upper or lower part of $f(\mathbb{D})$.

(d) Show that for r small,

$$v(r, \theta) = r \sin \theta [1 + O(r)],$$

and use the previous part to conclude that $v(r, \theta) \sin \theta \geq 0$ for all $0 < r < 1$ and $0 \leq \theta \leq \pi$.

(e) Prove that $|a_n r^n| \leq nr$, and let $r \to 1$ to conclude that $|a_n| \leq n$.

(f) Check that the function $f(z) = z/(1 - z)^2$ satisfies all the hypotheses and that $|a_n| = n$ for all n.

9.* Gauss found a connection between elliptic integrals and the familiar operations of forming arithmetic and geometric means.

We start with any pair (a, b) of numbers that satisfy $a \geq b > 0$, and form the arithmetic and geometric means of a and b, that is,

$$a_1 = \frac{a+b}{2} \qquad \text{and} \qquad b_1 = (ab)^{1/2}.$$

We then repeat these operations with a and b replaced by a_1 and b_1. Iterating this process provides two sequences $\{a_n\}$ and $\{b_n\}$ where a_{n+1} and b_{n+1} are the arithmetic and geometric means of a_n and b_n, respectively.

(a) Prove that the two sequences $\{a_n\}$ and $\{b_n\}$ have a common limit. This limit, which we denote by $M(a, b)$, is called the **arithmetic-geometric mean** of a and b. [Hint: Show that $a \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq b_n \geq \cdots \geq b_1 \geq b$ and $a_n - b_n \leq (a-b)/2^n$.]

(b) Gauss's identity states that

$$\frac{1}{M(a,b)} = \frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{(a^2 \cos^2\theta + b^2 \sin^2\theta)^{1/2}}.$$

To prove this relation, show that if $I(a, b)$ denotes the integral on the right-hand side, then it suffices to establish the invariance of I, namely

$$I(a,b) = I\left(\frac{a+b}{2}, (ab)^{1/2}\right). \tag{6}$$

Then, observe that the connection with elliptic integrals takes the form

$$I(a,b) = \frac{1}{a}K(k) = \frac{1}{a}\int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \qquad \text{where } k^2 = 1 - b^2/a^2,$$

and that the relation (6) is a consequence of the identity in (b) of Exercise 24.

Chapter 9. An Introduction to Elliptic Functions

1. Besides the approach in Section 1.2, there are several alternate ways of dealing with the sum $\sum 1/(z+\omega)^2$, where $\omega = n + m\tau$. For example, one may sum either (a) circularly, (b) first in n then in m, (c) or first in m then in n.

(a) Prove that if $z \notin \Lambda$, then

$$\lim_{R\to\infty} \sum_{n^2+m^2\le R^2} \frac{1}{(z+n+m\tau)^2} = S_1(z)$$

exists and $S_1(z) = \wp(z) + c_1$.

(b) Similarly,

$$\sum_m \left(\sum_n \frac{1}{(z+n+m\tau)^2} \right) = S_2(z)$$

exists and $S_2(z) = \wp(z) + c_2$, where $c_2 = F(\tau)$, and F is the forbidden Eisenstein series.

(c) Also

$$\sum_n \left(\sum_m \frac{1}{(z+n+m\tau)^2} \right) = S_3(z)$$

exists with $S_3(z) = \wp(z) + c_3$, and $c_3 = \tilde{F}(\tau)$, the reverse of F.

[Hint: To prove (a), it suffices to show that $\lim_{R\to\infty}, \sum_{1\le n^2+m^2\le R^2} 1/(n+m\tau)^2 = c_1$ exists. This is proved by a comparision with $\int_{1\le x^2+y^2\le R^2} \frac{dx}{(x+y\tau)^2} = I(R)$. It can be shown that $I(R) = 0$, which follows because $(x+y\tau)^{-2} = -(\partial/\partial x\,)(x+y\tau)^{-1}$.]

2. Show that

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z + m\tau)\pi)}$$

where c is an appropriate constant. In fact, by part (b) of the previous problem $c = -F(\tau)$.

3.* Suppose Ω is a simply connected domain that excludes the three roots of the polynomial $4z^3 - g_2 z - g_3$. For $\omega_0 \in \Omega$and ω_0 fixed, define the function I on Ω by

$$I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} \qquad \omega \in \Omega.$$

Then the function I has an inverse given by $\wp(z + \alpha)$ for some constant α; that is,

$$I(\wp(z + \alpha)) = z$$

for appropriate α.

[Hint: Prove that $(I(\wp(z + \alpha)))' = \pm 1$, and use the fact that $\wp$ is even.]

4.* Suppose τ is purely imaginary, say $\tau = it$ with $t > 0$. Consider the division of the complex plane into congruent rectangles obtained by considering the lines $x = n/2$, $y = tm/2$ as n and m range over the integers. (An example is the rectangle whose vertices are $0, 1/2, 1/2 + \tau/2$, and $\tau/2$.)

(a) Show that $\wp$ is real-valued on all these lines, and hence on the boundaries of all these rectangles.

(b) Prove that $\wp$ maps the interior of each rectangle conformally to the upper (or lower) half-plane.

Chapter 10. Applications of Theta Functions

1.* Suppose n is of the form $n = 4^a(8k + 7)$, where a and k are positive integers. Show that n cannot be written as the sum of three-squares. The converse, that every n that is *not* of that form *can* be written as the sum of three-squares, is a difficult theorem of Legendre and Gauss.

2. Let $\mathrm{SL}_2(\mathbb{Z})$ denote the set of 2×2 matrices with integer entries and determinant 1, that is,

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \text{ and } ad - bc = 1 \right\}.$$

This group acts on the upper half-plane by the fractional linear transformation $g(\tau) = (a\tau + b)/(c\tau + d)$. Together with this action comes the so-called fundamental domain $\mathcal{F}_1$ in the complex plane defined by

$$\mathcal{F}_1 = \{\tau \in \mathbb{C} : |\tau| \geq 1, \quad |\mathrm{Re}(\tau)| \leq 1/2 \text{ and } |\mathrm{Im}(\tau)| \geq 0\}.$$

It is illustrated in Figure 3.

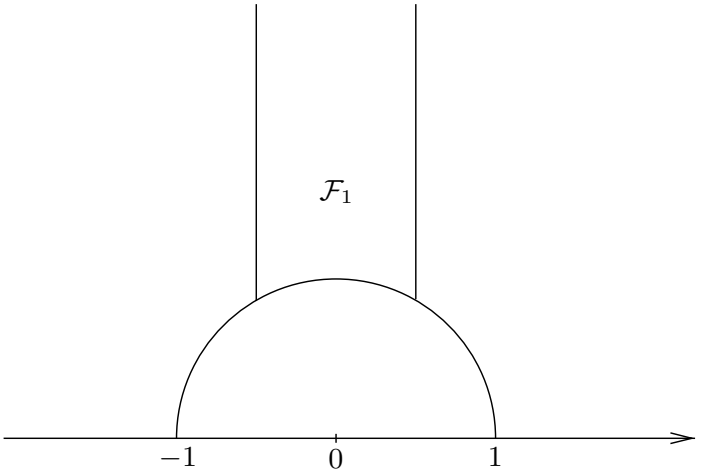


Figure 3. The fundamental domain $\mathcal{F}_1$

Consider the two elements in $\mathrm{SL}_2(\mathbb{Z})$ defined by $S(\tau) = -1/\tau$ and $T_1(\tau) = \tau + 1$. These correspond (for example) to the matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

respectively. Let $\mathfrak{g}$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by S and T_1.

(a) Show that for every $\tau \in \mathbb{H}$ there exists $g \in \mathfrak{g}$ such that $g(\tau) \in \mathcal{F}_1$.

(b) We say that two points τ and τ' are congruent if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g(\tau) = w$. Prove that if $\tau, w \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re}(\tau) = \pm 1/2$ and $\tau' = \tau \mp 1$ or $|\tau| = 1$ and $\tau' = -1/\tau$. [Hint: Say $\tau' = g(\tau)$. Why can one assume that $\mathrm{Im}(\tau') \geq \mathrm{Im}(\tau)$, and therefore $|c\tau + d| \leq 1$? Now consider separately the possibilities $c = -1$, $c = 0$, or $c = 1$.]

(c) Prove that S and T_1 generate the modular group in the sense that every fractional linear transformation corresponding to $g \in \mathrm{SL}_2(\mathbb{Z})$ is a composition of finitely many S's and T_1's, and their inverses. Strictly speaking, the matrices associated to S and T_1 generate the projective special linear group $\mathrm{PSL}_2(\mathbb{Z})$, which equals $\mathrm{SL}_2(\mathbb{Z})$ modulo $\pm I$. [Hint: Observe that $2i$ is in the interior of $\mathcal{F}_1$. Now map $g(2i)$ back into $\mathcal{F}_1$ by using part (a). Use part (b) to conclude.]

3. In this problem, consider the group G of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integer entries, determinant 1, and such that a and d have the same parity, b and c have the same parity, and c and d have opposite parity. This group also acts on the upper half-plane by fractional linear transformations. To the group G corresponds the fundamental domain $\mathcal{F}$ defined by $|\tau| \geq 1$, $|\mathrm{Re}(\tau)| \leq 1$, and $\mathrm{Im}(\tau) \geq 0$ (see Figure 1). Also, let

$$S(\tau) = -1/\tau \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T_2(\tau) = \tau + 2 \leftrightarrow \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Prove that every fractional linear transformation corresponding to $g \in G$ is a composition of finitely many S, T_2 and their inverses, in analogy with the previous problem.

4. Let G denote the group of matrices given in the previous problem. Here we give an alternate proof of Theorem 3.4, that states that a function in $\mathbb{H}$ which is holomorphic, bounded, and invariant under G must be constant.

(a) Suppose that $f : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and that there exists a sequence of complex numbers $\tau_k = x_k + iy_k$ such that

$$f(\tau_k) = 0, \qquad \sum_{k=1}^{\infty} y_k = \infty, \qquad 0 < y_k \leq 1, \qquad \text{and} \qquad |x_k| \leq 1.$$

Then $f = 0$. [Hint: When $x_k = 0$ see Problem 5 in Chapter 8.]

(b) Given two relatively prime integers c and d with different parity, show that there exist integers a and b such that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$. [Hint: All the solutions of $xc + dy = 1$ take the form $x_0 + dt$ and $y_0 - ct$ where x_0, y_0 is a particular solution and $t \in \mathbb{Z}$.]

(c) Prove that $\sum 1/(c^2 + d^2) = \infty$ where the sum is taken over all c and d that are relatively prime and of opposite parity. [Hint: Suppose not, and prove that $\sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ where the sum is over all relatively prime integers a and b. To do so, note that if a and b are both odd and relatively prime, then the two numbers c and d defined by $c = (a + b)/2$ and $d = (a - b)/2$ are relatively prime and of opposite parity. Moreover, $c^2 + d^2 \leq A(a^2 + b^2)$ for some universal constant A. Therefore

$$\sum_{n \neq 0} \frac{1}{n^2} \sum_{(a,b)=1} \frac{1}{a^2 + b^2} < \infty,$$

hence $\sum 1/(k^2 + \ell^2) < \infty$, where the sum is over all integers k and ℓ such that $k, \ell \neq 0$. Why is this a contradiction?]

(d) Prove that if $F : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and invariant under G, then F is constant. [Hint: Replace $F(\tau)$ by $F(\tau) - F(i)$ so that we can assume $F(i) = 0$ and prove $F = 0$. For each relatively prime c and d with opposite parity, choose $g \in G$ so that $g(i) = x_{c,d} + i/(c^2 + d^2)$ with $|x_{c,d}| \leq 1$.]

5.* In Chapter 9 we proved that the Weierstrass $\wp$ function satisfies the cubic equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2 = 60E_4$, $g_3 = 140E_6$, with E_k is the Eisenstein series of order k. The discriminant of the cubic $y^2 = 4x^3 - g_2x - g_3$ is defined by $\triangle = g_2^3 - 27g_3^2$. Prove that

$$\triangle(\tau) = (2\pi)^{12}\eta^{24}(\tau) \qquad \text{for all } \tau \in \mathbb{H}.$$

[Hint: $\triangle$ and η^{24} satisfy the same transformation laws under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. Because of the fundamental domain described in Problem 2, it suffices then to investigate the behavior at the only cusp, which is at infinity.]

6.* Here we will deduce the formula for $r_8(n)$, which counts the number of representations of n as a sum of eight squares. The method is parallel to that of $r_4(n)$, but the details are less delicate.

Theorem: $r_8(n) = 16\sigma_3^*(n)$.

Here $\sigma_3^*(n) = \sigma_3(n) = \sum_{d|n} d^3$, when n is odd. Also, when n is even

$$\sigma_3^*(n) = \sum_{d|n} (-1)^d d^3 = \sigma_3^e(n) - \sigma_3^o(n),$$

where $\sigma_3^e(n) = \sum_{d|n,\ d \text{ even}} d^3$ and $\sigma_3^o(n) = \sum_{d|n,\ d \text{ odd}} d^3$.

Consider the appropriate Eisenstein series

$$E_4^*(\tau) = \sum \frac{1}{(n+m\tau)^4},$$

where the sum is over integers n and m with opposite parity. Recall the standard Eisenstein series

$$E_4(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^4}.$$

Notice that the series defining E_4^* is absolutely convergent, in distinction to $E_2^*(\tau)$, which arose when considering $r_4(n)$. This makes some of the considerations below quite a bit simpler.

(a) Prove that $r_8(n) = 16\sigma_3^*(n)$ is equivalent to the identity $\theta(\tau)^8 = 48\pi^{-4}E_4^*(\tau)$. [Hint: Use the fact that $E_4(\tau) = 2\zeta(4) + \frac{(2\pi)^4}{3}\sum_{k=1}^{\infty}\sigma_3(k)e^{2\pi ik\tau}$ and $\zeta(4) = \pi^4/90$.]

(b) Note that $E_4^*(\tau) = E_4(\tau) - 2^{-4}E_4((\tau-1)/2)$.

(c) $E_4^*(\tau+2) = E_4^*(\tau)$.

(d) $E_4^*(\tau) = \tau^{-4}E_4^*(-1/\tau)$.

(e) $(48/\pi^4)E_4^*(\tau) \to 1$ as $\tau \to \infty$.

(f) $|E_4^*(1-1/\tau)| \approx |\ \tau|^4 |e^{2\pi i\tau}|$, as $\mathrm{Im}(\tau) \to \infty$. [Hint: Verify that $E_4^*(1-1/\tau) = \tau^4(E_4(\tau) - E_4(2\tau))$.]

Since $\theta(\tau)^8$ satisfies properties similar to (c), (d), (e) and (f) above, it follows that the invariant function $48\pi^{-4}E_4^*(\tau)/\theta(\tau)^8$ is bounded and hence a constant, which must be 1. This gives the desired result.

Appendix A. Asymptotics

1. Let η be an indefinitely differentiable function supported in a finite interval, so that $\eta(x) = 1$ for x near 0. Then, if $m > -1$ and $N > 0$,

$$\int_0^\infty e^{-sx} x^m \eta(x)\, dx = s^{-m-1}\Gamma(m+1) + O(s^{-N})$$

for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

(a) Consider first the case $-1 < m \leq 0$. It suffices to see that

$$\int_0^\infty e^{-sx} x^m (1 - \eta(x))\, dx = O(s^{-N}),$$

and this can be done by repeated integration by parts since $e^{-sx} = (-1)^N s^{-N} \left(\frac{d}{dx}\right)^N (e^{-sx})$.

(b) To extend this to all m, find an integer k so that $k - 1 < m \leq k$, write

$$\int \left[\left(\frac{d}{dx} \right)^k (x^m) \right] e^{-sx} \eta(x)\, dx = c_{k,m} s^{-m+k-1} + O(s^{-N}),$$

and integrate by parts k times.

2. The following is a more precise version of Stirling's formula. There are real constant $a_1 = 1/12$, a_2, ..., a_n, ..., so that for every $N > 0$

$$\Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + \sum_{j=1}^{N} a_j s^{-j} + O(s^{-N}) \right) \qquad \text{when } s \in S_\delta.$$

This can be proved by using the results of Problem 1 in place of Proposition 1.3.

3. The Bessel functions and Airy function have the following power series expansions:

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{x^2}{4}\right)^m}{m!\Gamma(\nu+m+1)},$$

$$\mathrm{Ai}(-x) = \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{x^n}{n!} \sin(2\pi(n+1)/3) 3^{n/3-2/3} \Gamma(n/3+1/3).$$

(a) From this, verify that when $x > 0$,

$$\mathrm{Ai}(-x) = \frac{x^{1/2}}{3}\left(J_{1/3}\left(\frac{2}{3}x^{3/2}\right) + J_{-1/3}\left(\frac{2}{3}x^{3/2}\right)\right).$$

(b) The function $\mathrm{Ai}(x)$ extends to an entire function of order 3/2.

[Hint: For (b), use (a), or alternatively, apply Problem 4 in Chapter 5 to the power series for Ai. Compare also with Problem 1, Chapter 4.]

4. Suppose $F(z) = \sum_{n=0}^{\infty} F_n w^n$ is meromorphic in a region containing the closed unit disc, and the only poles of F are on the unit circle at the points $\alpha_1, \ldots, \alpha_k$, and their orders are $r_1, \ldots, r_k$ respectively. Then for some $\epsilon > 0$

$$F_n = \sum_{j=1}^{k} P_j(n) + O(e^{-\epsilon n}) \qquad \text{as } n \to \infty.$$

Here

$$P_j(n) = \frac{1}{(r_j - 1)!} \left(\frac{d}{dw} \right)^{r_j - 1} \left[(w - \alpha_j)^{r_j} w^{-n-1} F(w) \right]_{w=\alpha_j}.$$

Note that each P_j is of the form $P_j(n) = A_j(\alpha_j^{-1} n)^{r_j - 1} + O(n^{r_j - 2})$.

To prove this, use the residue formula (Theorem 1.4, Chapter 3).

5.* The one shortcoming in our derivation of the asymptotic formula for $p(n)$ arose from the fact that while $f_1(z) = \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ is a good approximation to the generating function $f(z)$ near $z = 0$, this fails near other points on the real axis, since f_1 is regular there, but f is not.

However, using the transformation law (24) and the identity $f(z+1) = f(z)$, one can derive the following generalization of (24): whenever p/q is a rational number in lowest form (so p and q are relatively prime) then

$$f\left(z - \frac{p}{q}\right) = \omega_{p/q}\sqrt{\frac{zq}{i}}\ e^{\frac{i\pi}{12zq^2}} e^{-\frac{i\pi z}{12}} f\left(-\frac{1}{zq^2} - \frac{p'}{q}\right),$$

where $pp' = 1 \mod q$. Here $\omega_{p/q}$ is an appropriate 24th root of unity. This formula leads to an analogous $f_{p/q}$, approximating f at $z = p/q$.

From this one can obtain for each p/q a contribution of the form

$$c_{p/q}\frac{1}{2\pi\sqrt{2}}\frac{d}{dn}\left(\frac{e^{\frac{K}{q}(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\right)$$

to the asymptotic formula for $p(n)$. When suitably modified, the resulting series, summed over all proper fractions p/q in $[0, 1)$, actually converges and gives an exact formula for $p(n)$.

Real Analysis: Measure Theory, Integration, and Hilbert Spaces

Chapter 1. Measure Theory

1. Given an irrational x, one can show (using the pigeon-hole principle, for example) that there exists infinitely many fractions p/q, with relatively prime integers p and q such that

$$\left| x - \frac{p}{q} \right| \leq \frac{1}{q^2}.$$

However, prove that the set of those $x \in \mathbb{R}$ such that there exist infinitely many fractions p/q, with relatively prime integers p and q such that

$$\left| x - \frac{p}{q} \right| \leq \frac{1}{q^3} \qquad (\text{or} \leq 1/q^{2+\epsilon}),$$

is a set of measure zero.

[Hint: Use the Borel-Cantelli lemma.]

2. Any open set Ω can be written as the union of closed cubes, so that $\Omega = \bigcup Q_j$ with the following properties

(i) The Q_j's have disjoint interiors.

(ii) $d(Q_j, \Omega^c) \approx$ side length of Q_j. This means that there are positive constants c and C so that $c \leq d(Q_j, \Omega^c)/\ell(Q_j) \leq C$, where $\ell(Q_j)$ denotes the side length of Q_j.

3. Find an example of a measurable subset C of $[0,1]$ such that $m(C) = 0$, yet the difference set of C contains a non-trivial interval centered at the origin. Compare with the result in Exercise 29.

[Hint: Pick the Cantor set $C = \mathcal{C}$. For a fixed $a \in [-1,1]$, consider the line $y = x + a$ in the plane, and copy the construction of the Cantor set, but in the cube $Q = [0,1] \times [0,1]$. First, remove all but four closed cubes of side length $1/3$, one at each corner of Q; then, repeat this procedure in each of the remaining cubes (see Figure 6). The resulting set is sometimes called a Cantor dust. Use the property of nested compact sets to show that the line intersects this Cantor dust.]

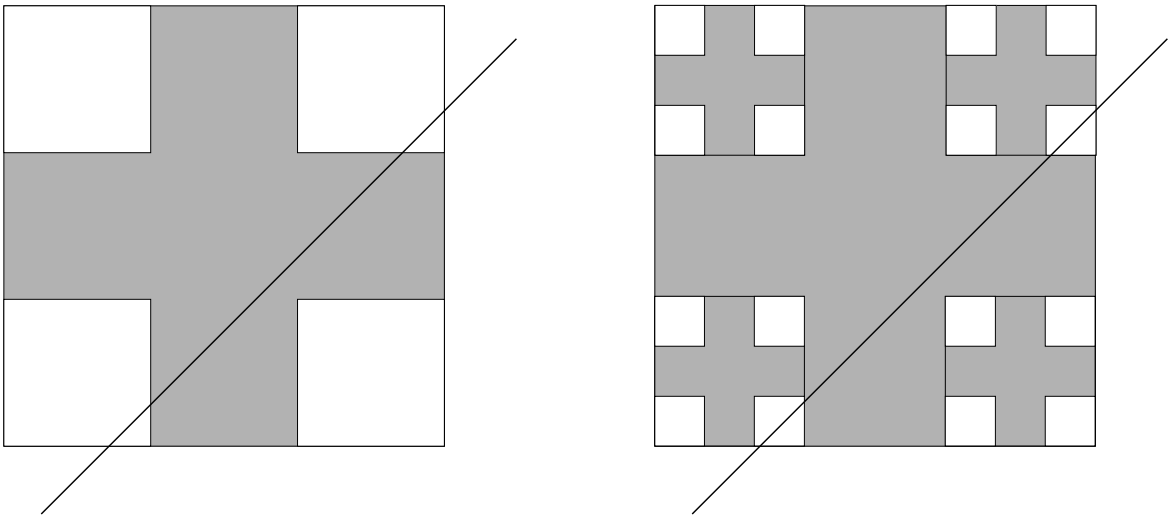


Figure 6. Construction of the Cantor dust

4. Complete the following outline to prove that a bounded function on an interval $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero. This argument is given in detail in the appendix to Book I.

Let f be a *bounded* function on a compact interval J, and let $I(c, r)$ denote the open interval centered at c of radius $r > 0$. Let $\operatorname{osc}(f, c, r) = \sup |f(x) - f(y)|$, where the supremum is taken over all $x, y \in J \cap I(c, r)$, and define the oscillation of f at c by $\operatorname{osc}(f, c) = \lim_{r \to 0} \operatorname{osc}(f, c, r)$. Clearly, f is continuous at $c \in J$ if and only if $\operatorname{osc}(f, c) = 0$.

Prove the following assertions:

(a) For every $\epsilon > 0$, the set of points c in J such that $\operatorname{osc}(f, c) \geq \epsilon$ is compact.

(b) If the set of discontinuities of f has measure 0, then f is Riemann integrable.

[Hint: Given $\epsilon > 0$ let $A_\epsilon = \{c \in J : \operatorname{osc}(f, c) \geq \epsilon\}$. Cover A_ϵ by a finite number of open intervals whose total length is $\leq \epsilon$. Select an appropriate partition of J and estimate the difference between the upper and lower sums of f over this partition.]

(c) Conversely, if f is Riemann integrable on J, then its set of discontinuities has measure 0.

[Hint: The set of discontinuities of f is contained in $\bigcup_n A_{1/n}$. Choose a partition P such that $U(f, P) - L(f, P) < \epsilon/n$. Show that the total length of the intervals in P whose interior intersect $A_{1/n}$ is $\leq \epsilon$.]

5. Suppose E is measurable with $m(E) < \infty$, and

$$E = E_1 \cup E_2, \qquad E_1 \cap E_2 = \emptyset.$$

If $m(E) = m_*(E_1) + m_*(E_2)$, then E_1 and E_2 are measurable.

In particular, if $E \subset Q$, where Q is a finite cube, then E is measurable if and only if $m(Q) = m_*(E) + m_*(Q - E)$.

6.* The fact that the axiom of choice and the well-ordering principle are equivalent is a consequence of the following considerations.

One begins by defining a partial ordering on a set E to be a binary relation $\leq$ on the set E that satisfies:

(i) $x \leq x$ for all $x \in E$.

(ii) If $x \leq y$ and $y \leq x$, then $x = y$.

(iii) If $x \leq y$ and $y \leq z$, then $x \leq z$.

If in addition $x \leq y$ or $y \leq x$ whenever $x, y \in E$, then $\leq$ is a linear ordering of E.

The axiom of choice and the well-ordering principle are then logically equivalent to the Hausdorff maximal principle:

> *Every non-empty partially ordered set has a (non-empty) maximal linearly ordered subset.*

In other words, if E is partially ordered by $\leq$, then E contains a non-empty subset F which is linearly ordered by $\leq$ and such that if F is contained in a set G also linearly ordered by $\leq$, then $F = G$.

An application of the Hausdorff maximal principle to the collection of all well-orderings of subsets of E implies the well-ordering principle for E. However, the proof that the axiom of choice implies the Hausdorff maximal principle is more complicated.

7.* Consider the curve $\Gamma = \{y = f(x)\}$ in $\mathbb{R}^2$, $0 \le x \le 1$. Assume that f is twice continuously differentiable in $0 \le x \le 1$. Then show that $m(\Gamma + \Gamma) > 0$ if and only if $\Gamma + \Gamma$ contains an open set, if and only if f is not linear.

8.* Suppose A and B are open sets of finite positive measure. Then we have equality in the Brunn-Minkowski inequality (8) if and only if A and B are convex and similar, that is, there are a $\delta > 0$ and an $h \in \mathbb{R}^d$ such that

$$A = \delta B + h.$$

Chapter 2. Integration Theory

1. If f is integrable on $[0, 2\pi]$, then $\int_0^{2\pi} f(x)e^{-inx}\,dx \to 0$ as $|n| \to \infty$.
Show as a consequence that if E is a measurable subset of $[0, 2\pi]$, then

$$\int_E \cos^2(nx + u_n)\,dx \to \frac{m(E)}{2}, \qquad \text{as } n \to \infty$$

for any sequence $\{u_n\}$.
[Hint: See Exercise 22.]

2. Prove the Cantor-Lebesgue theorem: if

$$\sum_{n=0}^{\infty} A_n(x) = \sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx)$$

converges for x in a set of positive measure (or in particular for all x), then $a_n \to 0$ and $b_n \to 0$ as $n \to \infty$.

[Hint: Note that $A_n(x) \to 0$ uniformly on a set E of positive measure.]

3. A sequence $\{f_k\}$ of measurable functions on $\mathbb{R}^d$ is **Cauchy in measure** if for every $\epsilon > 0$,

$$m(\{x : |f_k(x) - f_\ell(x)| > \epsilon\}) \to 0 \qquad \text{as } k, \ell \to \infty.$$

We say that $\{f_k\}$ **converges in measure** to a (measurable) function f if for every $\epsilon > 0$

$$m(\{x : |f_k(x) - f(x)| > \epsilon\}) \to 0 \qquad \text{as } k \to \infty.$$

This notion coincides with the "convergence in probability" of probability theory.

Prove that if a sequence $\{f_k\}$ of integrable functions converges to f in L^1, then $\{f_k\}$ converges to f in measure. Is the converse true?

We remark that this mode of convergence appears naturally in the proof of Egorov's theorem.

4. We have already seen (in Exercise 8, Chapter 1) that if E is a measurable set in $\mathbb{R}^d$, and L is a linear transformation of $\mathbb{R}^d$ to $\mathbb{R}^d$, then $L(E)$ is also measurable, and if E has measure 0, then so has $L(E)$. The quantitative statement is

$$m(L(E)) = |\det(L)|\, m(E).$$

As a special case, note that the Lebesgue measure is invariant under rotations. (For this special case see also Exercise 26 in the next chapter.)

The above identity can be proved using Fubini's theorem as follows.

(a) Consider first the case $d = 2$, and L a "strictly" upper triangular transformation $x' = x + ay$, $y' = y$. Then

$$\chi_{L(E)}(x, y) = \chi_E(L^{-1}(x, y)) = \chi_E(x - ay, y).$$

Hence

$$\begin{aligned} m(L(E)) &= \int_{\mathbb{R}\times\mathbb{R}} \left(\int \chi_E(x - ay, y) \right) dy \\ &= \int_{\mathbb{R}\times\mathbb{R}} \left(\int \chi_E(x, y)\, dx \right) dy \\ &= m(E), \end{aligned}$$

by the translation-invariance of the measure.

(b) Similarly $m(L(E)) = m(E)$ if L is strictly lower triangular. In general, one can write $L = L_1 \Delta L_2$, where L_j are strictly (upper and lower) triangular and Δ is diagonal. Thus $m(L(E)) = |\det(L)| m(E)$, if one uses Exercise 7 in Chapter 1.

5. There is an ordering $\prec$ of $\mathbb{R}$ with the property that for each $y \in \mathbb{R}$ the set $\{x \in \mathbb{R} : \ x \prec y\}$ is at most countable.

The existence of this ordering depends on the **continuum hypothesis**, which asserts: whenever S is an infinite subset of $\mathbb{R}$, then either S is countable, or S has the cardinality of $\mathbb{R}$ (that is, can be mapped bijectively to $\mathbb{R}$).[6]

[Hint: Let $\prec$ denote a well-ordering of $\mathbb{R}$, and define the set X by $X = \{y \in \mathbb{R} : \text{the set } \{x : x \prec y\} \text{ is not countable}\}$. If X is empty we are done. Otherwise, consider the smallest element $\overline{y}$ in X, and use the continuum hypothesis.]

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[6]This assertion, formulated by Cantor, is like the well-ordering principle independent of the other axioms of set theory, and so we are also free to accept its validity.

Chapter 3. Differentiation and Integration

1. Prove the following variant of the **Vitali covering lemma**: If E is covered in the Vitali sense by a family $\mathcal{B}$ of balls, and $0 < m_*(E) < \infty$, then for every $\eta > 0$ there exists a disjoint collection of balls $\{B_j\}_{j=1}^{\infty}$ in $\mathcal{B}$ such that

$$m_*\left(E / \bigcup_{j=1}^{\infty} B_j\right) = 0 \qquad \text{and} \qquad \sum_{j=1}^{\infty} |B_j| \le (1+\eta) m_*(E).$$

2. The following simple one-dimensional covering lemma can be used in a number of different situations.

Suppose $I_1, I_2, \ldots, I_N$ is a given finite collection of open intervals in $\mathbb{R}$. Then there are two finite sub-collections $I_1', I_2', \ldots, I_K'$, and $I_1'', I_2'', \ldots, I_L''$, so that each sub-collection consists of mutually disjoint intervals and

$$\bigcup_{j=1}^{N} I_j = \bigcup_{k=1}^{K} I_k' \cup \bigcup_{\ell=1}^{L} I_\ell''.$$

Note that, in contrast with Lemma 1.2, the full union is covered and not merely a part.

[Hint: Choose I_1' to be an interval whose left end-point is as far left as possible. Discard all intervals contained in I_1'. If the remaining intervals are disjoint from I_1', select again an interval as far to the left as possible, and call it I_2'. Otherwise choose an interval that intersects I_1', but reaches out to the right as far as possible, and call this interval I_1''. Repeat this procedure.]

3.* There is no direct analogue of Problem 2 in higher dimensions. However, a full covering is afforded by the Besicovitch covering lemma. A version of this lemma states that there is an integer N (dependent only on the dimension d) with the following property. Suppose E is any bounded set in $\mathbb{R}^d$ that is covered by a collection $\mathcal{B}$ of balls in the (strong) sense that for each $x \in E$, there is a $B \in \mathcal{B}$ whose center is x. Then, there are N sub-collections $\mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_N$ of the original collection $\mathcal{B}$, such that each $\mathcal{B}_j$ is a collection of disjoint balls, and moreover,

$$E \subset \bigcup_{B \in \mathcal{B}'} B, \qquad \text{where } \mathcal{B}' = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \cdots \cup \mathcal{B}_N.$$

4. A real-valued function φ defined on an interval (a, b) is **convex** if the region lying above its graph $\{(x, y) \in \mathbb{R}^2 : y > \varphi(x),\ a \leq x \leq b\}$ is a convex set, as defined in Section 5*, Chapter 1. Equivalently, φ is convex if

$$\varphi(\theta x_1 + (1-\theta)x_2) \leq \theta\varphi(x_1) + (1-\theta)\varphi(x_2)$$

for every $x_1, x_2 \in (a, b)$ and $0 \leq \theta \leq 1$. One can also observe as a consequence that we have the following inequality of the slopes:

$$\frac{\varphi(x+h) - \varphi(x)}{h} \leq \frac{\varphi(y) - \varphi(x)}{y - x} \leq \frac{\varphi(y) - \varphi(y-h)}{h},$$

whenever $x < y$, $h > 0$, and $x + h < y$.

The following can then be proved.

(a) φ is continuous on (a, b).

(b) φ satisfies a Lipschitz condition of order 1 in any proper closed sub-interval $[a', b']$ of (a, b). Hence φ is absolutely continuous in each sub-interval.

(c) φ' exists at all but an at most denumerable number of points, and $\varphi' = D^+\varphi$ is an increasing function with

$$\varphi(y) - \varphi(x) = \int_x^y \varphi'(t)\, dt.$$

(d) Conversely, if ψ is any increasing function on (a, b), then $\varphi(x) = \int_c^x \psi(t)\, dt$ is a convex function in (a, b) (for $c \in (a, b)$).

5. Suppose that F is continuous on $[a, b]$, $F'(x)$ exists for *every* $x \in (a, b)$, and $F'(x)$ is integrable. Then F is absolutely continuous and

$$F(b) - F(a) = \int_a^b F'(x)\, dx.$$

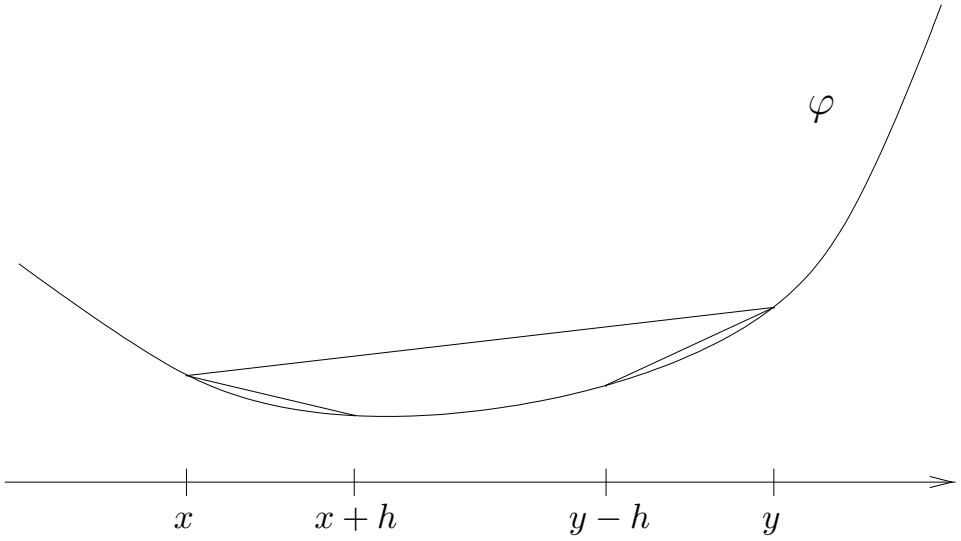


Figure 13. A convex function

[Hint: Assume $F'(x) \geq 0$ for a.e. x. We want to conclude that $F(b) \geq F(a)$. Let E be the set of measure 0 of those x such that $F'(x) < 0$. Then according to Exercise 25, there is a function Φ which is increasing, absolutely continuous, and for which $D^+\Phi(x) = \infty$, $x \in E$. Consider $F + \delta\Phi$, for each δ and apply the result (a) in Exercise 23.]

6.* The following converse to Exercise 30 characterizes functions of bounded variation.

Suppose F is a bounded measurable function on $\mathbb{R}$. If F satisfies either of conditions (a) or (b) in that exercise, then F can be modified on a set of measure zero so as to become a function of bounded variation on $\mathbb{R}$.

Moreover, on $\mathbb{R}^d$ we have the following assertion. Suppose F is a bounded measurable function on $\mathbb{R}^d$. Then the following two conditions on F are equivalent:

(a′) $\int_{\mathbb{R}^d} |F(x+h) - F(x)|\, dx \leq A|h|$, for all $h \in \mathbb{R}^d$.

(b′) $|\int_{\mathbb{R}^d} F(x) \frac{\partial \varphi}{\partial x_j}\, dx| \leq A$, for all $j = 1, \ldots, d$,

for all $\varphi \in C^1$ that have bounded support, and for which $\sup_{x \in \mathbb{R}^d} |\varphi(x)| \leq 1$.

The class of functions that satisfy either (a′) or (b′) is the extension to $\mathbb{R}^d$ of the class of functions of bounded variation.

7. Consider the function

$$f_1(x) = \sum_{n=0}^{\infty} 2^{-n} e^{2\pi i 2^n x}.$$

(a) Prove that f_1 satisfies $|f_1(x) - f_1(y)| \leq A_\alpha |x-y|^\alpha$ for each $0 < \alpha < 1$.

(b)* However, f_1 is nowhere differentiable, hence not of bounded variation.

8.* Let $\mathcal{R}$ denote the set of all rectangles in $\mathbb{R}^2$ that contain the origin, and with sides parallel to the coordinate axis. Consider the maximal operator associated to this family, namely

$$f^*_{\mathcal{R}}(x) = \sup_{R\in\mathcal{R}} \frac{1}{m(R)} \int_R |f(x-y)|\, dy.$$

(a) Then, $f \mapsto f^*_{\mathcal{R}}$ does not satisfy the weak type inequality

$$m(\{x : f^*_{\mathcal{R}}(x) > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1}$$

for all $\alpha > 0$, all integrable f, and some $A > 0$.

(b) Using this, one can show that there exists $f \in L^1(\mathbb{R})$ so that for $R \in \mathcal{R}$

$$\limsup_{\mathrm{diam(R)}\to 0} \frac{1}{m(R)} \int_R f(x-y)\, dy = \infty \qquad \text{for almost every } x.$$

Here $\mathrm{diam(R)} = \sup_{x,y\in R} |x-y|$ equals the diameter of the rectangle.

[Hint: For part (a), let B be the unit ball, and consider the function $\varphi(x) = \chi_B(x)/m(B)$. For $\delta > 0$, let $\varphi_\delta(x) = \delta^{-2}\varphi(x/\delta)$. Then

$$(\varphi_\delta)^*_{\mathcal{R}}(x) \to \frac{1}{|x_1|\,|x_2|} \qquad \text{as } \delta \to 0,$$

for every (x_1, x_2), with $x_1 x_2 \neq 0$. If the weak type inequality held, then we would have

$$m(\{|x| \leq 1 :\ |x_1 x_2|^{-1} > \alpha\}) \leq \frac{A}{\alpha}.$$

This is a contradiction since the left-hand side is of the order of $(\log \alpha)/\alpha$ as α tends to infinity.]

Chapter 4. Hilbert Spaces: An Introduction

1. Let $\mathcal{H}$ be an infinite-dimensional Hilbert space. There exists a linear functional ℓ defined on $\mathcal{H}$ that is not bounded (and hence not continuous).

[Hint: Using the axiom of choice (or one of its equivalent forms), construct an **algebraic basis** of $\mathcal{H}$, $\{e_\alpha\}$; it has the property that every element of $\mathcal{H}$ is uniquely a finite linear combination of the $\{e_\alpha\}$. Select a denumerable collection $\{e_n\}_{n=1}^{\infty}$, and define ℓ to satisfy the requirement that $\ell(e_n) = n\|e_n\|$ for all $n \in \mathbb{N}$.]

2.* The following is an example of a non-separable Hilbert space. We consider the collection of exponentials $\{e^{i\lambda x}\}$ on $\mathbb{R}$, where λ ranges over the real numbers. Let $\mathcal{H}_0$ denote the space of finite linear combinations of these exponentials. For $f, g \in \mathcal{H}_0$, we define the inner product as

$$(f, g) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} f(x)\overline{g(x)}\, dx.$$

(a) Show that this limit exists, and

$$(f, g) = \sum_{k=1}^{N} a_{\lambda_k} \overline{b_{\lambda_k}}$$

if $f(x) = \sum_{k=1}^{N} a_{\lambda_k} e^{i\lambda_k x}$ and $g(x) = \sum_{k=1}^{N} b_{\lambda_k} e^{i\lambda_k x}$.

(b) With this inner product $\mathcal{H}_0$ is a pre-Hilbert space. Notice that $\|f\| \leq \sup_x |f(x)|$, if $f \in \mathcal{H}_0$, where $\|f\|$ denotes the norm $\langle f, f\rangle^{1/2}$. Let $\mathcal{H}$ be the completion of $\mathcal{H}_0$. Then $\mathcal{H}$ is not separable because $e^{i\lambda x}$ and $e^{i\lambda' x}$ are orthonormal if $\lambda \neq \lambda'$.

A continuous function F defined on $\mathbb{R}$ is called **almost periodic** if it is the uniform limit (on $\mathbb{R}$) of elements in $\mathcal{H}_0$. Such functions can be identified with (certain) elements in the completion $\mathcal{H}$: We have $\mathcal{H}_0 \subset AP \subset \mathcal{H}$, where AP denotes the almost periodic functions.

(c) A continuous function F is in AP if for every $\epsilon > 0$ we can find a length $L = L_\epsilon$ such that any interval $I \subset \mathbb{R}$ of length L contains an "almost period" τ satisfying

$$\sup_x |F(x + \tau) - F(x)| < \epsilon.$$

(d) An equivalent characterization is that F is in AP if and only if every sequence $F(x + h_n)$ of translates of F contains a subsequence that converges uniformly.

3. The following is a direct generalization of Fatou's theorem: if $u(re^{i\theta})$ is harmonic in the unit disc and bounded there, then $\lim_{r\to 1} u(re^{i\theta})$ exists for a.e. θ.

[Hint: Let $a_n(r) = \frac{1}{2\pi}\int_0^{2\pi} u(re^{i\theta})e^{-in\theta}\,d\theta$. Then $a_n''(r) + \frac{1}{r}a_n'(r) - \frac{n^2}{r^2}a_n(r) = 0$, hence $a_n(r) = A_n r^n + B_n r^{-n}$, $n \neq 0$, and as a result[6] $u(re^{i\theta}) = \sum_{-\infty}^{\infty} a_n r^{|n|} e^{in\theta}$. From this one can proceed as in the proof of Theorem 3.3.]

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[6]See also Section 5, Chapter 2 in Book I.

4.* This problem provides some examples of functions that fail to have radial limits almost everywhere.

(a) At almost every point of the boundary unit circle, the function $\sum_{n=0}^{\infty} z^{2^n}$ fails to have a radial limit.

(b) More generally, suppose $F(z) = \sum_{n=0}^{\infty} a_n z^{2^n}$. Then, if $\sum |a_n|^2 = \infty$ the function F fails to have radial limits at almost every boundary point. However, if $\sum |a_n|^2 < \infty$, then $F \in H^2(\mathbb{D})$, and we know by the proof of Theorem 3.3 that F does have radial limits almost everywhere.

5.* Suppose F is holomorphic in the unit disc, and

$$\sup_{0\le r<1} \frac{1}{2\pi}\int_{-\pi}^{\pi} \log^+ |F(re^{i\theta})|\, d\theta < \infty,$$

where $\log^+ u = \log u$ if $u \ge 1$, and $\log^+ u = 0$ if $u < 1$.

Then $\lim_{r\to 1} F(re^{i\theta})$ exists for almost every θ.

The above condition is satisfied whenever (say)

$$\sup_{0\le r<1} \frac{1}{2\pi}\int_{-\pi}^{\pi} |F(re^{i\theta})|^p\, d\theta < \infty, \qquad \text{for some } p > 0,$$

(since $e^{pu} \ge pu$, $u \ge 0$).

Functions that satisfy the latter condition are said to belong to the Hardy space $H^p(\mathbb{D})$.

6.* If T is compact, and $\lambda \neq 0$, show that

(a) $\lambda I - T$ is injective if and only if $\overline{\lambda} I - T^*$ is injective.

(b) $\lambda I - T$ is injective if and only if $\lambda I - T$ is surjective.

This result, known as the **Fredholm alternative**, is often combined with that in Exercise 29.

7. Show that the identity operator on $L^2(\mathbb{R}^d)$ cannot be given as an (absolutely) convergent integral operator. More precisely, if $K(x, y)$ is a measurable function on $\mathbb{R}^d \times \mathbb{R}^d$ with the property that for each $f \in L^2(\mathbb{R}^d)$, the integral $T(f)(x) = \int_{\mathbb{R}^d} K(x, y)f(y)\,dy$ converges for almost every x, then $T(f) \neq f$ for some f.

[Hint: Prove that otherwise for any pair of disjoint balls B_1 and B_2 in $\mathbb{R}^d$, we would have that $K(x, y) = 0$ for a.e. $(x, y) \in B_1 \times B_2$.]

8.* Suppose $\{T_k\}$ is a collection of bounded opeartors on a Hilbert space $\mathcal{H}$. Assume that

$$\|T_k T_j^*\| \leq a_{k-j} \qquad \text{and} \qquad \|T_k^* T_j\| \leq a_{k-j}^*,$$

for positive constants $\{a_n\}$ with the property that $\sum_{-\infty}^{\infty} a_n = A < \infty$. Then $S_N(f)$ converges as $N \to \infty$, for every $f \in \mathcal{H}$, with $S_N = \sum_{-N}^{N} T_k$. Moreover, $T = \lim_{N\to\infty} S_N$ satisfies $\|T\| \leq A$.

9. A discussion of a class of regular Sturm-Liouville operators follows. Other special examples are given in the problems below.

Suppose $[a, b]$ is a bounded interval, and L is defined on functions f that are twice continuously differentiable in $[a, b]$ (we write, $f \in C^2([a, b])$) by

$$L(f)(x) = \frac{d^2 f}{dx^2} - q(x)f(x).$$

Here the function q is continuous and real-valued on $[a, b]$, and we assume for simplicity that q is non-negative. We say that $\varphi \in C^2([a, b])$ is an **eigenfunction** of L with eigenvalue μ if $L(\varphi) = \mu\varphi$, under the assumption that φ satisfies the boundary conditions $\varphi(a) = \varphi(b) = 0$. Then one can show:

(a) The eigenvalues μ are strictly negative, and the eigenspace corresponding to each eigenvalue is one-dimensional.

(b) Eigenvectors corresponding to distinct eigenvalues are orthogonal in $L^2([a, b])$.

(c) Let $K(x, y)$ be the "Green's kernel" defined as follows. Choose $\varphi_-(x)$ to be a solution of $L(\varphi_-) = 0$, with $\varphi_-(a) = 0$ but $\varphi'_-(a) \neq 0$. Similarly, choose $\varphi_+(x)$ to be a solution of $L(\varphi_+) = 0$ with $\varphi_+(b) = 0$, but $\varphi'_+(b) \neq 0$. Let $w = \varphi'_+(x)\varphi_-(x) - \varphi'_-(x)\varphi_+(x)$, be the "Wronskian" of these solutions, and note that w is a non-zero constant.

Set

$$K(x, y) = \begin{cases} \frac{\varphi_-(x)\varphi_+(y)}{w} & \text{if } a \leq x \leq y \leq b, \\ \frac{\varphi_+(x)\varphi_-(y)}{w} & \text{if } a \leq y \leq x \leq b. \end{cases}$$

Then the operator T defined by

$$T(f)(x) = \int_a^b K(x, y)f(y)\, dy$$

is a Hilbert-Schmidt operator, and hence compact. It is also symmetric. Moreover, whenever f is continuous on $[a, b]$, Tf is of class $C^2([a, b])$ and

$$L(Tf) = f.$$

(d) As a result, each eigenvector of T (with eigenvalue λ) is an eigenvector of L (with eigenvalue $\mu = 1/\lambda$). Hence Theorem 6.2 proves the completeness of the orthonormal set arising from normalizing the eigenvectors of L.

10.* Let L be defined on $C^2([-1,1])$ by

$$L(f)(x) = (1-x^2)\frac{d^2f}{dx^2} - 2x\frac{df}{dx}.$$

If φ_n is the nth Legendre polynomial, given by

$$\varphi_n(x) = \left(\frac{d}{dx}\right)^n (1-x^2)^n, \qquad n = 0, 1, 2, \ldots,$$

then $L\varphi_n = -n(n+1)\varphi_n$.

When normalized the φ_n form an orthonormal basis of $L^2([-1,1])$ (see also Problem 2, Chapter 3 in Book I, where φ_n is denoted by L_n.)

11.* The **Hermite functions** $h_k(x)$ are defined by the generating identity

$$\sum_{k=0}^{\infty} h_k(x)\frac{t^k}{k!} = e^{-(x^2/2-2tx+t^2)}.$$

(a) They satisfy the "creation" and "annihilation" identities $\left(x - \frac{d}{dx}\right) h_k(x) = h_{k+1}(x)$ and $\left(x + \frac{d}{dx}\right) h_k(x) = h_{k-1}(x)$ for $k \geq 0$ where $h_{-1}(x) = 0$. Note that $h_0(x) = e^{-x^2/2}$, $h_1(x) = 2xe^{-x^2/2}$, and more generally $h_k(x) = P_k(x)e^{-x^2/2}$, where P_k is a polynomial of degree k.

(b) Using (a) one sees that the h_k are eigenvectors of the operator $L = -d^2/dx^2 + x^2$, with $L(h_k) = \lambda_k h_k$, where $\lambda_k = 2k + 1$. One observes that these functions are mutually orthogonal. Since

$$\int_{\mathbb{R}} [h_k(x)]^2\, dx = \pi^{1/2} 2^k k! = c_k,$$

we can normalize them obtaining a orthonormal sequence $\{H_k\}$, with $H_k = c_k^{-1/2} h_k$. This sequence is complete in $L^2(\mathbb{R}^d)$ since $\int_{\mathbb{R}} f H_k\, dx = 0$ for all k implies $\int_{-\infty}^{\infty} f(x) e^{-\frac{x^2}{2}+2tx}\, dx = 0$ for all $t \in \mathbb{C}$.

(c) Suppose that $K(x,y) = \sum_{k=0}^{\infty} \frac{H_k(x)H_k(y)}{\lambda_k}$, and also $F(x) = T(f)(x) = \int_{\mathbb{R}} K(x,y) f(y)\, dy$. Then T is a symmetric Hilbert-Schmidt operator, and if $f \sim \sum_{k=0}^{\infty} a_k H_k$, then $F \sim \sum_{k=0}^{\infty} \frac{a_k}{\lambda_k} H_k$.

One can show on the basis of (a) and (b) that whenever $f \in L^2(\mathbb{R})$, not only is $F \in L^2(\mathbb{R})$, but also $x^2 F(x) \in L^2(\mathbb{R})$. Moreover, F can be corrected on a set of measure zero, so it is continuously differentiable, F' is absolutely continuous, and $F'' \in L^2(\mathbb{R})$. Finally, the operator T is the inverse of L in the sense that

$$LT(f) = LF = -F'' + x^2 F = f \qquad \text{for every } f \in L^2(\mathbb{R}).$$

(See also Problem 7* in Chapter 5 of Book I.)

Chapter 5. Hilbert Spaces: Several Examples

1. Suppose $F_0(x) \in L^2(\mathbb{R})$. Then a necessary and sufficient condition that there exists an entire analytic function F, such that $|F(z)| \leq Ae^{a|z|}$ for all $z \in \mathbb{C}$, and $F_0(x) = F(x)$ a.e. $x \in \mathbb{R}$, is that $\hat{F}_0(\xi) = 0$ whenever $|\xi| > a/2\pi$.

[Hint: Consider the regularization $F^\epsilon(z) = \int_{-\infty}^{\infty} F(z-t)\varphi_\epsilon(t)\,dt$ and apply to it the considerations in Theorem 3.3 of Chapter 4 in Book II.]

2. Suppose Ω is an open bounded subset of $\mathbb{R}^2$. A **boundary Lipschitz arc** γ is a portion of $\partial\Omega$ which after a rotation of the axes is represented as

$$\gamma = \{(x_1, x_2) : \, x_2 = \eta(x_1), \; a \leq x_1 \leq b\},$$

where $a < b$ and $\gamma \subset \partial\Omega$. It is also supposed that

$$|\eta(x_1) - \eta(x_1')| \leq M|x_1 - x_1'|, \qquad \text{whenever } x_1, x_1' \in [a, b], \tag{35}$$

and moreover if $\gamma_\delta = \{(x_1, x_2) : \; x_2 - \delta \leq \eta(x_1) \leq x_2\}$, then $\gamma_\delta \cap \Omega = \emptyset$ for some $\delta > 0$. (Note that the condition (35) is satisfied if $\eta \in C^1([a, b])$.)

Suppose Ω satisfies the following condition. There are finitely many open discs $D_1, D_2, \ldots, D_N$ with the property that $\bigcup_j D_j$ contains $\partial\Omega$ and for each j, $\partial\Omega \cap D_j$ is a boundary Lipschitz arc (see Figure 10). Then Ω verifies the outside-triangle condition of Theorem 4.12, guaranteeing the solvability of the boundary value problem.

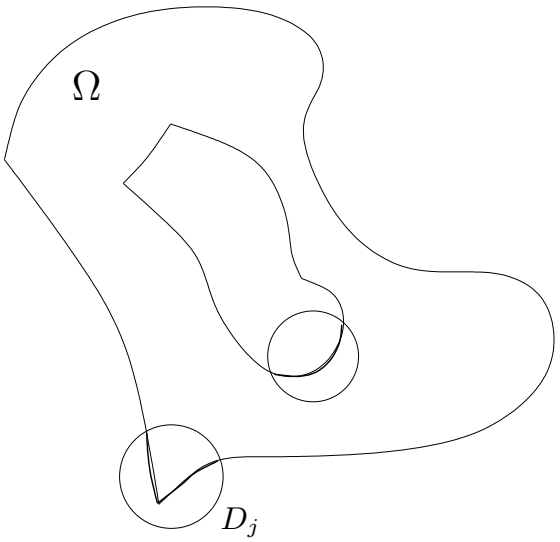
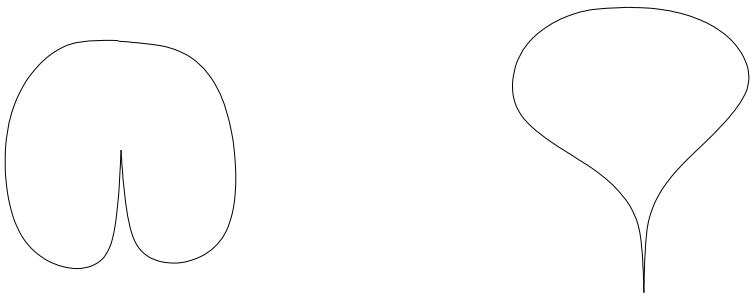


Figure 10. A domain with boundary Lipschitz arcs

3.* Suppose the bounded domain Ω has as its boundary a closed simple continuous curve. Then the boundary value problem is solvable for Ω. This is because there exists a conformal map Φ of the unit disc $\mathbb{D}$ to Ω that extends to a continuous bijection from $\overline{\mathbb{D}}$ to $\overline{\Omega}$. (See Section 1.3 and Problem 6^* in Chapter 8 of Book II.)

4. Consider the two domains Ω in $\mathbb{R}^2$ given by Figure 11.



Domain I Domain II

Figure 11. Domains with a cusp

The set I has as its boundary a smooth curve, with the exception of an (inside) cusp. The set II is similar, except it has an outside cusp. Both I and II fall within the scope of the result of Problem 3, and hence the boundary value problem is solvable in each case. However, II satisfies the outside-triangle condition while I does not.

5. Let T be a Fourier multiplier operator on $L^2(\mathbb{R}^d)$. That is, suppose there is a bounded function m such that $\widehat{(Tf)}(\xi) = m(\xi)\hat{f}(\xi)$, all $f \in L^2(\mathbb{R}^d)$. Then T commutes with translations, $\tau_h T = T\tau_h$, where $\tau_h(f)(x) = f(x-h)$, for all $h \in \mathbb{R}^d$.

Conversely any bounded operator on $L^2(\mathbb{R}^d)$ that commutes with translations is a Fourier multiplier operator.

[Hint: It suffices to prove that if a bounded operator $\hat{T}$ commutes with multiplication by exponentials $e^{2\pi i\xi\cdot h}$, $h \in \mathbb{R}^d$, then there is an m so that $\hat{T}g(\xi) = m(\xi)g(\xi)$ for all $g \in L^2(\mathbb{R}^d)$. To do this, show first that

$$\hat{T}(\Phi g) = \Phi\hat{T}(g), \qquad \text{all } g \in L^2(\mathbb{R}^d), \text{ whenever } \Phi \in C_0^\infty(\mathbb{R}^d).$$

Next, for large N, choose Φ so that it equals 1 in the ball $|\xi| \le N$. Then $m(\xi) = \hat{T}(\Phi)(\xi)$ for $|\xi| \le N$.]

As a consequence of this theorem show that if T is a bounded operator on $L^2(\mathbb{R})$ that commutes with translations and dilations (as in Exercise 9 above), then

(a) If $(Tf)(-x) = T(f(-x))$ it follows $T = cI$, where c is an appropriate constant and I the identity operator.

(b) If $(Tf)(-x) = -T(f(-x))$, then $T = cH$, where c is an appropriate constant and H the Hilbert transform.

6. This problem provides an example of the contrast between analysis on $L^1(\mathbb{R}^d)$ and $L^2(\mathbb{R}^d)$.

Recall that if f is locally integrable on $\mathbb{R}^d$, the maximal function f^* is defined by

$$f^*(x) = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)|\, dy,$$

where the supremum is taken over all balls containing the point x.

Complete the following outline to prove that there exists a constant C so that

$$\|f^*\|_{L^2(\mathbb{R}^d)} \leq C\|f\|_{L^2(\mathbb{R}^d)}.$$

In other words, the map that takes f to f^* (although not linear) is bounded on $L^2(\mathbb{R}^d)$. This differs notably from the situation in $L^1(\mathbb{R}^d)$, as we observed in Chapter 3.

(a) For each $\alpha > 0$, prove that if $f \in L^2(\mathbb{R}^d)$, then

$$m(\{x :\ f^*(x) > \alpha\}) \leq \frac{2A}{\alpha} \int_{|f|>\alpha/2} |f(x)|\, dx.$$

Here, $A = 3^d$ will do.

[Hint: Consider $f_1(x) = f(x)$ if $|f(x)| \geq \alpha/2$ and 0 otherwise. Check that $f_1 \in L^1(\mathbb{R}^d)$, and

$$\{x :\ f^*(x) > \alpha\} \subset \{x :\ f_1^*(x) > \alpha/2\}.]$$

(b) Show that

$$\int_{\mathbb{R}^d} |f^*(x)|^2 dx = 2\int_0^\infty \alpha m(E_\alpha) d\alpha,$$

where $E_\alpha = \{x :\ f^*(x) > \alpha\}$.

(c) Prove that $\|f^*\|_{L^2(\mathbb{R}^d)} \leq C\|f\|_{L^2(\mathbb{R}^d)}$.

Chapter 6. Abstract Measure and Integration Theory

1. Suppose Φ is a C^1 bijection of an open set $\mathcal{O}$ in $\mathbb{R}^d$ onto another open set $\mathcal{O}'$ in $\mathbb{R}^d$.

(a) If E is a measurable subset of $\mathcal{O}$, then $\Phi(E)$ is also measurable.

(b) $m(\Phi(E)) = \int_E |\det \Phi'(x)|\, dx$, where Φ' is the Jacobian of Φ.

(c) $\int_{\mathcal{O}'} f(y)\, dy = \int_{\mathcal{O}} f(\Phi(x))\, |\det \Phi'(x)|\, dx$ whenever f is integrable on $\mathcal{O}'$.

[Hint: To prove (a) follow the argument in Exercise 8, Chapter 1. For (b) assume E is a bounded open set, and write E as $\bigcup_{j=1}^{\infty} Q_j$, where Q_j are cubes whose interiors are disjoint, and whose diameters are less than ϵ. Let z_k be the center of Q_k. Then if $x \in Q_k$,

$$\Phi(x) = \Phi(z_k) + \Phi'(z_k)(x - z_k) + o(\epsilon),$$

hence $\Phi(Q_k) = \Phi(z_k) + \Phi'(z_k)(Q_k - z_k) + o(\epsilon)$, and as a result $(1 - \eta(\epsilon))\Phi'(z_k)(Q_k - z_k) \subset \Phi(Q_k) - \Phi(z_k) \subset (1 + \eta(\epsilon))\Phi'(z_k)(Q_k - z_k)$, where $\eta(\epsilon) \to 0$ as $\epsilon \to 0$. This means that

$$m(\Phi(\mathcal{O})) = \sum_k m(\Phi(Q_k)) = \sum_k |\det(\Phi'(z_k))|\, m(Q_k) + o(1) \qquad \text{as } \epsilon \to 0$$

on account of the linear transformation property of the Lebesgue measure given in Problem 4 of Chapter 2. Note that (b) is (c) for $f(\Phi(x)) = \chi_E(x)$.]

2. Show as a consequence of the previous problem: the measure $d\mu = \frac{dxdy}{y^2}$ in the upper half-plane $\mathbb{R}^2_+ = \{z = x + iy,\ \ y > 0\}$ is preserved by any fractional linear transformation $z \mapsto \frac{az+b}{cz+d}$, where $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belongs to $\mathrm{SL}_2(\mathbb{R})$.

3. Let S be a hypersurface in $\mathbb{R}^d = \mathbb{R}^{d-1} \times \mathbb{R}$, given by

$$S = \{(x, y) \in \mathbb{R}^{d-1} \times \mathbb{R} : y = F(x)\},$$

with F a C^1 function defined on an open set Ω in $\mathbb{R}^{d-1}$. For each subset $E \subset \Omega$ we write $\widehat{E}$ for the corresponding subset of S given by $\widehat{E} = \{(x, F(x))\, x \in E\}$. We note that the Borel sets of S can be defined in terms of the metric on S (which is the restriction of the Euclidean metric on $\mathbb{R}^d$). Thus if E is a Borel set in Ω, then $\widehat{E}$ is a Borel subset of S.

(a) Let μ be the Borel measure on S given by

$$\mu(\widehat{E}) = \int_E \sqrt{1 + |\nabla F|^2}\, dx.$$

If B is a ball in Ω, let $\widehat{B}^\delta = \{(x, y) \in \mathbb{R}^d,\ d((x, y), \widehat{B}) < \delta\}$. Show that

$$\mu(\widehat{B}) = \lim_{\delta \to 0} \frac{1}{2\delta} m((\widehat{B})^\delta),$$

where m denotes the d-dimensional Lebesgue measure. This result is analogous to Theorem 4.4 in Chapter 3.

(b) One may apply (a) to the case when S is the (upper) half of the unit sphere in $\mathbb{R}^d$, given by $y = F(x)$, $F(x) = (1 - |x|^2)^{1/2}$, $|x| < 1$, $x \in \mathbb{R}^{d-1}$. Show that in this case $d\mu = d\sigma$, the measure on the sphere arising in the polar coordinate formula in Section 3.2.

(c) The above conclusion allows one to write an explicit formula for $d\sigma$ in terms of spherical coordinates. Take, for example, the case $d = 3$, and write $y = \cos\theta$, $x = (x_1, x_2) = (\sin\theta \cos\varphi, \sin\theta \sin\varphi)$ with $0 \le \theta < \pi/2$, $0 \le \varphi < 2\pi$. Then according to (a) and (b) the element of area $d\sigma$ equals $(1 - |x|^2)^{-1/2}\, dx$. Use the change of variable theorem in Problem 1 to deduce that in this case $d\sigma = \sin\theta\, d\theta\, d\varphi$. This may be generalized to d dimensions, $d \ge 2$, to obtain the formulas in Section 2.4 of the appendix in Book I.

4.* Let μ be a Borel measure on the sphere S^{d-1} which is rotation-invariant in the following sense: $\mu(r(E)) = \mu(E)$, for every rotation r of $\mathbb{R}^d$ and each Borel subset E of S^{d-1}. If $\mu(S^{d-1}) < \infty$, then μ is a constant multiple of the measure σ arising in the polar coordinate integration formula.

[Hint: Show that

$$\int_{S^{d-1}} Y_k(x)\, d\mu(x) = 0$$

for every surface spherical harmonic of degree $k \geq 1$. As a result, there is a constant c so that

$$\int_{S^{d-1}} f\, d\mu = c \int_{S^{d-1}} f\, d\sigma$$

for every continuous function f on S^{d-1}.]

5.* Suppose X is a metric space, and μ is a Borel measure on X with the property that $\mu(B) < \infty$ for every ball B. Define $C_0(X)$ to be the vector space of continuous functions on X that are each supported in some closed ball. Then $\ell(f) = \int_X f \, d\mu$ defines a linear functional on $C_0(X)$ that is positive, that is, $\ell(f) \geq 0$ if $f \geq 0$.

Conversely, for any positive linear functional ℓ on $C_0(X)$, there exists a unique Borel measure μ that is finite on all balls, such that $\ell(f) = \int f \, d\mu$.

6. Consider an automorphism A of $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, that is, A is a linear isomorphism of $\mathbb{R}^d$ that preserves the lattice $\mathbb{Z}^d$. Note that A can be written as a $d \times d$ matrix whose entries are integers, with $\det A = \pm 1$. Define the mapping $\tau : \mathbb{T}^d \to \mathbb{T}^d$ by $\tau(x) = A(x)$.

(a) Observe that τ is a measure-preserving isomorphism of $\mathbb{T}^d$.

(b) Show that τ is ergodic (in fact, mixing) if and only if A has no eigenvalues of the form $e^{2\pi ip/q}$, where p and q are integers.

(c) Note that τ is never uniquely ergodic.

[Hint: The condition (b) is the same as $(A^t)^q$ has no invariant vectors, where A^t is the transpose of A. Note also that $f(\tau^k(x)) = e^{2\pi i(A^t)^k(n)\cdot x}$ where $f(x) = e^{2\pi in\cdot x}$.]

7.* There is a version of the maximal ergodic theorem that is akin to the "rising sun lemma" and Exercise 6 in Chapter 3.

Suppose f is real-valued, and $f^{\#}(x) = \sup_m \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$. Let $E_0 = \{x : f^{\#}(x) > 0\}$. Then

$$\int_{E_0} f(x)\, dx \geq 0.$$

As a result (when we apply this to $f(x) - \alpha$), we get when $f \geq 0$ that

$$\mu\{x : \ f^*(x) > \alpha\} \leq \frac{1}{\alpha} \int_{\{f^*(x) > \alpha\}} f(x)\, dx.$$

In particular, the constant A in Theorem 5.3 can be taken to be 1.

8. Let $X = [0, 1)$, $\tau(x) = \langle 1/x \rangle$, $x \neq 0$, $\tau(0) = 0$. Here $\langle x \rangle$ denotes the fractional part of x. With the measure $d\mu = \frac{1}{\log 2} \frac{dx}{1+x}$, we have of course $\mu(X) = 1$.

Show that τ is a measure-preserving transformation.

[Hint: $\sum_{k=1}^{\infty} \frac{1}{(x+k)(x+k+1)} = \frac{1}{1+x}$.]

9.* The transformation τ in the previous problem is ergodic.

10.* The connection between continued fractions and the transformation $\tau(x) = \langle 1/x \rangle$ will now be described. A **continued fraction**, $a_0 + 1/(a_1 + 1/a_2)\cdots$, also written as $[a_0 a_1 a_2 \cdots]$, where the a_j are positive integers, can be assigned to any positive real number x in the following way. Starting with x, we successively transform it by two alternating operations: reducing it modulo 1 to lie in $[0, 1)$, and then taking the reciprocal of that number. The integers a_j that arise then define the continued fraction of x.

Thus we set $x = a_0 + r_0$, where $a_0 = [x]$ = the greatest integer in x, and $r_0 \in [0, 1)$. Next we write $1/r_0 = a_1 + r_1$, with $a_1 = [1/r_0]$, $r_1 \in [0, 1)$, to obtain successively $1/r_{n-1} = a_n + r_n$, where $a_n = [1/r_{n-1}]$, $r_n \in [0, 1)$. If $r_n = 0$ for some n, we write $a_k = 0$ for all $k > n$, and say that such a continued fraction **terminates**.

Note that if $0 \leq x < 1$, then $r_0 = x$ and $a_1 = [1/x]$, while $r_1 = \langle 1/x \rangle = \tau(x)$. More generally then, $a_k(x) = [1/\tau^{k-1}(x)] = a_1 \tau^{k-1}(x)$. The following properties of continued fractions of positive real numbers x are known:

(a) The continued fraction of x terminates if and only if x is rational.

(b) If $x = [a_0 a_1 \cdots a_n \cdots]$, and $x_N = [a_0 a_1 \cdots a_N 00 \cdots]$, then $x_N \to x$ as $N \to \infty$. The sequence $\{x_N\}$ gives essentially an optimal approximation of x by rationals.

(c) The continued fraction is periodic, that is, $a_{k+N} = a_k$ for some $N \geq 1$, and all sufficiently large k, if and only if x is an algebraic number of degree ≤ 2 over the rationals.

(d) One can conclude that $\frac{a_1 + a_2 + \cdots + a_n}{n} \to \infty$ as $n \to \infty$ for almost every x. In particular, the set of numbers x whose continued fractions $[a_0 a_1 \cdots a_n \cdots]$ are bounded has measure zero.

[Hint: For (d) apply a consequence of the pointwise ergodic theorem, which is as follows: Suppose $f \geq 0$, and $\int f \, d\mu = \infty$. If τ is ergodic, then $\frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x)) \to \infty$ for a.e. x as $m \to \infty$. In the present case take $f(x) = [1/x]$.]

Chapter 7. Hausdorff Measure and Fractals

1. Carry out the construction below of two sets U and V so that

$$\dim U = \dim V = 0 \qquad \text{but} \qquad \dim(U \times V) \geq 1.$$

Let $I_1, \ldots, I_n, \ldots$ be given as follows:

- Each I_j is a finite sequence of consecutive positive integers; that is, for all j

$$I_j = \{n \in \mathbb{N} : A_j \leq n \leq B_j\} \qquad \text{for some given } A_j \text{ and } B_j.$$

- For each j, I_{j+1} is to the right of I_j; that is, $A_{j+1} > B_j$.

Let $U \subset [0,1]$ consist of all x which when written dyadically $x = .a_1 a_2 \cdots a_n \cdots$ have the property that $a_n = 0$ whenever $n \in \bigcup_j I_j$. Assume also that A_j and B_j tend to infinity (as $j \to \infty$) rapidly enough, say $B_j/A_j \to \infty$ and $A_{j+1}/B_j \to \infty$.

Also, let J_j be the complementary blocks of integers, that is,

$$J_j = \{n \in \mathbb{N} : B_j < n < A_{j+1}\}.$$

Let $V \subset [0,1]$ consist of those $x = .a_1 a_2 \cdots a_n \cdots$ with $a_n = 0$ if $n \in \bigcup_j J_j$.

Prove that U and V have the desired property.

2.* The iso-diametric inequality states the following: If E is a bounded subset of $\mathbb{R}^d$ and diam $E = \sup\{|x-y| : x, y \in E\}$, then

$$m(E) \leq v_d \left(\frac{\text{diam } E}{2}\right)^d,$$

where v_d denotes the volume of the unit ball in $\mathbb{R}^d$. In other words, among sets of a given diameter, the ball has maximum volume. Clearly, it suffices to prove the inequality for $\overline{E}$ instead of E, so we can assume that E is compact.

(a) Prove the inequality in the special case when E is symmetric, that is, $-x \in E$ whenever $x \in E$.

In general, one reduces to the symmetric case by using a technique called Steiner symmetrization. If e is a unit vector in $\mathbb{R}^d$, and $\mathcal{P}$ is a plane perpendicular to e, the Steiner symmetrization of E with respect to E is defined by

$$S(E,e) = \{x + te : x \in \mathcal{P},\ |t| \leq \frac{1}{2} L(E;e;x)\},$$

where $L(E;e;x) = m\left(\{t \in \mathbb{R} :\ x + t \cdot e \in E\}\right)$, and m denotes the Lebesgue measure. Note that $x + te \in S(E,e)$ if and only if $x - te \in S(E,e)$.

(b) Prove that $S(E,e)$ is a bounded measurable subset of $\mathbb{R}^d$ that satisfies $m(S(E,e)) = m(E)$.

[Hint: Use Fubini's theorem.]

(c) Show that diam $S(E,e) \leq$ diam E.

(d) If ρ is a rotation that leaves E and $\mathcal{P}$ invariant, show that $\rho S(E,e) = S(E,e)$.

(e) Finally, consider the standard basis $\{e_1, \ldots, e_d\}$ of $\mathbb{R}^d$. Let $E_0 = E$, $E_1 = S(E_0, e_1)$, $E_2 = S(E_1, e_2)$, and so on. Use the fact that E_d is symmetric to prove the iso-diametric inequality.

(f) Use the iso-diametric inequality to show that $m(E) = \frac{v_d}{2^d} m_d(E)$ for any Borel set E in $\mathbb{R}^d$.

3. Suppose S is a similarity.

(a) Show that S maps a line segment to a line segment.

(b) Show that if L_1 and L_2 are two segments that make an angle α, then $S(L_1)$ and $S(L_2)$ make an angle α or $-\alpha$.

(c) Show that every similarity is a composition of a translation, a rotation (possibly improper), and a dilation.

4.* The following gives a generalization of the construction of the Cantor-Lebesgue function.

Let F be the compact set in Theorem 2.9 defined in terms of m similarities $S_1, S_2, \ldots, S_m$ with ratio $0 < r < 1$. There exists a unique Borel measure μ supported on F such that $\mu(F) = 1$ and

$$\mu(E) = \frac{1}{m} \sum_{j=1}^{m} \mu(S_j^{-1}(E)) \qquad \text{for any Borel set } E.$$

In the case when F is the Cantor set, the Cantor-Lebesgue function is $\mu([0, x])$.

5. Prove a theorem of Hausdorff: Any compact subset K of $\mathbb{R}^d$ is a continuous image of the Cantor set $\mathcal{C}$.

[Hint: Cover K by 2^{n_1} (some n_1) open balls of radius 1, say $B_1, \ldots, B_\ell$ (with possible repetitions). Let $K_{j_1} = K \cap \overline{B_{j_1}}$ and cover each K_{j_1} with 2^{n_2} balls of radius $1/2$ to obtain compact sets K_{j_1,j_2}, and so on. Express $t \in \mathcal{C}$ as a ternary expansion, and assign to t a unique point in K defined by the intersection $K_{j_1} \cap K_{j_1,j_2} \cap \cdots$ for appropriate $j_1, j_2, \ldots$. To prove continuity, observe that if two points in the Cantor set are close, then their ternary expansions agree to high order.]

6. A compact subset K of $\mathbb{R}^d$ is **uniformly locally connected** if given $\epsilon > 0$ there exists $\delta > 0$ so that whenever $x, y \in K$ and $|x - y| < \delta$, there is a continuous curve γ in K joining x to y, such that $\gamma \subset B_\epsilon(x)$ and $\gamma \subset B_\epsilon(y)$.

Using the previous problem, one can show that a compact subset K of $\mathbb{R}^d$ is the continuous image of the unit interval $[0, 1]$ if and only if K is uniformly locally connected.

7. Formulate and prove a generalization of Theorem 3.5 to the effect that once appropriate sets of measure zero are removed, there is a measure-preserving isomorphism of the unit interval in $\mathbb{R}$ and the unit cube in $\mathbb{R}^d$.

8.* There exists a *simple* continuous curve in the plane of positive two-dimensional measure.

9. Let E be a compact set in $\mathbb{R}^{d-1}$. Show that $\dim(E \times I) = \dim(E) + 1$, where I is the unit interval in $\mathbb{R}$.

10.* Let $\mathcal{C}_\xi$ be the Cantor set considered in Exercises 8 and 11. If $\xi < 1/2$, then $\mathcal{C}_\xi + \lambda\mathcal{C}_\xi$ has positive Lebesgue measure for almost every λ.

Functional Analysis: Introduction to Further Topics in Analysis

Chapter 1. L^p Spaces and Banach Spaces

1. The spaces L^∞ and L^1 play universal roles with respect to all Banach spaces in the following sense.

(a) If $\mathcal{B}$ is any separable Banach space, show that it can be realized without change of norm as a linear subspace of $L^\infty(\mathbb{Z})$. Precisely, prove that there is a linear operator i of $\mathcal{B}$ into $L^\infty(\mathbb{Z})$ so that $\|i(f)\|_{L^\infty(\mathbb{Z})} = \|f\|_{\mathcal{B}}$ for all $f \in \mathcal{B}$.

(b) Each such $\mathcal{B}$ can also be realized as a quotient space of $L^1(\mathbb{Z})$. That is, there is a linear surjection P of $L^1(\mathbb{Z})$ onto $\mathcal{B}$, so that if $\mathcal{S} = \{x \in L^1(\mathbb{Z}) : \ P(x) = 0\}$, then $\|P(x)\|_{\mathcal{B}} = \inf_{y\in\mathcal{S}} \|x + y\|_{L^1(\mathbb{Z})}$, for each $x \in L^1(\mathbb{Z})$. This gives an identification of $\mathcal{B}$ (and its norm) with the quotient space $L^1(\mathbb{Z})/\mathcal{S}$ (and its norm), as defined in Exercise 30.

Note that similar conclusions hold for $L^\infty(X)$ and $L^1(X)$ if X is a measure space that contains a countable disjoint collection of measurable sets of positive and finite measure.

[Hint: For (a), let $\{f_n\}$ be a dense set of non-zero vectors in $\mathcal{B}$, and let $\ell_n \in \mathcal{B}^*$ be such that $\|\ell_n\|_{\mathcal{B}^*} = 1$ and $\ell_n(f_n) = \|f_n\|$. If $f \in \mathcal{B}$, set $i(f) = \{\ell_n(f)\}_{-\infty}^{\infty}$. For (b), if $x = \{x_n\} \in L^1(\mathbb{Z})$, with $\sum_{-\infty}^{\infty} |x_n| = \|x\|_{L^1(\mathbb{Z})} < \infty$, define P by $P(x) = \sum_{-\infty}^{\infty} x_n f_n/\|f_n\|$.]

2. There is a "generalized limit" L defined on the vector space V of all real sequences $\{s_n\}_{n=1}^{\infty}$ that are bounded, so that:

(i) L is a linear functional on V.

(ii) $L(\{s_n\}) \geq 0$ if $s_n \geq 0$, for all n.

(iii) $L(\{s_n\}) = \lim_{n\to\infty} s_n$ if the sequence $\{s_n\}$ has a limit.

(iii) $L(\{s_n\}) = L(\{s_{n+k}\})$ for every $k \geq 1$.

(iii) $L(\{s_n\}) = L(\{s_{n'}\})$ if $s_n - s'_n \neq 0$ for only finitely many n.

[Hint: Let $p(\{s_n\}) = \limsup_{n\to\infty} \left(\frac{s_1+\cdots+s_n}{n}\right)$, and extend the linear functional L defined by $L(\{s_n\}) = \lim_{n\to\infty} s_n$, defined on the subspace consisting of sequences that have limits.]

3. Show that the closed unit ball in a Banach space $\mathcal{B}$ is compact (that is, if $f_n \in \mathcal{B}$, $\|f_n\| \leq 1$, then there is a subsequence that converges in the norm) if and only if $\mathcal{B}$ is finite dimensional.

[Hint: If $\mathcal{S}$ is a closed subspace of $\mathcal{B}$, then there exists $x \in \mathcal{B}$ with $\|x\| = 1$ and the distance between x and $\mathcal{S}$ is greater than $1/2$.]

4. Suppose X is a σ-compact measurable metric space, and $C_b(X)$ is separable, where $C_b(X)$ denotes the Banach space of bounded continuous functions on X with the sup-norm.

(a) If $\{\mu_n\}_{n=1}^{\infty}$ is a bounded sequence in $M(X)$, then there exists a $\mu \in M(X)$ and a subsequence $\{\mu_{n_j}\}_{j=1}^{\infty}$, so that μ_{n_j} converges to μ in the following (weak*) sense:

$$\int_X g(x)\, d\mu_{n_j}(x) \to \int_X g(x)\, d\mu(x), \qquad \text{for all } g \in C_b(X).$$

(b) Start with a $\mu_0 \in M(X)$ that is positive, and for each $f \in L^1(\mu_0)$ consider the mapping $f \mapsto f d\mu_0$. This mapping is an isometry of $L^1(\mu_0)$ to the subspace of $M(X)$ consisting of signed measures which are absolutely continuous with respect to μ_0.

(c) Hence if $\{f_n\}$ is a bounded sequence of functions in $L^1(\mu_0)$, then there exist a $\mu \in M(X)$ and a subsequence $\{f_{n_j}\}$ such that the measures $f_{n_j} d\mu_0$ converge to μ in the above sense.

5.* Let X be a measure space. Suppose φ and ψ are both continuous, strictly increasing functions on $[0, \infty)$ which are inverses of each other, that is, $(\varphi \circ \psi)(x) = x$ for all $x \geq 0$. Let

$$\Phi(x) = \int_0^x \varphi(u)\, du \qquad \text{and} \qquad \Psi(x) = \int_0^x \psi(u)\, du.$$

Consider the Orlicz spaces $L^\Phi(X)$ and $L^\Psi(X)$ introduced in Exercise 23.

(a) In connection with Exercise 22 the following Hölder-like inequality holds:

$$\int |fg| \leq C\|f\|_{L^\Phi}\|g\|_{L^\Psi} \qquad \text{for some } C > 0, \text{ and all } f \in L^\Phi \text{ and } g \in L^\Psi.$$

(b) Suppose there exists $c > 0$ so that $\Phi(2t) \leq c\Phi(t)$ for all $t \geq 0$. Then the dual of L^Φ is equivalent to L^Ψ.

6.* There are generalizations of the parallelogram law for L^2 (see Exercise 25) that hold for L^p. These are the Clarkson inequalities:

(a) For $2 \leq p \leq \infty$ the statement is that

$$\left\| \frac{f+g}{2} \right\|_{L^p}^p + \left\| \frac{f-g}{2} \right\|_{L^p}^p \leq \frac{1}{2} \left(\|f\|_{L^p}^p + \|g\|_{L^p}^p \right).$$

(b) For $1 < p \leq 2$ the statement is that

$$\left\| \frac{f+g}{2} \right\|_{L^p}^q + \left\| \frac{f-g}{2} \right\|_{L^p}^q \leq \frac{1}{2} \left(\|f\|_{L^p}^p + \|g\|_{L^p}^p \right)^{q/p},$$

where $1/p + 1/q = 1$.

(c) As a result, L^p is **uniformly convex** when $1 < p < \infty$. This means that there is a function $\delta = \delta(\epsilon) = \delta_p(\epsilon)$, with $0 < \delta < 1$, (and $\delta(\epsilon) \to 0$ as $\epsilon \to 0$), so that whenever $\|f\|_{L^p} = \|g\|_{L^p} = 1$, then $\|f - g\|_{L^p} \geq \epsilon$ implies that $\left\| \frac{f+g}{2} \right\| \leq 1 - \delta$.

This is stronger than the conclusion of strict convexity in Exercise 27.

(d) Using the result in (c), prove the following: suppose $1 < p < \infty$, and the sequence $\{f_n\}$, $f_n \in L^p$, converges weakly to f. If $\|f_n\|_{L^p} \to \|f\|_{L^p}$, then f_n converges to f strongly, that is, $\|f_n - f\|_{L^p} \to 0$ as $n \to \infty$.

7.* An important notion is that of the equivalence of Banach spaces. Suppose $\mathcal{B}_1$ and $\mathcal{B}_2$ are a pair of Banach spaces. We say that $\mathcal{B}_1$ and $\mathcal{B}_2$ are **equivalent** (also said to be "isomorphic") if there is a linear bijection T between $\mathcal{B}_1$ and $\mathcal{B}_2$ that is bounded and whose inverse is also bounded. Note that any pair of finite-dimensional Banach spaces are equivalent if and only if their dimensions are the same.

Suppose now we consider $L^p(X)$ for a general class of X (which contains for instance, $X = \mathbb{R}^d$ with Lebesgue measure). Then:

(a) L^p and L^q are equivalent if and only if $p = q$.

(b) However, for any p with $1 \leq p \leq \infty$, L^2 is equivalent with a closed infinite-dimensional subspace of L^p.

8.* There is no finitely-additive rotationally-invariant measure extending Lebesgue measure to all subsets of the sphere S^d when $d \geq 2$, in distinction to what happens on the torus $\mathbb{R}^d/\mathbb{Z}^d$ when $d \geq 2$. (See Exercise 36). This is due to a remarkable construction of Hausdorff that uses the fact that the corresponding rotation group of S^d is non-commutative. In fact, one can decompose S^2 into four disjoint sets A, B, C and Z so that (i) Z is denumerable, (ii) $A \sim B \sim C$, but $A \sim (B \cup C)$.

Here the notation $A_1 \sim A_2$ means that A_1 can be transformed into A_2 via a rotation.

9.* As a consequence of the previous problem one can show that it is not possible to extend Lebesgue measure on $\mathbb{R}^d$, $d \geq 3$, as a finitely-additive measure on all subsets of $\mathbb{R}^d$ so that it is both translation and rotation invariant (that is, invariant under Euclidean motions). This is graphically shown by the "Banach-Tarski paradox": There is a finite decomposition of the unit ball $B_1 = \bigcup_{j=1}^N E_j$, with the sets E_j disjoint, and there are corresponding sets $\tilde{E}_j$ that are each obtained from E_j by a Euclidean motion, with the $\tilde{E}_j$ also disjoint, so that $\bigcup_{j=1}^N \tilde{E}_j = B_2$ the ball of radius 2.

Chapter 2. L^p Spaces in Harmonic Analysis

1. Another way $\mathbf{H}^1_r$ is an improvement over L^1 is in its weak compactness of the unit ball. The following can be proved. Suppose $\{f_n\}$ is a sequence in $\mathbf{H}^1_r$ with $\|f_n\|_{\mathbf{H}^1_r} \leq A$. Then we can select a subsequence $\{f_{n_k}\}$ and find an $f \in \mathbf{H}^1_r$ so that $\int f_{n_k}(x)\varphi(x)\,dx \to \int f(x)\varphi(x)\,dx$, as $k \to \infty$, for every φ that is a continuous function of compact support.

This is to be compared with L^1, and the failure there of weak compactness as described in Exercises 12 and 13 in the previous chapter.

[Hint: Apply the result in Problem 4 (c) of the previous chapter to obtain a subsequence $\{f_{n_k}\}$ and a finite measure μ so that $f_{n_k} \to \mu$ in the weak* sense. Next use the fact that if $\sup_{\epsilon>0} |\mu * \varphi_\epsilon| \in L^1$, for an appropriate φ, then μ is absolutely continuous.]

2. Suppose H^p is the complex Hardy space defined in Section 5. For $1 < p < \infty$, prove the following:

(a) If $F \in H^p$, then $\lim_{y\to 0} F(x+iy) = F_0(x)$ exists in the $L^p(\mathbb{R})$ norm.

(b) $\|F\|_{H^p} = \|F_0\|_{L^p}$.

(c) One has $2F_0 = f + iH(f)$, with f real-valued in $L^p(\mathbb{R})$, and $\|F_0\|_{L^p} \approx \|f\|_{L^p}$. Moreover, every F_0 (and thus F) arises this way. This gives a linear isomorphism (over the reals) of H^p, with L^p with an equivalence of norms.

[Hint: Here is an outline of the proof. For each $y_1 > 0$, write $F_{y_1}(z) = F(z + iy_1)$ and $F^\epsilon_{y_1}(z) = F_{y_1}(z)/(1 - i\epsilon z)$, $\epsilon > 0$. One has that F_{y_1} is bounded $\overline{\mathbb{R}^2_+}$ (see Section 2, in Chapter 5, Book III). Thus by Exercise 9, $F^\epsilon_{y_1}(z) = (F^\epsilon_{y_1} * \mathcal{P}_y)(x)$. Now using the weak compactness of the unit ball in L^p, (Exercise 12 in Chapter 1), we can find $F_0 \in L^p$ so that $F^\epsilon_{y_1}(x) \to F_0(x)$ weakly as ϵ and $y_1 \to 0$. Observe that this breaks down for $p = 1$. Conclusion (c) is then essentially a restatement of the boundedness of the Hilbert transform for $1 < p < \infty$.]

3. Let P be any non-zero polynomial of degree k in $\mathbb{R}^d$. Then $f = \log|P(x)|$ is in BMO and $\|f\|_{\mathrm{BMO}} \leq c_k$, where c_k depends only on the degree k of the polynomial.

[Hint: Verify the result first when $d = 1$. Then use induction in the dimension and the following assertion, stated for $\mathbb{R}^2$. Suppose $f(x, y)$, $(x, y) \in \mathbb{R}^2$ is for each y a $\mathrm{BMO}(\mathbb{R})$ function in x, uniformly in y. Assume also that this holds when the roles of x and y are interchanged. Then $f \in \mathrm{BMO}(\mathbb{R}^2)$.]

4. Prove the following John-Nirenberg inequalities for every $f \in \mathrm{BMO}(\mathbb{R}^d)$:

(a) For every $q < \infty$ there is a bound b_q so that

$$\sup_B \frac{1}{m(B)} \int_B |f - f_B|^q \, dx \leq b_q^q \|f\|_{\mathrm{BMO}}^q.$$

(b) There are positive constants μ and A, so that

$$\sup_B \frac{1}{m(B)} \int_B e^{\mu|f-f_B|} \, dx \leq A, \qquad \text{whenever } \|f\|_{\mathrm{BMO}} \leq 1.$$

[Hint: For (a) test f against p-atoms, where p is dual to q. For (b) use the bound $c_p = O(1/(p-1))$ as $p \to 1$ (in (38)) to obtain $b_q = O(q)$, as $q \to \infty$. Then write $e^u = \sum_{q=0}^{\infty} u^q/q!$.]

5. The Hilbert transform of a bounded function is in BMO. Show this in two different ways.

(a) Directly: Suppose f is bounded (and belongs to some L^p, $1 \leq p < \infty$). Then $H(f) \in \text{BMO}$ with

$$\|H(f)\|_{\text{BMO}} \leq A\|f\|_{L^\infty},$$

with A not depending on the L^p norm of f.

(b) By duality, using Theorem 5.4.

[Hint: For (a), fix any ball B, and let B_1 be its double. Consider separately, $f\chi_{B_1}$ and $f\chi_{B_1^c}$.]

6.* The following is the maximal characterization of $\mathbf{H}^1_r(\mathbb{R}^d)$. Suppose Φ belongs to the Schwartz space $\mathcal{S}$ and $\int \Phi(x)\,dx \neq 0$. Let $M(f)(x) = \sup_{\epsilon>0} |(f * \Phi_\epsilon)(x)|$, for $f \in L^1$. Then

(a) f is in $\mathbf{H}^1_r$ if and only if $M(f)$ belongs to L^1.

(b) The condition $\Phi \in \mathcal{S}$ can be relaxed to require only

$$|\partial_x^\alpha \Phi(x)| \leq c_\alpha (1+|x|)^{-d-1-|\alpha|}.$$

(c) Note two interesting examples, first $\Phi_{t^{1/2}}(x) = (4\pi t)^{-d/2} e^{-|x|^2/(4t)}$: then $u(x,t) = (f * \Phi_{t^{1/2}})(x)$ is the solution of the Heat equation $\triangle_x u = \partial_t u$, with initial data $u(x,0) = f(x)$. Also, $\Phi_t(x) = \frac{c_d t}{(t^2+|x|^2)^{\frac{d+1}{2}}}$ with $c_d = \Gamma(\frac{d+1}{2})/\pi^{\frac{d+1}{2}}$ so that $u(x,t) = (f * \Phi_t)(x)$ is the solution to Laplace's equation $\triangle_x u + \partial_t^2 u = 0$, with initial data $u(x,0) = f(x)$. (Here Γ denotes the gamma function.)

7.* H^p, when $p = 1$. The results in (a) and (b) of Problem 2 also hold for $p = 1$, but require a different proof. The analog of (c) is as follows. One has $2F_0 = f + iH(f)$, where f belongs to the real Hardy space $\mathbf{H}^1_r$. Also $\|F_0\|_{L^1} \approx \|f\|_{\mathbf{H}^1_r}$. As a consequence, a necessary and sufficient condition that $f \in \mathbf{H}^1_r$ is that both f and $H(f)$ are in L^1.

The conclusions (a) and (b) may be proved by showing that any $F \in H^1$ can be written as $F = F_1 \cdot F_2$ with $F_j \in H^2$ and $\|F_j\|^2_{H^2} = \|F\|_{H^1}$, and then using the corresponding results in H^2.

8.* Suppose $f \in L^1(\mathbb{R})$. Then we can define $H(f) \in L^1(\mathbb{R})$ in the weak sense to mean that there exists $g \in L^1(\mathbb{R})$ so that

$$\int_{\mathbb{R}} g\varphi \, dx = \int_{\mathbb{R}} f H(\varphi) \, dx, \qquad \text{for all functions } \varphi \text{ in the Schwartz space.}$$

Then we say $g = H(f)$ in the weak sense.

As a consequence of Problem 7*, one has that $f \in \mathbf{H}^1_r(\mathbb{R})$ if and only if $f \in L^1(\mathbb{R})$ and $H(f)$, taken in the weak sense, also belongs to $L^1(\mathbb{R})$.

9.* Let $\{f_n\}$ be a sequence of elements in $\mathbf{H}^1_r$ so that $\|f_n\|_{\mathbf{H}^1_r} \leq M < \infty$ for all n. Assume that f_n converges to f almost everywhere. Then:

(a) $f \in \mathbf{H}^1_r$.

(b) $\int f_n g \to \int fg$, as $n \to \infty$, for all g continuous with compact support.

A corresponding result holds for L^p, $p > 1$, but fails for $p = 1$. See Exercise 14 in Chapter 1.

10.* The following result illustrates the application of $\mathbf{H}^1_r$ to the theory of compensated compactness.

Suppose $A = (A_1, \ldots, A_d)$ and $B = (B_1, \ldots, B_d)$ are vector fields in $\mathbb{R}^d$ with $A_i, B_i \in L^2(\mathbb{R}^d)$ for all i. The divergence of A is defined by

$$\operatorname{div}(A) = \sum_{k=1}^{d} \frac{\partial A_k}{\partial_{x_k}},$$

and the curl of B is the $d \times d$ matrix whose ij-entry is

$$(\operatorname{curl}(B))_{ij} = \frac{\partial B_i}{\partial x_j} - \frac{\partial B_j}{\partial x_i}.$$

(The derivatives here are taken in the sense of distributions, as in the next chapter.)
If $\operatorname{div}(A) = 0$ and $\operatorname{curl}(B) = 0$ then $\sum_{k=1}^{d} A_k B_k \in \mathbf{H}^1_r$.
This is in contrast with the result that in general, if $f, g \in L^2$, then one only has $fg \in L^1$.

Chapter 3. Distributions: Generalized Functions

1. We consider **periodic distributions** and their Fourier series.

(a) The notion of a periodic distribution on $\mathbb{R}^d$ can be defined in two equivalent ways:

First, one can consider distributions F on $\mathbb{R}^d$ which are periodic in the sense that $\tau_h(F) = F$ for all $h \in \mathbb{Z}^d$;

Alternatively, one can consider the continuous linear functionals on $\mathcal{D}(\mathbb{T}^d)$, the space of C^∞ periodic functions on $\mathbb{R}^d$. (Here $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ denotes the d-dimensional torus.)

(b) Note that if $\varphi \in \mathcal{D}(\mathbb{T}^d)$, then φ has a Fourier series expansion

$$\varphi(x) = \sum_n a_n e^{2\pi i n\cdot x},$$

where the Fourier coefficients $a_n = \int_{\mathbb{T}^d} f(x)e^{-2\pi i n\cdot x}\,dx$ are rapidly decreasing, that is, for every $N > 0$, $|a_n| \le O(|n|^{-N})$ as $|n| \to \infty$.

Similarly, if F is a periodic distribution, and $a_n = F(e^{-2\pi i n x})$ denote its Fourier coefficients, then a_n are slowly increasing in the sense that for some $N > 0$, $|a_n| \le O(|n|^N)$ as $|n| \to \infty$.

Moreover, the Fourier series $\sum a_n e^{2\pi i n x}$ converges to F in the sense of distributions.

[Hint: To prove the equivalence in (a), consider the "periodization" operator $P : \mathcal{D}(\mathbb{R}^d) \to \mathcal{D}(\mathbb{T}^d)$,

$$P(\varphi)(x) = \sum_{h\in\mathbb{Z}^d} \tau_h(\varphi)(x) = \sum_{h\in\mathbb{Z}^d} \varphi(x-h).$$

Then find $\gamma \in \mathcal{D}(\mathbb{R}^d)$ so that $P(\gamma) = 1$. This allows to prove that P is surjective, and that, in the same way, its dual $P^* : \mathcal{D}_2^* \to \mathcal{D}_1^*$ is also surjective. (Here $\mathcal{D}_1^*$ and $\mathcal{D}_2^*$ denote, respectively, the two spaces of distributions described in (a).) To construct γ, pick $\psi \in \mathcal{D}(\mathbb{R}^d)$ so that $\psi \ge 0$ and $\psi = 1$ on $\{0 \le x_j < 1,\ \ 1 \le j \le d\}$, and let $\gamma = \psi/P(\psi)$.]

2. Suppose $Tf = f * K$ is a singular integral operator as in Theorem 3.2 of Section 3. Then the mapping $f \mapsto T(f)$ is bounded on the Hardy space $\mathbf{H}^1_r$, and in particular maps $\mathbf{H}^1_r$ to L^1.

[Hint: Consider first a 2-atom $\mathfrak{a}$ associated to the unit ball B. Then for an appropriate constant c (bounded independently of $\mathfrak{a}$) we have $T(\mathfrak{a}) = c(a_* + \Phi)$. Here a_* is a 2-atom for the ball $B^a st$, the double of B, and Φ satisfies $|\Phi(x)| \leq (1 + |x|)^{-d-1}$, $\int_{\mathbb{R}^d} \Phi(x)\, dx = 0$. With this apply Exercise 21 in Chapter 2. Then obtain the analog for 2-atoms $\mathfrak{a}$, after rescaling and translation.]

3. Prove the following interior estimates for an elliptic operator L of order m with constant coefficients.

Suppose $\mathcal{O}$ and $\mathcal{O}_1$ are bounded subsets of $\mathbb{R}^d$ with $\overline{\mathcal{O}} \subset \mathcal{O}_1$. Assume u and f are L^p functions in $\mathcal{O}_1$ with $Lu = f$ in $\mathcal{O}_1$ in the sense of distributions. Then if $1 < p < \infty$ and k is a non-negative integer, we have

$$\sum_{|\alpha| \leq m+k} \|\partial_x^\alpha u\|_{L^p(\mathcal{O})} \leq c \left(\sum_{|\beta| \leq k} \|\partial_x^\beta f\|_{L^p(\mathcal{O}_1)} + \|u\|_{L^p(\mathcal{O}_1)} \right)$$

where the derivatives are taken in the sense of distributions.

[Hint: Consider the parametrix $Q_\epsilon = \eta_\epsilon Q$ given in Corollary 2.13 which is supported in $|x| \leq \epsilon$. Here ϵ is chosen so that $\overline{\mathcal{O}}_\epsilon \subset \mathcal{O}_1$, where $\mathcal{O}_\epsilon$ are the points of distance $\leq \epsilon$ from $\mathcal{O}$.

Set $U = \psi u$, with ψ a C^∞ function that is 1 near $\overline{\mathcal{O}}_\epsilon$ but vanishes outside $\mathcal{O}_1$. Then

$$L(U) = \psi L(u) + \sum_{|\gamma| < m} \psi_\gamma \partial_x^\gamma u,$$

and what is important is that the ψ_γ vanishes in $\overline{\mathcal{O}}_\epsilon$. Now $U + r_\epsilon * U = Q_\epsilon * L(U)$, where $r_\epsilon \in \mathcal{S}$. This gives

$$\psi u = Q_\epsilon * (\psi f) - r_\epsilon(\psi u) + \sum_r Q_\epsilon * (\psi_\gamma \partial_x^\gamma u).$$

As has been pointed out, $\partial_x^\alpha Q$ are Calderón-Zygmund distributions whenever $|\gamma| \leq m$, so the same is true for Q_ϵ. Then using Theorem 3.2, the result follows.]

4.* Let $P(x)$ be any real polynomial in $\mathbb{R}^d$, and k a homogeneous function of degree $-d$ with $\int_{|x|=1} k(x)\, d\sigma(x) = 0$.

(a) One can define the tempered distribution $\mathrm{pv}\left(e^{iP(x)}k(x)\right) = K$ by

$$K(\varphi) = \lim_{\epsilon \to 0} \int_{|x|\geq \epsilon} e^{iP(x)} k(x)\varphi(x)\, dx.$$

(b) Then the Fourier transform of K is a bounded function (with bound independent of the coefficients of P).

5.* Let Q be a fixed real-valued polynomial on $\mathbb{R}^d$. Consider the distributions initially defined for $\mathrm{Re}(s) > 0$ by

$$I(s)(\varphi) = \int_{Q(x)>0} |Q(x)|^s \varphi(x)\, dx, \qquad \text{where } \varphi \in \mathcal{S}.$$

Then $I(s)(\varphi)$ has a meromorphic continuation to the whole complex s-plane, with poles at most at $s = -k/m$, where m is a positive integer determined by Q, and k is any positive integer. The order of the poles do not exceed d.

6.* As a consequence of the results in Problem 5*, one may prove the following.

(a) Suppose $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ is a non-zero partial differential operator on $\mathbb{R}^d$ with a_α complex constants. Then L has a *tempered* fundamental solution. As an immediate corollary we also have:

(b) Suppose P is a complex-valued polynomial on $\mathbb{R}^d$. Then there exists a tempered distribution F that agrees with $1/P$ where $P(x) \neq 0$.

In fact, let P be the characteristic polynomial of L and apply the result of the previous problem to $Q = |P|^2$. Suppose $I(s)$ has a pole of order r at $s = 1$, then define the tempered distribution F by

$$F = \overline{P} \, \frac{1}{r!} \frac{d^r}{ds^r} (s+1)^r I(s) \Big|_{s=-1}.$$

Consequently, $PF = 1$, and the inverse Fourier transform of F is the desired fundamental solution of L.

7.* Suppose $L = \sum_{|\alpha| \le m} a_\alpha \partial_x^\alpha$ is a partial differential operator on $\mathbb{R}^d$, with a_α complex constants. Then L is hypo-elliptic if and only if for each $\alpha \neq 0$

$$\frac{\partial_\xi^\alpha P(\xi)}{P(\xi)} \to 0 \qquad \text{as } |\xi| \to \infty,$$

where P denotes the characteristic polynomial of L.

8.* We describe several fundamental solutions of the wave operator

$$\Box = \frac{\partial^2}{\partial t^2} - \triangle_x,$$

where $(x,t) \in \mathbb{R}^d \times \mathbb{R}$ and $\triangle_x = \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2}$.

We let Γ_+ be the forward open cone $= \{(x,t) : \ t > |x|\}$, and $\Gamma_- = -\Gamma_+$, the backward cone. For each s with $\mathrm{Re}(s) > -1$ we define the function F_s by

$$F_s(x,t) = \begin{cases} a_s(t^2 - |x|^2)^{s/2}, & \text{if } (x,t) \in \Gamma_+ \\ 0 & \text{otherwise.} \end{cases} \tag{29}$$

Here $a_s^{-1} = 2^{s+d}\pi^{\frac{d-1}{2}}\Gamma\left(\frac{s+d+1}{2}\right)\Gamma\left(s/2+1\right)$. Then $s \mapsto F_s$ has an analytic continuation in the complex s plane as an entire (tempered) distribution-valued function. Moreover, one can prove that $F_+ = F_s|_{s=-d+1}$ is a fundamental solution of $\Box$.

Note that F_-, obtained from F_+ by mapping $t \mapsto -t$, is also a fundamental solution, and F_+ and F_- are supported in $\overline{\Gamma}_+$ and $\overline{\Gamma}_-$ respectively. In addition, if d is odd and $d \geq 3$, then a_s vanishes for $s = -d+1$, so both F_+ and F_- are supported on the boundary of their cones, which is a reflection of the Huygens' principle.

Finally, a third fundamental solution F_0 of interest is given by

$$F_0^\wedge = \lim_{\epsilon \to 0,\, \epsilon > 0} \frac{1}{4\pi^2}\left(\frac{1}{|\xi|^2 - \tau^2 + i\epsilon}\right),$$

with the limit taken in the sense of distribution, and (ξ, τ) representing the dual variables to (x,t). The fundamental solutions F_+, F_-, and F_0 are each homogeneous of degree -2, and invariant under the Lorentz group of linear transformations of determinant 1 that preserves Γ_+. Also each fundamental solution of $\Box$ with these invariance properties can be written as $c_1F_+ + c_2F_- + c_3F_0$, with $c_1 + c_2 + c_3 = 1$.

Chapter 4. Applications of the Baire Category Theorem

1. Let $T : \mathcal{B}_1 \to \mathcal{B}_2$ be a bounded linear transformation of a Banach space $\mathcal{B}_1$ to a Banach space $\mathcal{B}_2$.

(a) Prove that either T is surjective, or the image $T(\mathcal{B}_1)$ is of the first category in $\mathcal{B}_2$.

(b) As a consequence, prove the following: Suppose (X, μ) is a finite measure space, and $1 \leq p_1 < p_2 \leq \infty$. One has of course $L^{p_2}(X) \subset L^{p_1}(X)$. Show that $L^{p_2}(X)$ is a set of the first category in $L^{p_1}(X)$ (except in the trivial case for which each element of L^{p_1} belongs to L^{p_2}).

[Hint: For (a), assume that $T(\mathcal{B}_1)$ is of the second category and use an argument similar to the proof of Theorem 3.1 to show that the image under T of a ball centered at the origin of $\mathcal{B}_1$ contains a ball centered at the origin in $\mathcal{B}_2$.]

2. For each integer $n \geq 2$, let Λ_n denote the set of real numbers x so that there exists infinitely many distinct fractions p/q so that

$$|x - p/q| \leq 1/q^n.$$

Show that:

(a) Λ_n is a generic set in $\mathbb{R}$.

(b) However, the Hausdorff dimension of Λ_n equals $2/n$.

(c) Hence $m(\Lambda_n) = 0$, if $n > 2$, where m denotes the Lebesgue measure.

The elements of $\Lambda = \bigcap_{n \geq 2} \Lambda_n$ are called the **Liouville numbers**. While it is not difficult to see that every element of Λ is transcendental, it is a deeper fact that the same holds for each element of Λ_n when $n > 2$. (Note that in the case $n = 2$, the set Λ consists of the irrationals.)

3. Consider the Banach space $\mathcal{B}$ of continuous functions on the circle (with the sup-norm). Prove that the set of f in $\mathcal{B}$ whose Fourier series diverges in a generic set on the circle, is itself a generic set in $\mathcal{B}$.

[Hint: Choose $\{x_i\}$ dense in $[0,1]$, let $E_i = \{f \in \mathcal{B} : \sup_N |S_N(f)(x_i)| = \infty\}$, and $E = \cap E_i$. Then E is generic. For each $f \in E$, define $\mathcal{O}_n = \{x : |S_N(f)(x)| > n \text{ some } N\}$. Show that $\cap \mathcal{O}_n$ is generic.]

4. Let $\mathbb{D}$ denote the open unit disc in the complex plane, and let $\mathcal{A}$ be the Banach space of all continuous complex-valued functions on $\overline{\mathbb{D}}$ that are holomorphic on $\mathbb{D}$,

equipped with the sup-norm. Then, the space of functions in $\mathcal{A}$ which cannot be extended analytically past any point of the boundary of $\mathbb{D}$ is generic. To prove this statement establish the following:

(a) The set $\mathcal{A}_N = \{f \in \mathcal{A} : \ |f(e^{i\theta}) - f(1)| \leq N|\theta|\}$ is closed.

(b) $\mathcal{A}_N$ is nowhere dense.

[Hint: For (b) use the function $f_0(z) = (1-z)^{1/2}$ and consider $f + \epsilon f_0$.]

5. Let $I = [0, 1]$ denote the unit interval, and $C^\infty(I)$ the vector space of all smooth functions on I equipped with the metric d given by

$$d(f, g) = \sum_{n=0}^{\infty} \frac{1}{2^n} \frac{\rho_n(f - g)}{1 + \rho_n(f - g)},$$

where $\rho_n(h) = \sup_{x \in I} |h^{(n)}(x)|$. A function $f \in C^\infty(I)$ is analytic at a point $x_0 \in I$, if its Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

converges in a neighborhood of x_0 to the function f. The function f is said to be singular at x_0 if its Taylor series diverges at x_0.

(a) Show that $(C^\infty(I), d)$ is a complete metric space.

(b) Prove that the set of functions in $C^\infty(I)$ that are singular at every point is generic.

[Hint: For (b), consider the set F_K of smooth functions f that satisfy $|f^{(n)}(x^*)|/n! \leq K^n$ for some x^* and all n, and show that F_K is closed and nowhere dense.]

6. The space L^∞ in Theorem 4.2 cannot be replaced by any L^q, with $1 \leq q < \infty$. In fact there exists a closed infinite dimensional subspace of $L^1([0,1])$ consisting of functions that belong to L^q for all $1 \leq q < \infty$.

[Hint: One may use Exercise 19 in the next chapter.]

7.* As an application of Exercise 14, let $\mathcal{H}$ denote the vector space of entire functions, that is, the set of functions that are holomorphic in all of $\mathbb{C}$. Given a compact subset K of the complex plane and $f \in \mathcal{H}$, let $\|f\|_K = \sup_{z\in K} |f(z)|$. If K_n denotes the closed disc centered at the origin and of radius n, define

$$d(f,g) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{\|f-g\|_{K_n}}{1+\|f-g\|_{K_n}} \qquad \text{whenever } f, g \in \mathcal{H}.$$

Then d is a metric, and $\mathcal{H}$ is a complete metric space with respect to d. Also, $d(f_n, f) \to 0$ if and only if f_n converges to f uniformly on every compact subset of $\mathbb{C}$.

Birkhoff's theorem (Problem 5, Chapter 2, Book II) states that there exists an entire function F so that the set $\{F(z+n)\}_{n=1}^{\infty}$ is dense in $\mathcal{H}$. Also, MacLane's theorem (see the end of the same problem in Book II) says that there is an entire function G so that the set of its derivatives $\{G^{(n)}(z)\}_{n=1}^{\infty}$ is dense in $\mathcal{H}$.

By Exercise 14, the set of functions in $\mathcal{H}$ with either of these properties is generic in $\mathcal{H}$, hence the set of entire functions with *both* properties is also generic.

Chapter 5. Rudiments of Probability Theory

1. In the context of Bernoulli trials with probabilities $0 < p, q < 1$, where $p + q = 1$, let $D : \mathbb{Z}_2^\infty \to [0, 1]$, be given by

$$D(x) = \sum_{n=1}^{\infty} x_n/2^n \qquad \text{if } x = (x_1, \ldots, x_n, \ldots).$$

Under this mapping the measure m_p goes to the measure μ_p that can be written symbolically as a "Riesz product," $\mu_p = \prod_{n=1}^{\infty}(1 + (p - q)r_n(t))\, dt$. The meaning of this is as follows. For each N, let

$$F_N(t) = \int_0^t \prod_{n=1}^{N} (1 + (p - q)r_n(s))\, ds.$$

Then one can show that:

(a) Each F_N is increasing on $[0, 1]$.

(b) $F_N(0) = 0$, $F_N(1) = 1$.

(c) F_N converges uniformly to a function F, as $N \to \infty$.

(d) $\mu_p = dF$, in the sense that $\mu_p((a, b)) = F(b) - F(a)$.

(e) If $p \neq 1/2$, then μ_p is completely singular (that is $dF/dt = 0$ almost everywhere.)

[Hint: Show that if $I = (a, b)$ is a dyadic interval of length 2^{-n}, $a = \ell/2^n$ and $b = (\ell + 1)/2^n$, and $N \geq n$, then

$$F_N(b) - F_N(a) = p^{n_0} q^{n_1},$$

where n_0 is the number of zeroes in the first n terms of the dyadic expansion of $\ell/2^n$, and n_1 is the number of 1's, with $n_0 + n_1 = n$.]

2.* There is an analogy between the Walsh-Paley expansion (see Exercise 16) and the Fourier expansion, that is, between $\{w_n\}_{n=0}^{\infty}$ and $\{e^{in\theta}\}_{n=-\infty}^{\infty}$. In this analogy the Rademacher functions $r_k = w_{2^{k-1}}$ correspond to the lacunary frequencies $\{e^{i2^k\theta}\}_{k=0}^{\infty}$. In fact, the following is known:

(a) If $\sum_{k=0}^{\infty} c_k e^{i2^k\theta}$ is an $L^2([0,2\pi])$ function, then it belongs to L^p, for every $p < \infty$.

(b) If $\sum_{k=0}^{\infty} c_k e^{i2^k\theta}$ is the Fourier series of an integrable function, it belongs to L^2, and hence to L^p for every $p < \infty$.

(c) This function belongs to L^∞ if and only if $\sum_{k=0}^{\infty} |c_k| < \infty$.

(d) From (c) it follows that the conclusion (a) of Theorem 1.7 does not necessarily extend to $p = \infty$.

3. The following is a general form of the central limit theorem. Suppose $f_1, \ldots, f_n, \ldots$ are square integrable mutually independent functions on X, and assume for simplicity that each has mean equal to 0. Let μ_n be the distribution measure of f_n, and σ_n^2 the variance. Set $S_n^2 = \sum_{k=1}^n \sigma_n^2$. The critical assumption is that for every $\epsilon > 0$

$$\lim_{n\to\infty} \frac{1}{S_n^2} \sum_{k=1}^{n} \int_{|t|\geq \epsilon S_n} t^2 \, d\mu_n(t) = 0.$$

Under these conditions the distribution measures of $\frac{1}{S_n} \sum_{k=1}^n f_k$ converge weakly to the normal distribution ν with variance 1.

4.* Suppose $\{f_n\}$ are identically distributed, square integrable, mutually independent, have each mean 0 and variance 1. Let $s_n = \sum_{k=1}^{n} f_k$. Then for a.e. x

$$\limsup_{n\to\infty} \frac{s_n(x)}{(2n \log\log n)^{1/2}} = 1.$$

This is the law of the "iterated logarithm."

5. An interesting variant of the random walk in $\mathbb{R}^d$ (often referred to as a "random flight") arises if the motion of unit distance at the nth step is allowed to be in *any* direction (of the unit sphere). More precisely

$$s_n = f_1 + \cdots + f_n,$$

where the f_n are mutually independent, and each f_n is uniformly distributed on the unit sphere $S^{d-1} \subset \mathbb{R}^d$. The underlying probability space is defined as the infinite product $X = \prod_{j=1}^{\infty} S_j$, where each $S_j = S^{d-1}$ with the usual surface measure normalized to have integral 1.

(a) If μ is the distribution measure of each f_n, connect $\hat{\mu}(\xi)$ with Bessel functions.

(b) What is the covariance matrix?

(c) What is the limiting distribution of $s_n(x)/n^{1/2}$?

[Hint: Show that $\hat{\mu}(\xi) = \Gamma(d/2)(\pi|\xi|)^{(2-d)/2} J_{(d-2)/2}(2\pi|\xi|)$ by using the formulas in Problem 2, Chapter 6 of Book I.]

Chapter 6. An Introduction to Brownian Motion

1. The condition of continuity of Brownian paths B-3 is in effect a consequence of properties B-1 and B-2. This is implied by the following general theorem.

Suppose that for each $t \geq 0$, we are given an L^p function $F_t = F_t(x)$ on the space (X, m). Assume that $\|F_{t_1} - F_{t_2}\|_{L^p} \leq c|t_1 - t_2|^\alpha$, with $\alpha > 1/p$, and $1 \leq p \leq \infty$. Then there is a "corrected" $\tilde{F}_t$, so that for each t, $F_t = \tilde{F}_t$ (almost everywhere with respect to m), and so that $t \mapsto \tilde{F}_t(x)$ is continuous for all $t \geq 0$, for almost every $x \in X$. Moreover the functions $t \mapsto \tilde{F}_t(x)$ satisfy a Lipschitz condition of order γ if $\gamma < \alpha - 1/p$.

2. The proof of the Donsker invariance principle follows along the same lines as the proof of Theorem 3.1. Let $f_1, \ldots, f_n, \ldots$ be a sequence of identically distributed mutually independent square integrable $\mathbb{R}^d$-valued functions on a probability space (X, m), each having mean zero and the identity as its covariance matrix. Define

$$S_t^{(N)} = \frac{1}{N^{1/2}} \sum_{1 \le k \le [Nt]} f_k + \frac{(Nt - [Nt])}{N^{1/2}} f_{[Nt]+1},$$

and let $\{\mu_N\}$ be the corresponding measures on $\mathcal{P}$ induced via the measure m on X.

(a) Instead of Lemma 3.2 use Exercise 11 to show that for $T = 1$, $\eta > 0$ and $\sigma > 0$, there exists $0 < \delta < 1$ and an integer N_0 so that for all $0 \le t \le 1$ one has

$$m(\{x : \sup_{0<h<\delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \le \delta\eta, \qquad \text{for all } N \ge N_0.$$

(b) Deduce from the above that for all $T > 0$, $\epsilon > 0$, and $\sigma > 0$ there is a $\delta > 0$ so that

$$m(\{x : \sup_{0 \le t \le T,\ 0<h<\delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \le \epsilon, \qquad \text{for all } N \ge 1.$$

(c) Use the inequality in (b) to show that the sequence $\{\mu_N\}$ is tight.

(d) Conclude as before that $\{\mu_N\}$ converges weakly to W.

3. There are a number of other constructions of Brownian motion besides the one given in this chapter. A particularly elegant approach is based on simple Hilbert space ideas.

On (Ω, P) consider a sequence $\{f_n\}$ of independent, identically distributed $\mathbb{R}^d$-valued functions with Gaussian distribution of mean zero and covariance matrix equal to the identity. Observe that the sequence $\{f_n\}$ is an orthonormal sequence of $L^2(\Omega, \mathbb{R}^d)$. Let $\mathcal{H}$ denote the closed subspace of $L^2(\Omega, \mathbb{R}^d)$ spanned by $\{f_n\}$.

Observe that $\mathcal{H}$ is a separable infinite dimensional Hilbert space. Hence there is a unitary correspondence U between $L^2([0,\infty), dx)$ and $\mathcal{H}$. Let $B_t = U(\chi_t)$ where χ_t is the characteristic function of the interval $[0,t]$. Then, each B_t can be corrected as in Problem 1, so that the process $\{B_t\}$ becomes Brownian motion. In this connection see also Exercise 9 in Chapter 5.

Note that for instance, if $B_t = \sum c_n(t) f_n$, then $B_t - B_s = \sum [c_n(t) - c_n(s)] f_n$ with $\sum |c_n(t) - c_n(s)|^2 = t - s$.

4.* In the previous chapter, we noted that recurrence results for the (discrete) random walks depend on the dimension d, and in particular, whether $d \leq 2$ or $d \geq 3$ (see Theorem 2.18 in Chapter 5 and the remark that follows it).

One can establish the following results for the (continuous) Brownian motion B_t in $\mathbb{R}^d$.

(a) If $d = 1$, Brownian motion hits, almost surely, every point infinitely often, in the sense that for each $x \in \mathbb{R}$ and for any $t_0 > 0$,

$$P(\{\omega : B_t(\omega) = x \text{ for some } t \geq t_0\}) = 1.$$

Thus B_t is pointwise recurrent in $\mathbb{R}$.

(b) If $d \geq 2$, then for every point $x \in \mathbb{R}^d$, Brownian motion almost surely never hits that point, that is,

$$P(\{\omega : B_t(\omega) = x \text{ for some } t > 0\}) = 0.$$

So, in this case, Brownian motion is not pointwise recurrent.

(c) However if $d = 2$, then B_t is recurrent in every neighborhood of every point, that is, if D is any open disc with positive radius, and $t_0 > 0$, then

$$P(\{\omega : B_t(\omega) \in D \text{ for some } t \geq t_0\}) = 1.$$

(d) Finally, when $d \geq 3$, Brownian motion is transient, that is, it escapes to infinity in the sense that

$$P(\{\omega : \lim_{t\to\infty} |B_t(\omega)| = \infty\}) = 1.$$

5.* The law of the iterated logarithm describes the amplitude of the oscillations of Brownian motion as $t \to \infty$ and $t \to 0$: if B_t is an $\mathbb{R}$-valued Brownian motion process, then for almost all ω

$$\limsup_{t\to\infty} \frac{B_t(\omega)}{\sqrt{2t \log\log t}} \to 1, \qquad \liminf_{t\to\infty} \frac{B_t(\omega)}{\sqrt{2t \log\log t}} \to -1.$$

By Exercise 13, time inversion implies that for almost all ω

$$\limsup_{t\to 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} \to 1, \qquad \liminf_{t\to 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} \to -1.$$

6.* There is a converse to Theorem 6.1 when $d \geq 2$: if $u(x) \to f(y)$ as $x \to y$ with $x \in \mathcal{R}$, for each continuous function f, then y is a regular point.

[Hint: If y is not regular, then, using Problem 4* (b), show that $P(\{|B^y_{\tau_y} - y| > 0\}) = 1$, hence $P(\{|B^y_{\tau_y} - y| \geq \delta\}) > 1/2$ for some $\delta > 0$. If S_ϵ denotes the sphere centered at y of radius $\epsilon < \delta$, use the strong Markov property to prove that there exists $x_\epsilon \in S_\epsilon \cap \mathcal{R}$ so that $P(\{|B^{x_\epsilon}_{\tau^{x_\epsilon}} - y| \geq \delta\}) > 1/2$. Then, considering any continuous function $0 \leq f \leq 1$ on $\mathcal{R}$ with $f(y) = 1$, and $f(z) = 0$ whenever $|z - y| \geq \delta$, leads to a contradiction.]

7.* A simple example of a non-regular point arises when we remove from an open ball its center, with the center then becoming a non-regular point. A more interesting example of a non-regular point is given by Lebesgue's thorn with its cusp at the origin.

Suppose $d \geq 3$, and consider the ball $B = \{x \in \mathbb{R}^d : |x| < 1\}$ from which we remove the set

$$E = \{(x_1, \ldots, x_d) \in \mathbb{R}^d : \ 0 \leq x_1 \leq 1, \ x_2^2 + \cdots + x_d^2 \leq f(x_1)\}.$$

Here f is continuous and $f(x) > 0$ if $x > 0$. If $f(x)$ decreases sufficiently rapidly as $x \to 0$, then the origin is non-regular for the set $\mathcal{R} = B - E$. Clearly, $\mathcal{R}$ can be modified so that its boundary is smooth except at the origin.

Chapter 7. A Glimpse into Several Complex Variables

1.* Suppose $f = f(z_1, \ldots, z_n)$ is defined in a region $\Omega \subset \mathbb{C}^n$, and for each j, $1 \leq j \leq n$, the function f is holomorphic in z_j with the other variables fixed. Then f is holomorphic in Ω. This was shown at the start of the chapter when f is continuous, and the point of this problem is that no condition on f is required besides the analyticity in each separate variable.

An important ingredient in the proof of this result is an application of the Baire category theorem.

2.* Assume f is holomorphic in a neighborhood of the origin and $f(0) = 0$. Let $f(z) = \sum a_\alpha z^\alpha$ be the power series expansion of f valid near the origin. The order of the zero (at $z = 0$) is the integer k that is the smallest $|\alpha|$, for which $a_\alpha \neq 0$. Then, after a linear change of variables, we can write $f(z) = c(z)P(z)$ near the origin, where $P(z) = z_n^k + a_{k-1}(z')z_n^{k-1} + \cdots + a_0(z')$ with (z', z_n), and $c(z) \neq 0$ while $a_{k-1}(0) = \cdots = a_0(0) = 0$. This result is the **Weierstrass preparation** theorem.

[Hint: Assume that our coordinate system $(z', z_n) \in \mathbb{C}^{n-1} \times \mathbb{C}$ is such that $f(0, z_n) = z_n^k$. Then by Rouché's theorem we can choose $\epsilon, r > 0$, so that $z_k \to f(z', z_k)$ has k zeroes inside the disc $|z_k| \leq r$, but is non-vanishing on the boundary, for all $|z'| < \epsilon$. Let $\gamma_1(z')$, $\gamma_2(z')$, $\ldots$, $\gamma_k(z')$ be an arbitrary ordering of these zeroes. Then the symmetric functions $\sigma_1(z') = \sum_{\ell=1}^k \gamma_\ell(z')$, $\sigma_2(z') = \sum_{m<\ell} \gamma_\ell(z')\gamma_m(z'), \ldots$, are holomorphic in z', for $|z'| < \epsilon$. This follows since the sums $s_m(z') = \sum_{\ell=1}^k (\gamma_\ell(z'))^m$, $1 \leq m \leq k$, have this property because they are given by the formula

$$s_m(z') = \frac{1}{2\pi i} \int_{|w|=r} w^m \frac{(\partial f/\partial w)(z', w)}{f(z', w)}\, dw.$$

Now we need only take $a_{k-j}(z') = (-1)^j \sigma_j(z')$, and the result holds for $P(z) = z_n^k + a_{k-1}(z')z^{k-1} + \cdots + a_0(z')$.]

3.* The original proof of Theorem 4.1 represented F in terms of F_0 by Green's theorem via the "Bochner-Martinelli integral." The result then held for F_0 merely of class C^1.

4.* We are concerned with the problem

$$\overline{\partial} u = f, \qquad \text{on } \Omega, \tag{42}$$

where Ω is a bounded region in $\mathbb{C}^n$ with C^∞ boundary and f is given in Ω with $\overline{\partial} f = 0$ there.

(a) If Ω is pseudo-convex, and $f \in C^\infty(\overline{\Omega})$ then there is $u \in C^\infty(\overline{\Omega})$ that solves (42).

(b) The "normal" solution (if it exists) is defined as the (unique) solution u in $L^2(\Omega)$ for which

$$\int_\Omega u\overline{F}\, dm(z) = 0$$

for all F that are holomorphic in Ω and are in L^2 there. For Ω that are strongly pseudo-convex (and many other classes of Ω) whenever $f \in C^\infty(\overline{\Omega})$, the normal solution u also belongs to $C^\infty(\overline{\Omega})$. This results from the study of the "$\overline{\partial}$-Neumann problem."

5.* Domains of holomorphy. A domain of holomorphy is a region Ω with the property that there exists a holomorphic function F so that for every $z^0 \in \partial\Omega$, the function F cannot be continued into some ball centered at z^0. If Ω is a domain of holomorphy and has boundary of class C^2, then Ω is pseudo-convex, by Theorem 7.5. Conversely, it can be shown that if Ω is pseudo-convex, then it is a domain of holomorphy.

6.* The converse to Theorem 8.7 holds. If f is a distribution with compact support so that $C(f)(z)$ is analytic near $z = 0$ then $\overline{L}(U) = f$ is locally solvable near the origin.

This is proved by finding a kernel K so that the convolution operator $T(f) = f * K$ on the Heisenberg group is a relative inverse to $\overline{L}$ in the sense that $\overline{L}T(f) = f - C(f)$. Then write $f = f - C(f) + C(f) = f_1 + f_2$, with $f_1 = f - C(f)$ and $f_2 = C(f)$. We can solve $L(U_1) = f_1$ by what has just been asserted, and we can solve $L(Uu_2) = f_2$ locally by the Cauchy-Kowaleski theorem, since f_2 is real-analytic at the origin.

Chapter 8. Oscillatory Integrals in Fourier Analysis

1.* Suppose M is a local hypersurface in $\mathbb{R}^d$. In a neighborhood of a point $x_0 \in M$ one can choose a smooth vector field ν, defined in this neighborhood restricted to M, so that $\nu(x)$ is a unit normal vector of M at each $x \in M$. (There are two choices of this vector field, determined up to a sign.) The map $x \mapsto \nu(x)$ from M to S^{d-1} (with S^{d-1} the unit sphere in $\mathbb{R}^d$) is called the **Gauss map**.

One can prove that the Gauss curvature of M near x_0 is non-vanishing if and only if the Gauss map is a diffeomorphism near x_0. Moreover, if $d\sigma_M$ and $d\sigma_{S^{d-1}}$ are the induced Lebesgue measures of M and S^{d-1}, and $(d\sigma_{S^{d-1}})^*$ the pull-back of $d\sigma_{S^{d-1}}$ to M defined by

$$\int_M f\,(d\sigma_{S^{d-1}})^* = \int_{S^{d-1}} f(\nu^{-1}(x))\,d\sigma_{S^{d-1}}(x),$$

then $K d\sigma_M = (d\sigma_{S^{d-1}})^*$, with K the absolute value of the Gauss curvature.

2.* The spherical maximal function. Define

$$A_t(f)(x) = \frac{1}{\sigma(S^d)} \int_{S^d} f(x - ty)\, d\sigma(y)$$

for each $t \neq 0$, and $A^*(f)(x) = \sup_{t \neq 0} |A_t(f)(x)|$. Then

$$\|A^*(f)\|_{L^p} \leq c_p \|f\|_{L^p}, \qquad \text{if } p > d/(d-1) \text{ and } d \geq 2.$$

As a result, if $f \in L^p$, $p > d/(d-1)$, then $\lim_{t \to 0} A_t(f)(x) = f(x)$ a.e. Simple examples show that this fails if $p \leq d/(d-1)$.

A hint that there may be estimates for $\sup_t |A_t(f)|$ (and in particular that the result holds for $p = 2$ and $d \geq 3$) is the following simple observation for $d \geq 3$:

$$\| \sup_{1 \leq t \leq 2} |A_t(f)| \, \|_{L^2} \leq c \|f\|_{L^2}.$$

To establish this, one notes that

$$\int_1^2 \left| \frac{\partial A_s(f)(x)}{\partial s} \right|^2 dx\, ds \leq c' \|f\|_{L^2}^2$$

by using Theorem 3.1. However $\sup_{1 \leq t \leq 2} |A_t(f)| \leq \int_1^2 \left| \frac{\partial A_s(f)}{\partial s} \right| ds + |A_1(f)(x)|$, hence the assertion follows by using Schwartz's inequality.

Refinements of this argument prove the result for $\sup_{t>0} |A_t(f)|$, $p = 2$ and $d \geq 3$, and then also for $p > d/(d-1)$. Further ideas are needed for the case $d = 2$.

3.* There is a variant of Problem 2 that applies to the wave equation.

Suppose u solves $\triangle_x u = \frac{\partial^2 u}{\partial t^2}$ for $(x,t) \in \mathbb{R}^d \times \mathbb{R}$, with $u(x,0) = 0$, and $\frac{\partial u}{\partial t}(x,0) = f(x)$. If $f \in L^2$ we observe that $\frac{u(x,t)}{t} \to f(x)$ in the $L^2(\mathbb{R}^d)$ norm as $t \to 0$. One can show that $\lim_{t\to 0} \frac{u(x,t)}{t}$ exists and equals $f(x)$ a.e. if $f \in L^p$, $p > 2d/(d+1)$.

4.* The restriction phenomenon (inequality (31)) is valid in $\mathbb{R}^2$, for the full range $1 \leq p < 4/3$.

[Hint: One may dualize the assertion as in the proof of Theorem 5.2. Consider the operator $\mathcal{R}^*$ defined by

$$\mathcal{R}^*(F)(x) = \int_M e^{2\pi i x\cdot\xi} F(\xi)\, d\mu(\xi).$$

The desired result then becomes the inequality

$$\|\mathcal{R}^*(F)\|_{L^q(\mathbb{R}^2)} \leq A\|F\|_{L^p(d\mu)}$$

when $q = 3p'$ and $1 \leq p < 4$. Now the key point is that if we consider the singular measure $d\nu = Fd\mu$, then the convolution $\nu * \nu$ is actually an absolutely continuous measure $f\,dx$ with density f, a locally integrable function on $\mathbb{R}^2$. This fact reflects the assumed curvature of M. Indeed, it can be shown that $f \in L^p(\mathbb{R}^2)$, with $\frac{3}{r} = \frac{2}{p} + 1$, whenever $F \in L^p(d\mu)$ and $1 \leq p \leq 4$, and $\|f\|_{L^r(\mathbb{R}^2)} \leq c\|F\|^2_{L^p(d\mu)}$ and $1 \leq p < 4$. Now if this is so, then

$$\mathcal{R}^*(F)^2 = (\hat{\nu}(-x))^2 = (\nu * \nu)^\wedge(-x) = \hat{f}(x),$$

and by the Hausdorff-Young inequality,

$$\|\mathcal{R}^*(F)\|^2_{L^{2r'}} \leq \|(\mathcal{R}^*(F))^2\|_{L^{r'}} = \|\hat{f}\|_{L^{r'}} \leq c\|F\|^2_{L^p},$$

and this proves the assertion since $2r' = 3p'$.]

5.* An analog of Theorem 6.3 for the wave equation is as follows. Let $u(x,t)$ be the solution of the wave equation $\frac{\partial^2 u}{\partial t^2} = \triangle u$ for $(x,t) \in \mathbb{R}^d \times \mathbb{R}$, with initial data

$$\begin{cases} u(x,0) &= 0 \\ \frac{\partial u}{\partial t}(x,0) &= f(x). \end{cases}$$

Then $\|u(x,t)\|_{L^q(\mathbb{R}^d \times \mathbb{R})} \leq c\|f\|_{L^2(\mathbb{R}^d)}$ if $q = \frac{2d+2}{d-2}$ and $d \geq 3$.

6.* The following further results are known about $E(R) = N(R) - \pi R^2$, the error term appearing in Theorem 8.3.

(a) The Hardy series $R\sum_{k=1}^{\infty} \frac{r_2(k)}{k^{1/2}} J_1(2\pi k^{1/2} R)$ converges for each $R \geq 0$, and its sum equals $E(R)$ whenever $R \neq k^{1/2}$, for any positive integer k.

(b) The error $E(R)$ is on the average a multiple of $R^{1/2}$ in the sense that

$$\int_0^r E(R)^2 R\, dR = cr^3 + O(r^{2+\epsilon}),$$

for some $c > 0$ and every $\epsilon > 0$.

(c) However, $E(R)$ is not exactly $O(R^{1/2})$ since

$$\limsup_{R\to\infty} \frac{|E(R)|}{R^{1/2}} = \infty.$$

(d) It has been proved that $E(R) = O(R^{\alpha+\epsilon})$, for certain α, $1/2 < \alpha < 2/3$. A relatively recent result of this kind is for $\alpha = 131/208$.

7.* The oscillatory integral $\mathfrak{J}(\lambda)$ can be identified in terms of Bessel functions of the second and third kind. One has that

$$\mathfrak{J}(\lambda) = 4K_0(2\lambda) - 2\pi Y_0(2\lambda),$$

where Y_m and K_m are respectively Neumann and Macdonald functions.

8.* Consider the error term in the divisor problem

$$\Delta(\mu) = \sum_{k=1}^{\mu} d(k) - \mu \log \mu - (2\gamma - 1)\mu - 1/4.$$

It is given for μ not an integer by the convergent series

$$\frac{-2}{\pi}\mu^{1/2} \sum_{k=1}^{\infty} \frac{d(k)}{k^{1/2}} \left[K_1(4\pi k^{1/2}\mu^{1/2}) + \frac{\pi}{2} Y_1(4\pi k^{1/2}\mu^{1/2}) \right].$$

For Δ there are estimates, analogous to those for E in Problem 6, with $\Delta(\mu) = O(\mu^{\beta+\epsilon})$ and $\beta = \alpha/2$.